Microbiology

Career Avenues

Contents

Prokaryotes and Eukaryotes Cell Structure 4
Microbial Nutrition, Growth and Control 22
Microbial Metabolism 46
Nitrogen Fixation **59**
Mutations and their Chemical Basis 64
Microbial Genetics 84
Microbial Diversity 100
Viruses 105
Practice Questions 116

Career Avenues

PROKARYOTIC AND EUKARYOTIC CELL STRUCTURE

PROKARYOTES AND EUKARYOTES

"True" bacteria (which include all bacteria that infect man) are members of one kingdom (the eubacteria, bacteria). In addition, a group of organisms often found in extreme environments form a second kingdom (archaebacteria, *Archaea*). Morphologically, the two kingdoms of organisms appear similar, especially in the absence of a nucleus, and thus are classified together as prokaryotes. However, they have major biochemical differences. Most archaea live in environments such as hot sulfur springs where they experience temperatures as high as 80 degrees C and a pH of 2. These are called thermoacidophiles. Others live in methane-containing (methanogens) or high salt (extreme halophiles) environments.

Cells

- The smallest units of life
- Nothing smaller or simpler than a cell is considered alive
- All living things are made of one or more cells

Structures that all cells (eukaryotic and prokaryotic cells) contain:
• Cell membrane
• Cytoplasm
• Chromosome(s)
• Ribosomes

There are two basic cell types: Prokaryotic cells and Eukaryotic cells.

• **Prokaryotic cells:**
  - ✓ Cells with no membrane around the DNA(no nucleus).
  - ✓ Prokaryotic cells have no membranous organelles and are smaller than eukaryotic cells.
  - ✓ Bacteria and archaea domain species are prokaryotic cells.

• **Eukaryotic cells:**
  - ✓ Cells with a membrane around the DNA(a nucleus).
  - ✓ Eukaryotic cells have membranous organelles and are larger than prokaryotic cells
  - ✓ Eukarya domain species have eukaryotic cells

Organelles

- Structures inside the cell.
- Organelles carry out functions that keep the cell alive and operating.

Career Avenues

- Organelles can be membranous (made of a membrane (phospholipid bilayer)) or non-membranous (made of other molecules)

Plasma membrane (cell membrane)

- The phospholipid bilayer (and its associated proteins) that surround a cell.
- Its function is to act as a barrier and as a selective gateway (controlling the entry and exit of materials).
- The hydrophobic phospholipids are barriers to most solutes
- Transport proteins bring solutes through the membrane in a controlled manner.
- Receptors detect molecules outside the cell.

Cell wall

- A ridged structure outside the plasma membrane used for structural support of the cell.
- Made of cellulose in plant cells.
- Animal cells do not have cell walls

Cytoplasm/cytosol

- The semi-liquid that fills the cell.
- Organelles float in it and small molecules are dissolved in it

DNA

- The genetic molecule of the cell.
- Chromosome = A long strand of highly coiled double helix DNA, wrapped around histone proteins.
- Centromere = central region of the chromosome.
- Eukaryotes have one or more linear chromosomes that are located in the nucleus.
- Prokaryotes have a single circular chromosome located in the cytoplasm (prokaryotes have no nucleus)

Ribosome

- A protein-RNA organelle that constructs proteins by linking together amino acids

DNA function

- DNA contains the genetic information needed to operate the cell because it contains the instructions for making all the cell's proteins.

Gene

- A section of a DNA molecule that encodes one protein.
- Genes encode the amino acid sequence of each protein using codons

Career Avenues

Codon

- Three consecutive nucleotides.
- Each codon represents one of the amino acids.
- The genetic code = All 64 codons and the amino acid each one represents.
- Three of the codons represent stop (end-of-gene) signals

Steps from gene to protein:

1) Transcription of the gene into an mRNA (a temporary disposable RNA copy of the gene)
2) Translation of the mRNA into the protein

Transcription

- Making an RNA copy of one strand of a gene.
- The enzyme RNA Polymerase transcribes one DNA strand.
- The enzyme uses complementary base pairing with one DNA strand to generate an RNA copy of the other DNA strand.
- Messenger RNA (mRNA) = The single-stranded RNA copy of one DNA strand.
- In eukaryotes, the mRNA moves from the nucleus to the cytoplasm after transcription

Translation

- When a ribosome constructs a protein from amino acids using an mRNA as the instructions

Transfer RNA (tRNA)

- RNA molecules that bring amino acids to the ribosome.
- Each tRNA has an anticodon for the amino acid it is carrying.
- Anticodon = Three nucleotides complementary to an amino acid's codon.
- The anticodon binds to the mRNA's codon by complementary base pairing.
- This process matches each of the mRNA's codons to the correct amino acid, thus correctly assembling the protein

Nucleus

- A double membrane (**two** phospholpid bilayers) organelle that contains the chromosomes.
- Nuclear pores = Holes in the membrane to allow materials in and out of the nucleus

Endoplasmic reticulum (ER)

- A series of membranous compartments that is an extension of the nucleus' outer membrane.
- The rough ER (closest to the nucleus) is where the ribsomes make new proteins.
- The new proteins go inside the ER and travel to the smooth ER.

Career Avenues

- The smooth ER (farther from the nucleus) is where new lipids are made and where harmful substances are detoxified.
- The newly made proteins and lipids leave the ER in vesicles that bud off from the smooth ER

Vesicles

- Small membranous spheres that transport materials inside the cell

Golgi apparatus

- A stack of membranous sacs that is the cell's protein and lipid sorting and distribution center.
- The Golgi receives new proteins and lipids in vesicles from the smooth ER.
- They are modified and sorted as they move through the Golgi.
- They are distributed to other organelles in new vesicles that bud off from the Golgi

Exocytosis

- The expelling of substances from the cell by fusion of a vesicle with the plasma membrane

Endocytosis

- The formation of a vesicle by the budding inward of the plasma membrane.
- Example: Food vesicles bring macromolecules into the cell

Lysosome

- A membranous organelle used for the digestion of materials.
- Lysosomes bud directly from the Golgi.
- They have an acidic internal environment and contain digestive enzymes.
- Example: Lysosomes fuse with food vesicles to digest the vesicle's contents

Vacuoles

- Membranous spheres used for storage and other purposes.
- Central vacuole = A large vacuole found in plant cells that usually takes up most of the cell.

The Endomembrane System

- The membranous organelles that share materials with each other (by sending or receiving vesicles).
- Organelles of the endomembrane system are:
    - The nucleus
    - The endoplasmic reticulum
    - The Golgi apparatus
    - The cell membrane
    - Lysosomes

Career Avenues

- Vacuoles

- The following three membranous organelles are **not** part of the endomembrane system (They do **not** share materials. They do **not** send or receive vesicles)
  - Peroxysomes
  - Mitochondria
  - Chloroplasts

Peroxysome

- A membranous organelle that digests fatty acids using O_2

Mitochondria

- A double membrane organelle where ATP is made from the energy in glucose

Chloroplasts

- A triple membrane organelle where photosynthesis (making glucose using sunlight energy) occurs.
- Only present in the cells of plants (and some protistans)

Non-membranous organelles

- Organelles not made of membranes
  - The cytoskeleton
  - Centrioles
  - Ribosomes

Cytoskeleton

- A system of protein fibers that form a network throughout the entire cytoplasm.
- Holds organelles in place.
- Controls the shape of the cell.
- Example: Cilia and flagella.
- Forms "railroad tracks" for moving vesciles.
- Microtubules and actin filaments are examples of cytoskeletal fibers

Cilia and flagella

- Whip-like protrusions used for movement.
- Flagella = one long whip.
- Cilia = many tiny whips.
- Formed from microtubules inside cell

Centriole (basal body)

- A short cylinder made of microtubules.
- Centrioles anchor and organize microtubules.

Career Avenues

- Example: Microtubules in cilia and flagella are anchored to a basal body

DIFFERENCES BETWEEN EUKARYOTIC AND PROKARYOTIC CELLS

The difference between the structure of prokaryotes and eukaryotes is so great that it is considered to be the most important distinction among groups of organisms.

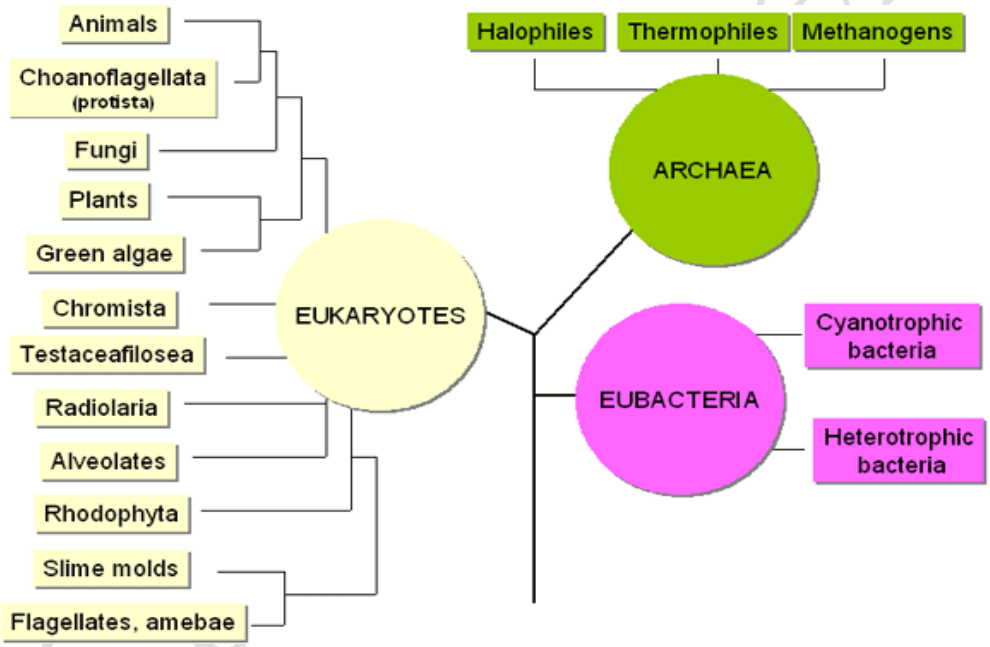
- The most fundamental difference is that eukaryotes do have "true" nuclei containing their DNA, whereas the genetic material in prokaryotes is not membrane-bound.
- In eukaryotes, the mitochondria and chloroplasts perform various metabolic processes and are believed to have been derived from endosymbiotic bacteria. In prokaryotes similar processes occur across the cell membrane; endosymbionts are extremely rare.
- The cell walls of prokaryotes are generally formed of a different molecule (peptidoglycan) to those of eukaryotes (many eukaryotes do not have a cell wall at all).
- Prokaryotes are usually much smaller than eukaryotic cells.
- Prokaryotes also differ from eukaryotes in that they contain only a single loop of stable chromosomal DNA stored in an area named the nucleoid, while eukaryote DNA is found on tightly bound and organised chromosomes. Although some eukaryotes have satellite DNA structures called plasmids, these are generally regarded as a prokaryote feature and many important genes in prokaryotes are stored on plasmids.
- Prokaryotes have a larger surface area to volume ratio giving them a higher metabolic rate, a higher growth rate and consequently a shorter generation time compared to Eukaryotes.
- Genes:
    - Prokaryotes also differ from eukaryotes in the structure, packing, density, and arrangement of their genes on the chromosome. Prokaryotes have incredibly compact genomes compared to eukaryotes, mostly because prokaryote genes lack introns and large non-coding regions between each gene.
    - Whereas nearly 95% of the human genome does not code for proteins or RNA or includes a gene promoter, nearly all of the prokaryote genome codes or controls something.
    - Prokaryote genes are also expressed in groups, known as operons, instead of individually, as in eukaryotes.
    - In a prokaryote cell, all genes in an operon(three in the case of the famous lac operon) are transcribed on the same piece of RNA and then made into separate proteins, whereas if these genes were native to eukaryotes, they each would have their own promoter and be transcribed on their own strand of mRNA. This lesser degree of control over gene expression contributes to the simplicity of the prokaryotes as compared to the eukaryotes.

Career Avenues

A BACTERIAL CELL

ARCHAEA

Based on DNA sequence similarities, it appears that the archaea and eukaryotes diverged from the eubacteria before they diverged from each other (see the figure given below) and in some ways, archaea are biochemically more like eukaryotes than they are the eubacteria. For example, the RNA polymerase of archaea is as complex, in terms of number of subunits, as the eukaryote nuclear polymerases and there is considerable amino acid homology with some of the eukaryotic subunits. Gene promoter structure in archaea is also more similar to that of eukaryotes than eubacteria, although, like the eubacteria, archaea have operons and transcribe these to polycistronic mRNA. Similarity also exists between the protein synthesis factors of archaea and eukaryotes suggesting that the overall protein synthesis mechanisms of eukaryotes and archaea may be similar. The 16S rRNAs of the eubacteria and the archaea are quite distinct in sequence.



Eubacteria (with the exception of the genera *Mycoplasma* and *Chlamydia*) possess peptidoglycan (synonyms: murein, mucopeptide, cell wall skeleton). Peptidoglycan, contains a unique sugar, muramic acid, not found elsewhere in nature. Archaebacteria contain a pseudomurein that is different in structure from eubacterial murein.

In view of the increasing number of similarities between the archaea and the eukaryotes, the term archaebacteria is no longer used. All other cellular forms of life (including plants, animals, and fungi) are referred to as eukaryotes.

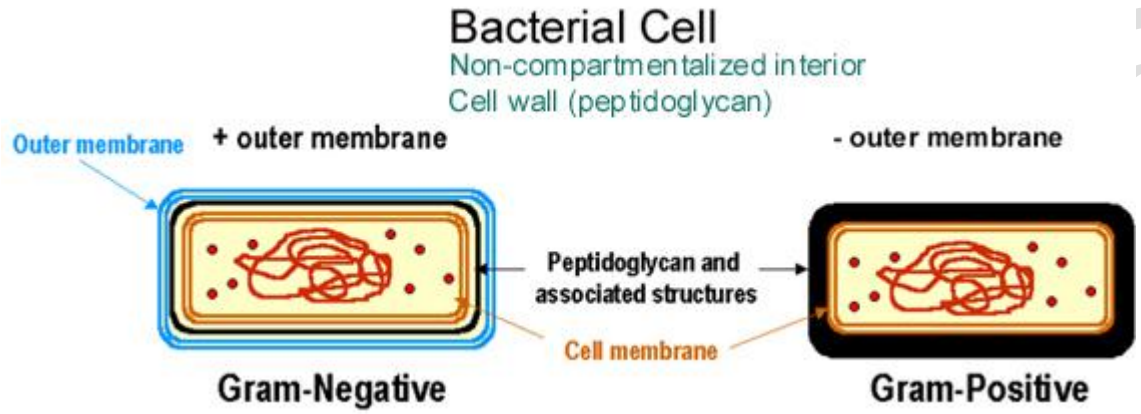
Members of the *Archaea* are not human pathogens and will not be discussed further.

Similarities between Archaea and Eukaryotes

	Eubacteria	Archaea	Eukaryotes
Nucleus	No	No	Yes: membrane-bound
Nucleosomes/histones	No	**Yes**	**Yes**
Operons/polycistronic mRNAs	Yes	Yes	No
Introns	No	No	Yes
TATA Box binding protein	No	**Yes**	**Yes**
Organelles	No	No	Yes: mitochondria, lysosomes, endoplasmic reticulum etc.
Chromosomes	One Circular	One Circular	More than one
RNA polymerase	One (simple)	**More than one (complex)**	**More than one (complex)**
Protein initiator amino acid	N-formyl methionine	**Methionine**	**Methionine**
Protein synthesis sensitivity to diphtheria toxin	Insensitive	**Sensitive**	**Sensitive**
Peptidoglycan	Yes	**No**	**No**
Protein synthesis	• initiation factors • ribosomal proteins • elongation factors of Archaea are more similar to those of eukaryotes than eubacteria		



Animal Cell
Compartmentalized interior
No cell wall



Any bacterial cell whether it is a coccus or a bacillus will have some structures common. These structures are cell wall, cell membrane, cytoplasm, ribosomes and the chromosome. Other intra-cellular structures such as plasmid, inclusion bodies and extra-cellular structures such as capsule, fimbriae and flagella are possessed only by some bacteria.

GLYCOCALYX/CAPSULE/SLIME

A gelatinous polysaccharide or polypeptide outer covering of certain bacteria is called glycocalyx. These are the structures that surround the outside of the cell envelope. The glycocalyx is referred to as a capsule if it is firmly attached to the cell wall, or as a slime layer if loosely attached. The chemical nature of bacterial capsules is diverse but majority of them are polysaccharides.

SIGNIFICANCE

- Virulence factor. Capsules of pathogenic bacteria inhibit ingestion and killing by phagocytes. It can also prevent complement-mediated bacterial cell lysis. Capsules protect the cells from lysozyme. Strains lacking capsule are avirulent.
- Permit bacteria to adhere to cell surfaces and structures such as medical implants and catheters. This is a first step in colonization and sometimes leads to disease.
- Prevent cell from drying out (desiccation)

- Toxicity to the host cell; capsule of *Bacteroides fragilis* induces abscess formation.
- Capsules may protect cells from bacteriophages.

CELL WALL

The layers of cell envelope lying between the cytoplasmic membrane and the capsule are referred to collectively as cell wall. In gram positive bacteria, the cell wall mainly consists of peptidoglycan and teichoic acid while the cell wall in gram negative bacteria includes peptidoglycan, lipoprotein, outer membrane and lipopolysaccharide layers. Cell wall does not take up any stain and hence are not seen by light microscope.

Most bacteria have a complex cell wall consisting of peptidoglycan (also called murein, mucopeptide). This complex polymer consists of three parts-

- A backbone consisting of alternating units of NAG (N-acetylglucosamine) and NAM (N-acetylmuramic acid).
- Tetrapeptide side chain attached to NAM
- Peptide cross-bridges, which are short chains of amino acids that crosslink the backbone.

SIGNIFICANCE OF CELL WALL

- Maintains cell shape, any cell that loses its cell wall, loses its shape as well.
- Protects bacteria from osmotic lysis
- Acts as a barrier, protects cell contents from external environment
- Determines reactivity to Gram stain, cells become gram negative if they lose cell wall
- Site of action of certain antimicrobial agents (E.g. Penicillins, Cephalosporins)
- Bacteria may attach to surface, produce slime, divide and produce microcolonies within the slime layer and construct a biofilm. E.g. formation of dental plaque mediated by the bacterium *Streptococcus mutans*.
- Confer specific antigenicity to a strain/species that can be exploited to detect and identify an isolate.

SUBSTANCES ACTING AGAINST CELL WALL

- Lysozyme, an enzyme found in tears and saliva breaks down a component of cell walls
- Antibiotics that inhibits cell wall synthesis such as Penicillins and cephalosporins
- Autolytic enzymes produced by some bacteria such as *Streptococcus pneumoniae*

Career Avenues

GRAM'S STAINING

Bacteria can be divided into two groups on the basis of staining with the Gram stain; Gram positive bacteria remain stained by crystal violet on washing, Gram negative do not.

It is based on the chemical and physical properties of their cell walls. Primarily, it detects peptidoglycan, which is present in a thick layer in Gram positive bacteria. A Gram positive results in a purple/blue color while a Gram negative results in a pink/red color.

Difference between Gram positive and Gram negative cell wall

Gram Positive	Gram negative
Thick layer of peptidoglycan	Thin layer of peptidoglycan
Teichoic acid	No teichoic acid
No Lipopolysaccharide	Lipopolysaccharide
No Periplasmic membrane	Periplasmic membrane
No Outer membrane	Outer membrane
High sensitivity to penicillin	Low sensitivity to penicillin
Low sensitivity to streptomycin	High sensitivity to streptomycin

STAINING MECHANISM

Gram-positive bacteria have a thick mesh-like cell wall made of peptidoglycan (50-90% of cell envelope), which are stained purple by crystal violet, whereas Gram-negative bacteria have a thinner layer (10% of cell envelope), which are stained pink by the counter-stain. There are four basic steps of the Gram stain:

- Applying a primary stain (crystal violet) to a heat-fixed smear of a bacterial culture. Heat fixing kills some bacteria but is mostly used to affix the bacteria to the slide so that they don't rinse out during the staining procedure.
- The addition of a mordant, which binds to crystal violet and traps it in the cell (Gram's iodine)
- Rapid decolorization with alcohol or acetone, and
- *Counterstaining* with safranin. Carbol fuchsin is sometimes substituted for safranin since it will more intensely stain anaerobic bacteria but it is much less commonly employed as a counterstain.

Crystal violet (CV) dissociates in aqueous solutions into CV^+ and chloride (Cl^-) ions. These ions penetrate through the cell wall and cell membrane of both Gram-positive and Gram-negative cells. The CV^+ ion interacts with negatively charged components of bacterial cells and stains the cells purple.

Iodine (I^- or $I{-}3$) interacts with CV^+ and forms large complexes of crystal violet and iodine (CV–I) within the inner and outer layers of the cell. Iodine is often referred to as a mordant,

Career Avenues

but is a trapping agent that prevents the removal of the CV–I complex and, therefore, color the cell.

When a decolorizer such as alcohol or acetone is added, it interacts with the lipids of the cell membrane. A Gram-negative cell will lose its outer lipopolysaccharide membrane, and the inner peptidoglycan layer is left exposed. The CV–I complexes are washed from the Gram-negative cell along with the outer membrane. In contrast, a Gram-positive cell becomes dehydrated from an ethanol treatment. The large CV–I complexes become trapped within the Gram-positive cell due to the multilayered nature of its peptidoglycan. The decolorization step is critical and must be timed correctly; the crystal violet stain will be removed from both Gram-positive and negative cells if the decolorizing agent is left on too long (a matter of seconds).

After decolorization, the Gram-positive cell remains purple and the Gram-negative cell loses its purple color. Counterstain, which is usually positively charged safranin or basic fuchsin, is applied last to give decolorized Gram-negative bacteria a pink or red color.

CELL MEMBRANE

Cell membrane or cytoplasmic membrane is a typical unit membrane composed of phospholipids (40%) and proteins (60%). It measures approximately 5-10 nm in thickness. It lies below the peptidoglycan layer of the cell wall and encloses the cytoplasm. The arrangement of proteins and lipids to form a membrane is called the fluid mosaic model. The membranes of bacteria (except Mycoplasma) do not contain sterols. It is a phospholipid bilayer with polar heads on either side of the membrane. Hydrophobic tails are oriented to the interior of the membrane. Specialized structures called mesosomes or chondroids are formed from the convoluted invaginations of cytoplasmic membrane. There are two types of mesosomes, septal mesosome and lateral mesosome. The bacterial chromosome is attached to the septal mesosome. During cell division, the septal mesosome participates in the formation of cross-walls. Mesosomes are more prominant in gram positive bacteria. They are believed to be analogous to eukaryotic mitochondria since they are rich in respiratory enzymes.

FUNCTIONS OF CELL MEMBRANE

- A selectively permeable barrier: substances that pass through the membrane are limited by pore sizes and the hydrophobic nature of the membrane.
- Integral (transmembrane) proteins form channels and act as carriers.
- Peripheral proteins can act as receptors and as enzymes for metabolic reactions.
- Electron transport and oxidative phosphorylation: cytochromes and dehydrogenases of respiratory chain are located in the cell membrane.
- Excretion of hydrolytic enzymes.
- Site of attachment of the chromosome.
- Bear receptors and proteins of sensory transduction system.

Career Avenues

SUBSTANCES ACTING ON CELL MEMBRANE

- Detergents that contain lipophilic and hydrophilic groups disrupt cytoplasmic membranes.
- Antibiotics such as Polymyxin B and Gramicidin selectively damages membrane.
- Ionophores (E.g. Valinomycin) are compounds that permit rapid diffusion of cations through the membrane.
- Chemical agents such as alcohols and quaternary ammonium compounds.

CYTOPLASM

The cytoplasm or protoplasm is the portion of the cell that lies within the cytoplasmic membrane. It is gel-like in consistency and includes the procaryotic chromosome and ribosomes. Constituents of cytoplasm include proteins (including enzymes), vitamins, ions, nucleic acids and their precursors, amino acids and their precursors, carbohydrates and their derivatives, fatty acids and their derivatives. The cytoplasm does not exhibit any internal mobility (cytoplasmic streaming). The cytoplasm also lacks organelles such as mitochondria, golgi apparatus or endoplasmic reticulum. Cytoplasm stains uniformly in young cultures.

CHROMOSOME

The chromosome in bacteria is typically a single, closed circle DNA that is concentrated in a nucleoid region. It is not membrane bound as in eukaryotes. Some bacteria possess smaller extrachromosomal pieces of DNA called plasmids. Plasmids replicate independently of the chromosome and carry genes that are not essential for cell survival but may give some advantage to an organism. The chromosome is attached to an invagination of the cytoplasmic membrane called mesosome. Mitotic apparatus and nuclear membrane are completely lacking. The length of *E.coli* chromosome is approximately 1.4 mm but is condensed inside the cell by supercoiling. DNA is mainly negatively charged hence bind readily to basic dyes. It can be demonstrated by Feulgen stain or by electron microscopy.

RIBOSOMES

Bacterial cells can contain thousands of ribosomes, which are the sites of protein synthesis. The distinct granular appearance of procaryotic cytoplasm is due to the presence and distribution of ribosomes. Often they aggregate to form structures known as polysomes. Bacterial ribosomes are termed 70 S (Svedberg units) and eukaryotic ribosomes are termed 80S. The difference between bacterial and eukaryotic ribosomes is often exploited during antibiotic therapy.

INCLUSION BODIES

Intracytoplasmic inclusions can be vacuoles, crystals or storage bodies. Bacteria often store reserve material in the form of insoluble cytoplasmic granules. Inclusions accumulate when a cell is grown in the presence of excess nutrients and they are often observed under laboratory conditions. Various examples of these bodies are:

Career Avenues

- Starch/Glycogen granules - blue-greens and enteric bacteria
- Poly-ß-hydroxybutyrate granules - Azotobacter and Rhizobium
- Nitrogen-reserve granules - blue-greens
- Sulphur inclusions – Thiotrix
- Lipid inclusions
- Volutin granules – *Corynebacterium diphtheriae*

The inclusion bodies can be appreciated using phase contrast microscope or using special stains such as Albert's stain (volutin granules) or Sudan black (lipid inclusion).

FLAGELLA

Some bacteria are motile and some are not. Almost all motile bacteria possess flagella as the organ of locomotion. Such bacteria tend to move towards or away from the source of stimulus. These stimuli can be chemicals (chemotaxis), light (phototaxis), air (aerotaxis) or magnetism (magnetotaxis).

STRUCTURE

Procaryotic flagella are much thinner than eukaryotic flagella and they lack the typical 9 + 2 arrangement of microtubules. Over 40 genes are involved in its assembly and function. They are approximately 3-20µm long and end in a square tip. Since flagella are very thin (20-30 nm in diameter), they are below the resolution limits of a normal light microscope and cannot be seen. The bacterial flagellum is a non-contractile, composed of single kind of protein subunit called flagellin. It is anchored to the bacterial cytoplasmic membrane and cell well by means of disk like structures. A flagellum comprises of three parts, filament, hook and basal body. The flagellum is attached to the cell body by hook and basal body. While the hook and basal body are embedded in the cell envelope, the filament is free. If a flagellum is cut off it will regenerate until reaches a maximum length. This is so because the growth is not from base, but from tip. The basal body bears a set of rings, one pair in gram positive bacteria and two pairs in gram negative bacteria. While the rings named S and M are common to both, the rings names P and L are found only in gram negative bacteria. Rings in the basal body rotate relative to each other causing the flagella to turn like a propeller. The energy to drive the basal body is obtained from the proton motive force. Bacteria move at average speed of 50µm/sec, the fastest being *Vibrio cholerae* that moves 200µm/sec. The numbers of flagella, as well as their location on the cell surface are characteristic of a species.

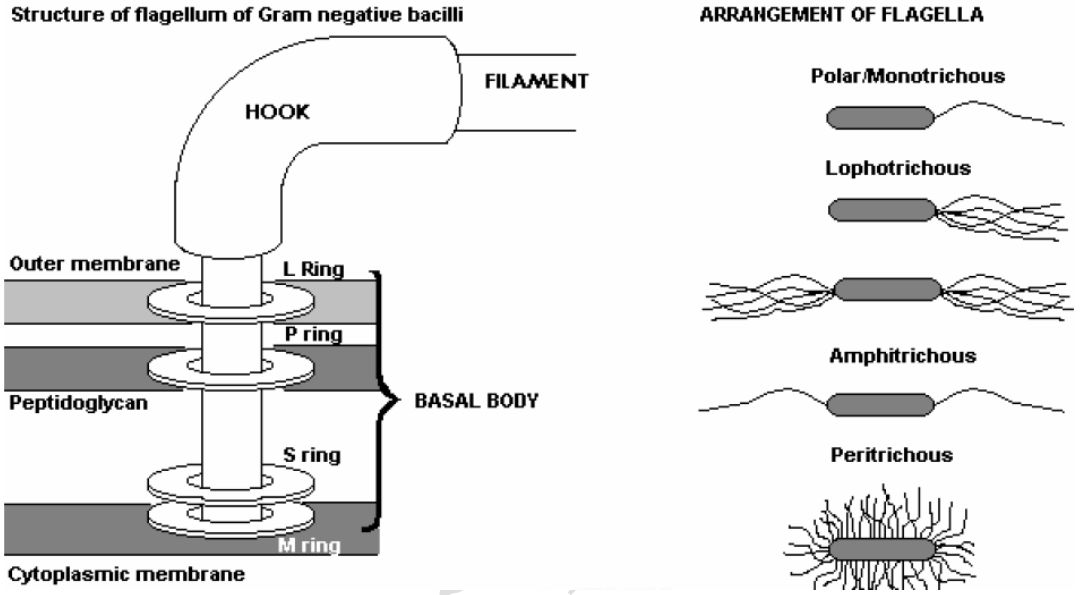
Flagella arrangements are

- **Monotrichous** - a single flagellum at one pole (also called polar flagellum) E.g. *Vibrio cholerae*
- **Amphitrichous** - single flagellum at both poles. Eg. Spirilla

Career Avenues

- **Lophotrichous** - two or more flagella at one or both poles of the cell E.g. *Spirillum undula*
- **Peritrichous** - completely surrounded by flagella E.g. *E.coli.*

Other mechanisms of bacterial locomotion include gliding and motion by axial filament contraction. Gliding is movement of bacteria along solid surfaces by an unknown mechanism. Spirochetes have internally-located axial filaments or endo flagella. Axial filaments wrap around the spirochete towards the middle from both ends. They are located above the peptidoglycan cell wall but below the outer membrane.



Significance of flagella

- Primarily function is motility (chemotaxis, aerotaxis, phototaxis etc). Positive taxis is movement toward a favorable environment whereas negative taxis is movement away from a repellent.
- Flagella can help in identifying certain types of bacteria. For example, Proteus species show 'swarming' type of growth on solid media.
- Flagellar antigens are used to distinguish different species and strains of bacteria (serovars). Variations in the flagellar H antigen are used in serotyping.

FIMBRIAE AND PILI

Fimbriae are short, hair-like structures made up of protein pilin and are present in many gram negative bacteria. Even though pili arise from plasma membrane they are not considered part of the plasma membrane. They are anchored in the membrane and protrude through the cell wall to the outside of the cell. Fimbriae are shorter and straighter than flagella and are more numerous. They are 0.5µm long and 10 nm thick. Since they are made up of protein, they are antigenic. Bacteria from different genera may possess common fimbrial antigens. Fimbriae are usually seen in young cultures and lost on subcultures on solid media. While some

authors use the two terms (fimbriae and pili) interchangeably, some restrict the term pili to denote sex pili. Sex pili acts to join bacterial cells for transfer of DNA from one cell to another by a process called conjugation.

SIGNIFICANCE

- They act as adhesins and allow bacteria to colonize cells. For example, *Neisseria gonorrhoea* uses its fimbriae to attach to the lining of the genital tract and initiate an infection.
- Fimbriae can also detect chemical signals and are important in bacterial cell communication and biofilm formation.
- Fimbriae also act as receptors for bacteriophages.
- Fimbriae of *Streptococcus pyogenes* are coated with M protein, which acts as an important virulence factor by adhering to host cells and resisting phagocytosis.
- Fimbriated bacteria form surface pellicle of liquid media.
- Some fimbriae can agglutinate RBC of guinea pigs, horses, pigs and fowls. This haemagglutination may or may not be inhibited by mannose.

SPORE

In poor growth conditions some bacteria such as Bacillus and Clostridium produce resistant survival forms termed endospores. This process is known as sporulation. Bacterial spores are endospores in contrast to fungal spores, which are usually exospores. Unlike the spores of fungi, bacterial spores do not serve reproductive function. They are resistant to extreme environmental conditions such as high temperatures, dryness, toxic chemicals (disinfectants, antibiotics), and UV radiation. Once the endospore is formed, the vegetative portion of the bacterium is degraded and the dormant endospore is released. The endospore is able to survive for long periods of time until environmental conditions again become favorable for growth. The endospore then germinates, producing a single vegetative bacterium. Spores can be killed by sterilization methods such as autoclave and hot air oven. Some chemical disinfectants such as formaldehyde and ethylene oxide can also kill spores.

SIGNIFICANCE OF SPORES

- Since they are resistant forms of bacteria, they can survive unfavourable conditions for long period.
- Since spores occur in soil, wounds contaminated by soil can lead to infections like gangrene or tetanus.
- Since spores survive ordinary disinfection, they may contaminate surgical wounds.
- Since spores are everywhere, they may contaminate bacterial culture media.

Career Avenues

- Since they are highly heat resistant, they can be used to monitor the efficacy of sterilization process in autoclave (*Bacillus stearothermophilus*) and hot air oven (*Clostridium tetani* var *niger*).

GATE BT 2010

Q.1 Which one of the following DOES NOT belong to the domain of Bacteria?

(A) Cyanobacteria
(B) Proteobacteria
(C) Bacteroides
(D) Methanobacterium

Q.2 Match Group I with Group II

Group I	Group II
P. Staphylococcus aureus	1. Biofilms
Q. Candida albicans	2. Bacteriocins
R. Mycobacterium tuberculosis	3. Methicillin resistance
S. Lactobacillus lactis	4. Isoniazid

(A) P-1, Q-4, R-2, S-3 (B) P-2, Q-3, R-1, S-4
(C) P-3, Q-1, R-4, S-2 (D) P-1, Q-2, R-4, S-3

GATE BT 2011

Q.3 The lipopolysaccharides present in bacterial cell wall has lipid A which is connected to

(A) O – Polysaccharide
(B) Core polysaccharide
(C) Both with O – polysaccharide and core polysaccharide
(D) Rhamnose – mannose disaccharide

Q.4 Match the products in Group I with their respective organisms in Group II

Group I	Group II
P. Glycerol	1.Corynebacterium glutamicum
Q. Glutamic acid	2. Alcaligenes faecalis
R. Curdlan	3. Dunaliella salina

Career Avenues

S. Amphotericin B — 4. Streptomyces nodosus

(A) P-2, Q-1, R-3, S-4 (B) P-4, Q-2, R-1, S-3

(C) P-3, Q-1, R-2, S-4 (D) P-2, Q-1, R-4, S-3

GATE BT 2012

Q.5 Match the organisms in Group I with the entries in Group II

Group I	Group II
P. Clostridium	1. Rods with teichoic acid in the cell wall
Q. Escherichia	2. Rods with endospores
R. Vibrio	3. Helical rods with flagella
S. Bacillus	4. Rods with LPS in the outer membrane
	5. Curved rods with polar flagella

(A) P-2, Q-4, R-5, S-1 (B) P-2, Q-1, R-5, S-4

(C) P-5 Q-4, R-2, S-3 (D) P-3 Q-2, R-1, S-4

GATE BT 2013

Q.6 In nature, the horizontal gene transfer across bacteria is mediated by

(A) gene cloning followed by transformation
(B)conjugation and transformation
(C) conjugation only
(D) transformation only

Answer:

- **Q.1 – A.**
- **Q.2 – C.**
- **Q.3 – A.**
- **Q.4 - C.**
- **Q.5 – A.**
- **Q.6 – B.**

Career Avenues

MICROBIAL NUTRITION, GROWTH AND CONTROL

THE COMMON NUTRIENT REQUIREMENTS

Analysis of microbial cell composition shows that over 95% of cell dry weight is made up of a few major elements: carbon, oxygen, hydrogen, nitrogen, sulfur, phosphorus, potassium, calcium, magnesium, and iron. These are called macroelements or macronutrients because they are required by microorganisms in relatively large amounts.

All organisms, including microorganisms, require several micronutrients or trace elements besides macroelements. The micronutrients — manganese, zinc, cobalt, molybdenum, nickel, and copper — are needed by most cells.

Note: Diatoms need silicic acid (H_4SiO_4) to construct their beautiful cell walls of silica [$(SiO_2)_n$]. Although most bacteria do not require large amounts of sodium, many bacteria growing in saline lakes and oceans depend on the presence of high concentrations of sodium ion (Na^+).

Prokaryotes can be divided into various physiological groups based on how they derive energy and assimilate carbon.

- Carbon source
    - Autotroph: carbon from CO2
    - Heterotroph: carbon from organic molecules
- Energy source
    - Phototroph: light
    - Chemotroph: oxidation of organic or inorganic compounds
- Electron source
    - Lithotroph: reduced inorganic molecules (lithos = stone)
    - Organotroph: organic molecules

Career Avenues

MAJOR NUTRITIONAL TYPES OF MICROORGANISMS

Major Nutritional Types	Sources of Energy, Hydrogen/Electrons, and Carbon	Representative Microorganisms
Photolithotrophic autotrophy (Photolithoautotrophy)	Light Energy Inorganic hydrogen/electron(H/e^-) donor CO_2 carbon source	Algae Purple and Green sulfur bacteria Cyanobacteria
Photoorganotrophic heterotrophy (Photoorganoheterotrophy)	Light energy Organic H/e^- donor Organic carbon source (CO_2 may also be used)	Purple nonsulfur bacteria Green nonsulfur bacteria
Chemolithotrophic autotrophy (Chemolithoautotrohy)	Chemical energy source(inorganic) Inorganic H/e^- donor CO_2 carbon source	Sulfur-oxidizing bacteria Hydrogen bacteria Nitrifying bacteria Iron-oxidizing bacteria
Chemoorganotrophic Heterotrophy (Chemoorganoheterotrophy)	Chemical energy source(organic) Organic H/e^- donor Organic carbon source	Protozoa Fungi Most nonphotosynthetic bacteria (including most pathogens)

THE MOVEMENT OF NUTRIENTS ACROSS MEMBRANES

SIMPLE DIFFUSION

Simple diffusion results from the molecular kinetic energy and random movement of particles. The role of diffusion in living cells depends on the size of particles, nature of membranes, and distances substances must move inside cells.

FACILITATED DIFFUSION

Facilitated diffusion uses protein carrier molecules or protein-lined pores in membranes in moving ions or molecules from high to low concentrations.

OSMOSIS

Osmosis is the net movement of water molecules through a **selective permeable** membrane from a region of higher concentration of water to a region of lower concentration. The **osmotic pressure** of a solution is the pressure required to prevent such a flow.

Career Avenues

ACTIVE TRANSPORT

Active processes that move substances across membranes generally result in movement from regions of lower concentration of the substances to regions of higher concentration and require the cell to expand energy.

Active transport requires a protein carrier molecule in a membrane, a siurce of ATP, and an enzyme that releases energy from ATP.

Active transport is important in cell functions because it allows cells to take up substances that are in low concentration in the environment and to concentrate those substances within the cell.

ENDOCYTOSIS AND EXOCYTOSIS

Endocytosis and exocytosis, which occur only in eukaryotic cells, involve formation of vesicles from fragments of plasma membrane and fusion of vesicles with the plasma membranes, respectively.

In endocytosis the vesicles enters the cell, as in phagocytosis. In exocytosis the vesicle leaves the cell, as in secretion.

Endocytosis and exocytosis are important because they allow the movement of relatively large quantities of materials across plasma membranes.

TYPES OF CULTURE MEDIA

Some microorganisms, particularly photolithotrophic autotrophs such as cyanobacteria and eucaryotic algae, can be grown on relatively simple media containing CO_2 as a carbon source (often added as sodium carbonate or bicarbonate), nitrate or ammonia as a nitrogen source, sulfate, phosphate, and a variety of minerals. Such a medium in which all components are known is a defined medium or synthetic medium. Many chemoorganotrophic heterotrophs also can be grown in defined media with glucose as a carbon source and an ammonium salt as a nitrogen source.

COMPLEX MEDIA

Media that contain some ingredients of unknown chemical composition are complex media. Complex media contain undefined components like peptones, meat extract, and yeast extract. Peptones are protein hydrolysates prepared by partial proteolytic digestion of meat, casein, soya meal, gelatin, and other protein sources. They serve as sources of carbon, energy, and nitrogen. Beef extract and yeast extract are aqueous extracts of lean beef and brewer's yeast, respectively. Beef extract contains amino acids, peptides, nucleotides, organic acids, vitamins, and minerals. Yeast extract is an excellent source of B vitamins as well as nitrogen and carbon compounds. Three commonly used complex media are (1) nutrient broth, (2) tryptic soy broth, and (3) MacConkey agar Selective media favor the growth of particular microorganisms. Bile salts or dyes like basic fuchsin and crystal violet favour the growth of

Career Avenues

gram-negative bacteria by inhibiting the growth of gram-positive bacteria without affecting gram-negative organisms. Endo agar, eosin methylene blue agar, and MacConkey agar three media widely used for the detection of E. coli and related bacteria.

Differential media are media that distinguish between different groups of bacteria and even permit tentative identification of microorganisms based on their biological characteristics. Blood agar is both a differential medium and an enriched one. It distinguishes between hemolytic and nonhemolytic bacteria. Hemolytic bacteria (e.g., many streptococci and staphylococci isolated from throats) produce clear zones around their colonies because of red blood cell destruction. MacConkey agar is both differential and selective. Since it contains lactose and neutral red dye, lactose-fermenting colonies appear pink to red in color and are easily distinguished from colonies of nonfermenters.

OXYGEN REQUIREMENTS

Toxic forms of oxygen (free radicals)

- **Singlet oxygen**: molecular O_2 with energized electrons
- **Superoxide radical** (O_2^-) : superoxide dismutase enzyme required to change superoxide radical into hydrogen peroxide and oxygen; anaerobes die because they lack this enzyme
- **Peroxide ion** (O_2^{2-}): - They are produced through following reactions:

$$20_2 + 2H^+ \rightarrow 0_2 + H_2O_2$$

$$O_2 + H_2O_2 \text{ Chelated iron } O_2 + OH + OH$$

OH radical are among the most reactive free radical known to organic chemistry and can damage almost every kind of molecule found in living cells.

It is not a free radical, but it is a powerful oxidizing agent that is toxic to many cells.
This is why hydrogen peroxide (H2O2) is antimicrobial;

Defense Mechanism: Catalase enzyme: converts H2O2 into H2O and O2,

- Peroxidase: H2O2 into water and NAD+ (oxidized)
- Enzyme superoxide dismutase (JOD) SOD convert the superoxide free radial into molecular oxygen (0_2) and H_2O_2
- **Hydroxyl radical**: most reactive of the four; enzymes above minimize production

Types of organisms and need for oxygen

- **Aerobes**: use oxygen; have enzymes for detoxifying
- **Facultative anaerobes**: can use anaerobic respiration or fermentation although prefer aerobic respiration; *Escherichia*

Career Avenues

- **Aerotolerant anaerobes**: don't use but can tolerate oxygen because they have some of the enzymes that detoxify it (often peroxidases since they don't make more O_2); *Lactobacilli*
- **Microaerophiles**: very limited ability to detoxify oxygen; can tolerate smaller amounts than aerotolerant anaerobes; *Helicobacter pylori*
- **Anaerobes**: killed by oxygen because they can't detoxify it; *Clostridia*
- Obligate aerobes: algae, most fungi, protozoa and many prokaryotes
- Facultative anaerobes: some yeast (fungi) and many prokaryotes
- Other oxygen use patterns: many prokaryotes and a few protozoa

METABOLIC USES FOR OXYGEN

- Structural component of organic compounds (C,H,O,N)
- Final electron acceptor in aerobic respiration pathway
- Part of bacteriocidal response in organisms with lysosomes, neutrophils, macrophages, etc.
- Photosynthesis waste product

NITROGEN REQUIREMENTS

- Growth limiting nutrient
- For use in metabolic reactions N must be in the form of ammonium or ammonia
- **Nitrogen fixation**: rare but essential function; N_2 (atmospheric gas) → NH_3

OTHER CHEMICAL REQUIREMENTS

- May be limiting (phosphorus) and may be accumulated by the organism (inclusion bodies: sulfur)
- Major elements: sulfur, magnesium, calcium, copper, iron, phosphorus, manganese
- **Trace elements**: selenium, zinc, cobalt

Note that because of our similar metabolic pathways, in general, a (trace element), or **micronutrient** for a prokaryote is a micronutrient for eukaryotes (check out your multivitamin bottle)

- **Growth factors**: some organisms need none (algae and some bacteria = lithotropic photoautotrpohs), a few (many prokaryotes synthesize their own vitamins), others require standard vitamins and still others are **fastidious** requiring many different and complex organic compounds

Career Avenues

TEMPERATURE

Effects on

- Shape of protein compounds
- Rigidity/fluidity of cell membranes & other lipid structures

Minimum growth temperature →optimum growth temperature → maximum growth temperature describes the temperature range of an organism

- Psychrophiles: optimum is a low temperature
- Mesophiles: medium temperatures; often our pathogens
- Thermophiles: grow well at high to boiling temperatures; include many Archaea and have been fruitful in providing unique enzymes important in genetic engineering (stable at high temperatures)

Cold temperatures do not kill organisms but do slow the growth rate; extreme cold and drying (= lyophilization) may even be used to preserve organisms in type culture (standard strains) collections.

PH

Effects on

- Shape of protein and nucleic acid compounds
- Availability for metabolic reactions

NEUTROPHILES

- pH range of 6.5 to 7.5
- Our pathogenic organisms are of this type
- Organisms producing acid waste products in a limited environment will eventually stop growing; has been used in food preservation (ex: sauerkraut & dill pickles)

ACIDOPHILES

- Obligate acidophiles: ex: chemoautotrophs that use sulfur as the acceptor of excess hydrogen ions to produce sulfuric acid (H2S)
- Acid-tolerant organisms:
    - May manipulate environment: *Helicobacter pylori* secretes bicarbonate and urease (urea → ammonia which is basic)

Career Avenues

ALKALINOPHILES

- Yeasts: disruption of normal acid-tolerant bacteria permits overgrowth of yeasts in more alkaline vaginal environment
- Increases in alkalinity of soil and water by natural changes in the environment encourage growth of the cholera bacterium Vibrio cholerae

GROWTH OF MICROBIAL POPULATIONS

- **Binary fission**: duplication of the chromosome followed by equal division of the cytoplasm between the daughter cells
- **Logarithmic** or **exponential growth**: number of cells in the population = 2^n, where n is the number of generations

GENERATION TIME

Time from one cell division event to the next, under optimal conditions

- Can have consequences in pathogenicity and lab culture
- *E. coli* and *S. aureus* = 20 minutes
- *Mycobacterium* = 10 days

PHASES OF MICROBIAL GROWTH

Growth Phases of a Simple Batch Culture:

- Lag phase
- Accelerating growth phase
- Log (exponential growth) phase
- Declining growth phase
- Stationary phase
- Death or declining death phase
- Log death phase

Lag phase:

- Adjustment to new medium.
- Adaptation (acclimation) period - the lag phase is the initial phase which represents the period (time) required for bacteria to adapt to their new environment.
- Constant number of cells - during this phase, the individual bacterial cells increase in size, but the number of cells remains unchanged.
- Physiologically active - they are very active physiologically and are synthesizing new enzymes and activating factors.

Career Avenues

Accelerating Growth Phase

- Transition period from the lag phase to the log phase.
- Cell is beginning to grow (increase in numbers) noticeably as enzyme systems are gearing up.

Log (Exponential Growth) phase

- Rapid constant log growth
- Most pathogenic in this phase
- Physiological traits usually expressed best (staining reactions, etc.)
- Can be maintained indefinitely if in a chemostat that replaces nutrients and removes wastes.
- Exponential growth - during this phase, the bacterial cells divide regularly at a constant rate.
- Straight line on semilog scale - The logarithms of the number of cells plotted against time results in a straight line.
- Maximum Rate of Substrate utilization - A maximum growth rate occurs under optimal conditions, and substrate is removed from the medium at the maximum rate.
    - ✓ The growth rate is limited only by the bacteria's ability to process the substrate.
    - ✓ Food is in excess (not limiting) so that the rate of growth is only limited by the ability to process the food.
    - ✓ Sometimes called "0-order growth" and growth rate is constant and maximum.

Declining Growth Phase

- Transition period from the log phase to the stationary phase.
- Decreasing growth rate
- Exhaustion of essential nutrients
- Accumulation of toxic metabolic products - the growth rate can be limited either by the exhaustion of essential nutrients or by the accumulation of toxic metabolic products.
    - ✓ Food becomes limiting factor and therefore growth rate and mass of bacteria are dependent on the amount of food present.

Stationary phase:

- Rate of growth = Rate of death.
- The number of cells remains constant perhaps as a results of complete cessation of division or the balancing of reproduction rate by an equivalent death rate.
- Growth of new cells is balanced by the death of old cells.
- No increase in cell mass
- Population is "stable"
- Net growth rate = 0

Career Avenues

Death phase or Declining Phase

- Death exceeds growth
- If microbes can form resistant stages (cysts, endospores,etc) these will be made in this phase.
- The number of viable cells decreases slowly while the total mass may remain constant due to the fact that the death rate exceeds the production rate of new cells.
- Depletion of essential nutrients
- Accumulation of inhibitory products. - Death occurs primarily as a result of depletion of essential nutrients and/or the accumulation of inhibitory products.

Log Death Phase

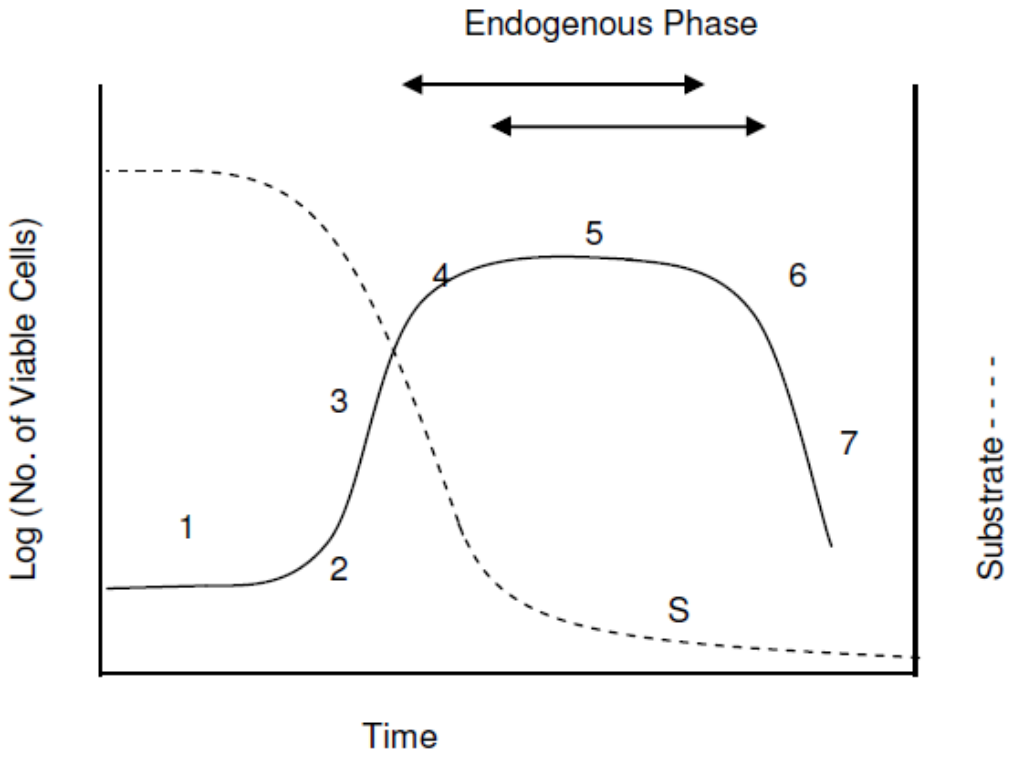
- Exponential death - "wholesale die-off"
- System is dead.
- Even if you add food, you will get no growth.

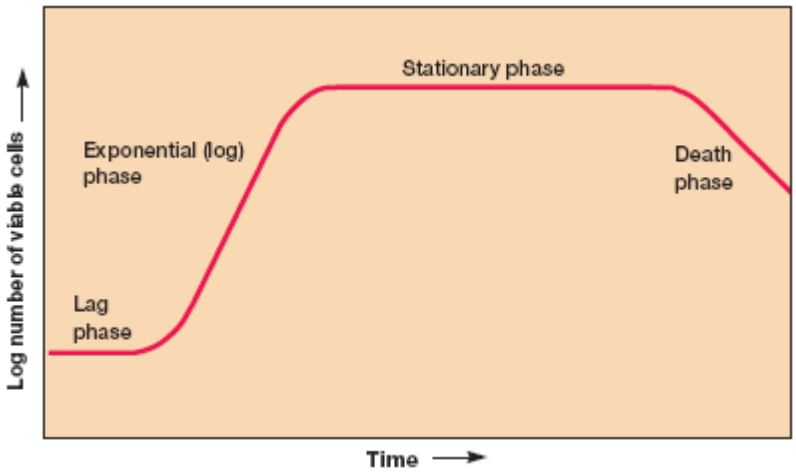
Endogenous Phase

- Near-starvation condition - That portion of the bacterial growth curve encompassing parts of the stationary and declining phases in which microorganisms are in near-starvation is more frequently called the endogenous phase.

Growth curve is for a batch system (fed once) and pure culture. - real biological treatment systems are typically continuously fed (and wasted). Growth of bacteria is dependent upon many other factors. Activated sludge and other BWT system contain many types of organisms - i.e., biological treatment systems are ecosystems and as such the growth characteristics of one organism may be affected by other organisms. In general, cells settle much better when they are in endogenous phases. In real biological treatment systems, you must operate at a positive growth rate or you will eventually lose your bacteria.

Career Avenues





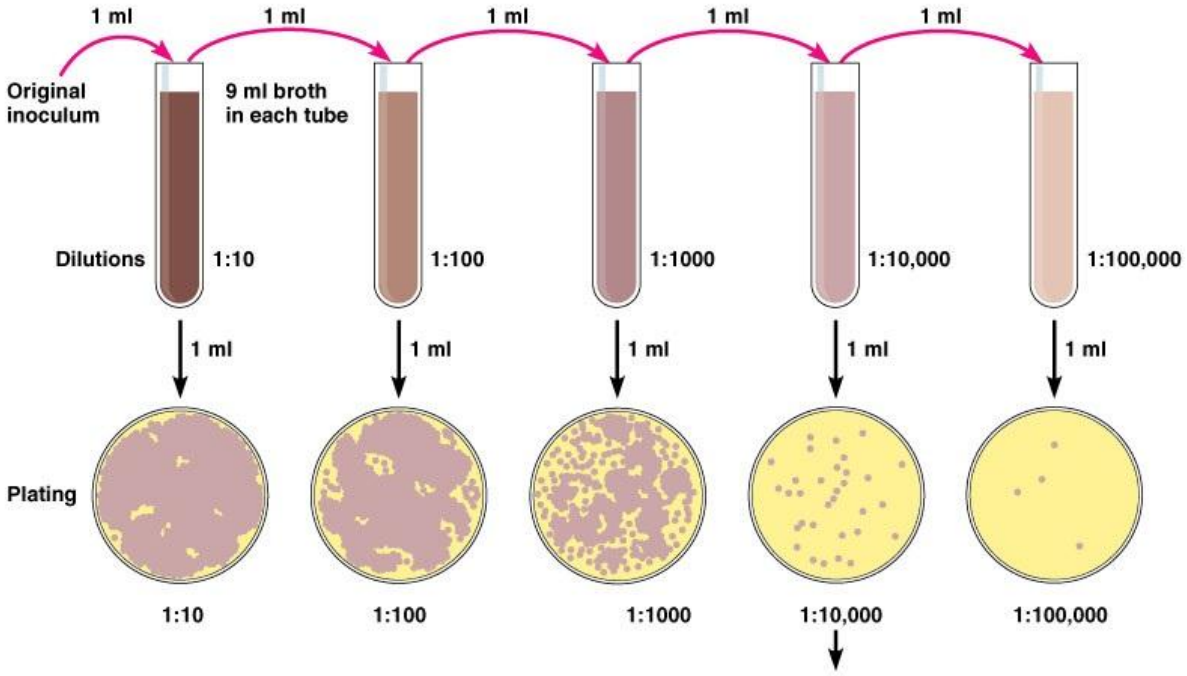
MEASURING MICROBIAL GROWTH

DIRECT METHODS

Plate counts

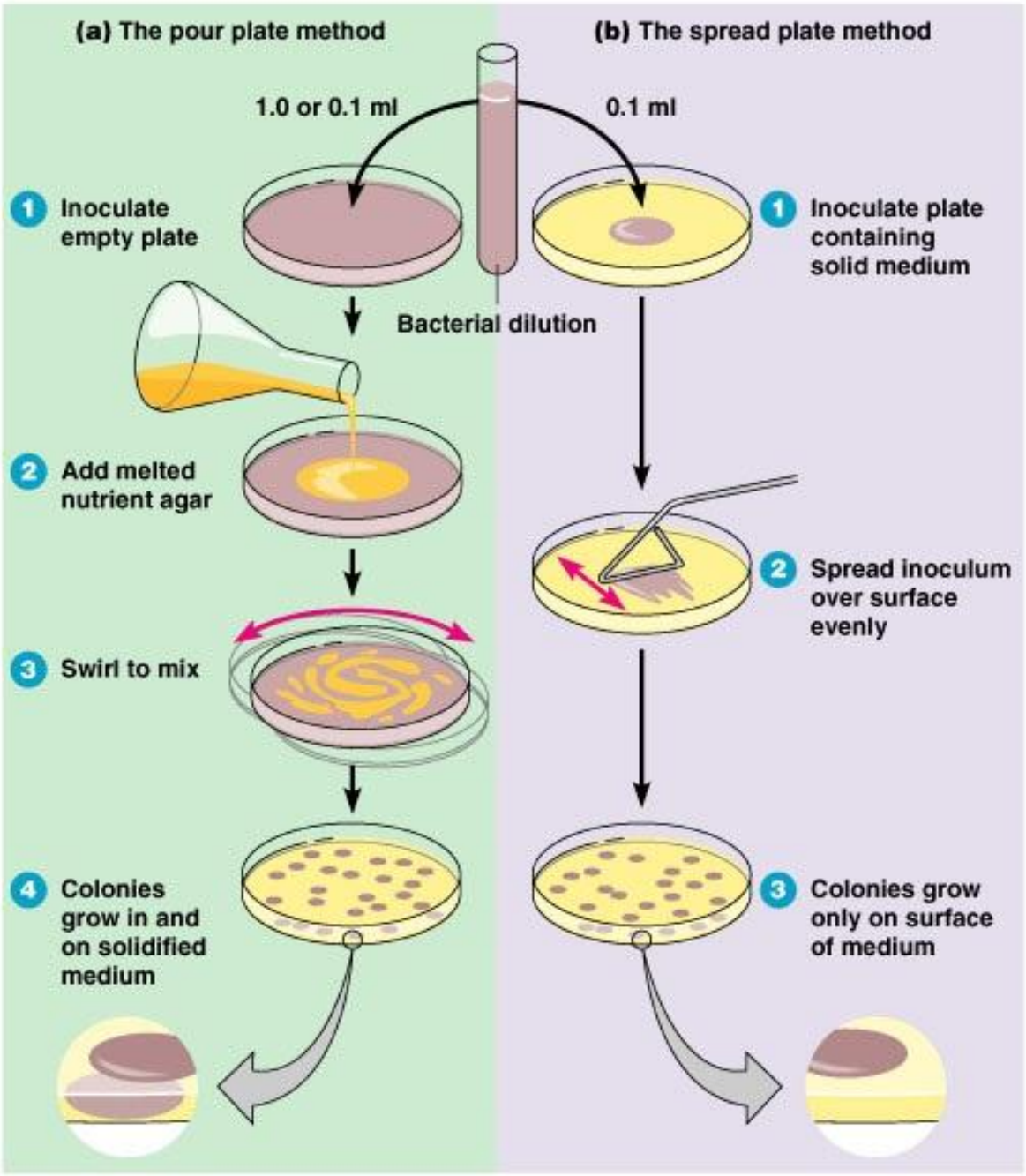
- Serial dilutions are plated out until colonies on one plate can be counted.
- Most frequently used method of measuring bacterial populations.
- Inoculate plate with a sample and count number of colonies.
- Each colony originates from a single bacterial cell.
- Original inoculum is homogeneous.
- No cell aggregates are present.

- **Advantages:** Measures **viable** cells.
- **Disadvantages:**
  - ✓ Takes 24 hours or more for visible colonies to appear.
  - ✓ Only counts between 25 and 250 colonies are accurate.
  - ✓ Must perform serial dilutions to get appropriate numbers/plate.
- Pour plate method, Spread plate method, etc.



Calculation: Number of colonies on plate × reciprocal of dilution of sample = number of bacteria/ml
(For example, if 32 colonies are on a plate of $^{1}/_{10,000}$ dilution, then the count is 32 × 10,000 = 320,000/ml in sample.)

Caree



Membrane filtration

- Known volume of sample is filtered through a membrane and the membrane is placed on a suitable culture medium.
- Used to measure small quantities of bacteria.
- **Example**: Fecal bacteria in a lake or in ocean water.
- A large sample (100 ml or more) is filtered to retain bacteria.
- Filter is transferred onto a Petri dish.
- Incubate and count colonies.

Direct counting

- On a grid or electronically with a **Coulter.**
- A specific volume of a bacterial suspension (0.01 ml) is placed on a microscope slide with a special grid.
- Stain is added to visualize bacteria.
- Cells are counted and multiplied by a factor to obtain concentration.
- **Advantages:** No incubation time required.
- **Disadvantages:**
  - ✓ Cannot always distinguish between live and dead bacteria.
  - ✓ Motile bacteria are difficult to count.
  - ✓ Requires a high concentration of bacteria (10 million/ml).

Counter

Most probable number (**MPN**)

- Number of samples showing growth at each dilution will give results that can be used to estimate density of original culture.
- Used mainly to measure bacteria that will not grow on solid medium.
- Dilute a sample repeatedly and inoculate several broth tubes for each dilution point.
- Count the number of positive tubes in each set.
- **Statistical method**: Determines 95% probability that a bacterial population falls within a certain range.

INDIRECT METHODS

Metabolism

- Rate of usage or production of measurable compound; can measure living microbes.
- As bacteria multiply in media, they produce certain products: Carbon dioxide and Acids.
- Measure metabolic products.
- Expensive

Dry weight

- Limited to measuring dead microbes.
- Bacteria or fungi in liquid media are centrifuged.
- Resulting cell pellet is weighed.
- Doesn't distinguish live and dead cells.

Turbidity

- Low tech (reading newspaper print) to high tech methods (spectrophotometer).
- As bacteria multiply in media, it becomes turbid.
- Use a spectrophotometer to determine % transmission or absorbance.

Career Avenues

- Multiply by a factor to determine concentration.
- **Advantages:** No incubation time required.
- **Disadvantages:**
  - ✓ Cannot distinguish between live and dead bacteria.
  - ✓ Requires a high concentration of bacteria (10 to 100 million cells/ml).

THE CONTROL OF MICROBIAL GROWTH

BASIC TERMINOLOGIES

STERILIZATION

It is the process of complete removal or destruction of all forms of microbial life.

COMMERCIAL STERILIZATION

The limited heat treatment of food which destroys the endospores of disease causing microorganism is called COMMERCIAL STERILIZATION

The endospores of a number of thermophillic bacteria, capable of causing food spoilage but not human disease survive as they are more resistant to C botulinum. Their presence is of no practical consequence as they will not grow at room temperature.

DISINFECTION

- It is control directed at destroying harmful microorganisms
- It usually refers to destruction of vegetative (non-endospore – forming) pathogens.
- It might make use of chemicals UV radiation boiling water or steam.

ANTISEPSIS

- When the disinfection treatment is directed at living tissues, it is called ANTISEPSIS
- The chemical used are called ANTISEPTIC
  - There are modification of disinfection and antisepsis **i) degerming (or degermation)**
- When someone is about to receive an injection, the skin is swabbed with alcohol, process is called degerming
- It results in mechanical removal, rather than the killing of most microbes in the limited area. **ii) Sanitization**
- It is intended to lower microbial counts to safe public health levels and minimize the chances of disease transmission from one user to another.

Career Avenues

- It is used for restaurant glassware, china and tableware.
- It is accomplished by high temperature washing, or, in the case of glassware in a bar, washing in a sink followed by use, e.g. bacteriostasis.

SEPSIS (GREEKS WORD FOR DECAY OR PUTRID)

It indicates bacterial contamination as in septic tanks for sewage treatment.

ASEPSIS

It means that an object or area is free of pathogens. It is the absence of significant contamination.

Actions of microbial control Agents

- Cruses alternation of membrane permeability by damaging lipids or proteins crusing leakage
- Damage to protein and nucleic acids.

BASIC TREATMENT METHODOLOGIES USED FOR CONTROL

Names of treatments that cause the outright death of microbes have the suffix – 'cide' meaning kill, e.g. Biocide or germicide – killing of microorganism (usually with exceptions e.g. endospares)

- Fungicide – Killing of fungi
- Virucide – Killing of virus

Other treatments which only inhibit the growth and multiplication of bacteria have suffix – stat or stasis.

PHYSICAL METHODS OF MICROBIAL CONTROL

Drying (desiccation) and salting (osmotic pressure) were probably among the earlier techniques

Heat

- Laboratory media and glassware, and hospital instrument are used.
- Heat kills microorganisms by denaturing their enzymes the resultant changes to the three dimensional shapes of these proteins inactivate them.
- The heat resistance may be expressed as THERMAL DEATH POINT (TDP). It is the lowest temperature at which al the microorganisms in a particular liquid suspension will be killed in 10 min.

Career Avenues

Thermal death time (tdt)

It is the minimum length of time for all bacteria in a particular liquid culture to be killed at a given temperature.

TDP and TDT are useful guidelines that indicates the severity of treatment require to kill a given population of bacteria.

Decimal Reduction Time (Drt Or D Value)

It is the time in minutes in which 90% of a population of bacteria at a given temperature would be killed.

These are various methods of sterilization through heat –

Moist Heat

Most heat primarily kills microorganism by coagulation of proteins (denaturation) which is caused by barrelage of hydrogen bonds that hold the proteins in their 3 – D structures.

Types of moist heating

- Boiling
    - Kills vegetative forms of bacterial pathogens almost all viruses and fungi and their spares with 10 minutes.
    - free flowing (unpressurized) steam is equivalent in temperature to boiling water, e.g. use of boiling to sanitize baby bottles.

Boiling is not always a reliable sterilization procedure because some viruses and endospores are not destroyed that quickly, e.g. hepatitis virus can survive for 30 min some bacterial endospores can resist boiling for 20 min.

- Autoclave
    - steam under pressure higher reliable sterilization temperature are attained above that of boiling water.
    - higher the pressure in the autoclave higher temperature.

Temperature	Pressure
100^0C	1 atm (psi)
121^0C	15 psi
126^0C	20 psi

Psi = pressure per square inch

At about 15 psi (121^0C) all organisms and their endospores are killed in 15 minutes

Career Avenues

Sterilization in autoclave is most effective when the organisms are either contracted in a small volume of aqueous (primarily water) liquid.

Autoclave is used to sterilize culture media instruments, dressings, intravenous equipment applicator, syringes, transfusion equipment and numerous other items which can withstand high temperature and pressure

* Large autoclaves are called RETORTS

* The same principle applies for common household pressure cookers used in how

Heat requires extra time to reach the centre of solid materials such as canned meats, because such materials do not develop the efficient heat – distributing convection currents that occur in liquids.

Heating large containers also requires extra time.

* To sterilize dry glassware bandages and the like are must be taken to ensure that steam contacts all surfaces, e.g. Aluminium foil is impervious to steam and should not be used to wrap dry materials that are to be sterilized.

Air care should be taken to avoid trapping of air in the bottom of dry container because dry air will not be replaced by steam which is lighter than air.

Trapped air is like a small hot air oven, which requires higher temperature and layer time to sterilize.

Products that do not permit penetration by moisture such as mineral oil or petroleum jelly are not sterilized by this method.

DRY HEAT STERILIZATION

Dry heat kills by oxidation effects Various methods used are:

- Flaming
    - Used to sterilize inoculated loops
    - Wire is heated to a red glow.

* A similar process called INCINERATION is an effective way to sterilize and dispose of contaminated paper cups, taps and dressings.

- Not air sterilization
    - Items to be sterilized are placed in an oven,
    - Generally a temperature of about 170^0 maintained for nearly 2 hours ensures sterilization.
    - Relative to moist air longer period and higher temperature are required because the heat in water is more readily transferred to a cool today than is the heat in air.

Career Avenues

→ Steam heating is also better than dry heat sterilization because steam has high latent heat and thus has higher penetration

INDICATIONS OF STERILIZATION

- Chemical Reactions
    - Indicator changes colour when proper temperature and times of sterilization reached.
    - In some designs the word sterile or autoclaved appears on wrappings or tapes.
- A pellet is contained within a glass vial melts.
- A preparation of specified species of bacterial endospores is impregnated into paper strips.
    - After autoclaving these can be aseptically inoculate into culture media.
    - Growth in culture media indicates survival of the endospores and there fore inadequate processing.
- Endospore suspension can be released after heating, into a surrounding & culture media within the same vital.

PASTEURIZATION

Pasteur used mild heating which was sufficient to kill the organisms that caused the particular spoilage problem without seriously damaging the task of the product.

The same principle is used for Pasteurization of milk

Pasteurization of milk intends to eliminate pathogenic microbes and lower microbial number which prolong best quality of milk under refrigeration (COMMERCIAL STERILIZATION)

Products other than milk such as ice cream, yogurt and beer all have their own pasteurization times and temperature

Reason:

- Heating is less efficient in foods that are more viscous
- Fats in food have protective effect on microorganisms

PHOSPHATEASE TEST

It is used determine pasteurization of dairy products. Phosphatise is an enzyme naturally present in the milk.

If the product has been pasteurized phosphatise will had been inactivated

There are three techniques for pasteurization:

Career Avenues

- Classic Pasteurization
    - Milk is exposed to 63^0C for 30 minutes
- High temperature short time (HTST) pasteurization uses higher temperature 70 72^0C for only 15 minutes
    - It is applied as the milk flows continuous past a heat exchanges.
    - In addition to killing pathogens, HTST pasteurization lowers total bacterial counts so milk keeps well under refrigeration.
- Ultra High Temperature Treatment (UHT)
    - Milk is so sterilized that it can be stored without refrigeration

* In USA, Sterilization is sometimes used on small containers of coffee creamers found in restaurants.

To avoid giving the milk a cooked task UFIT system is used in which the Liquid milk never touches surface hotter than milk itself while being heated by steam

The milk falls in a thin film through a chamber of superheated steam and reaches 140^0C in less than a second.

It is held for 3 seconds in a holding tube and then cooled in a vacuum chamber, where steam flashes

With this process in less than 5 seconds the milk temperature rises from 74^0C to 140^0C and drops back to 74^0C

CONCEPT OF EQUIVALENT TREATMENTS

As the temperature is increased much less time is needed to kill the same no of microbes, e.g. 70^0C minutes at 115^0C all needed for 7 minutes at 125^0C construction of highly resistant microbes. Therefore by

- Classical - 63^0C for 30 min
- HTST - 72^0C for 15 min
- UHT - 140^0C for less than a second

FILTRATION

Filtration is the passage of a liquid or gas through a screen like material with gas through a screen like material with pores small enough to retain microorganism.

Process

- The sample is placed into the upper chamber and forced through the membrane filter by a vacuum in the lower chamber

Career Avenues

- A vacuum that is created in the lower chamber receiving flask helps gravity pull the liquid through the filter.
- Pores in the membrane filter are smaller than the bacteria, so bacteria are retained on the filter.

* Similar equipment with removable filter disks is used to count bacteria in samples.

Filtration is used to sterilize heat – sensitive material, such as some culture media, enzymes vaccines and antibiotic solution.

FILTERS

- High efficiency particulate air (HEPA) filters are used to filter air for reducing the number of air borne microbes in operation theatres and rooms occupied by burn patients.
    - remove almost all microorganisms larger than about 0.3 Hm in diameter.
- Porcelain Filters
    - Was used in early days
    - Hollow candle shaped filters of unglazed porcelain were used to filter liquids
    - The long and indirect passage ways through the walls of the filter adsorbed the bacteria.

* Unseen pathogens that passed through the filters (e.g. rabies cruising) were called filterable viruses.

- Membrane Filters
    - Composed of substances such a cellulose esters (eg Nitrocellulose) or plastic polymers
    - Are used for industrial and laboratory uses
    - Are only 0.1 mm trick
    - The pores of membrane filters include for example 0.22 Hm and 0.45 Hm size which are intended for bacteria.

* Some very flexible bacteria, such as spirochetes or the wall less mycoplasma will sometimes pass through such filters.

    - Filters are available with pores as small as 0.01 Hm, a size that will retain viruses and even some large protein molecules.

- Low Temperatures
    - It is mostly used for preservation and commercial sterilization of food rather than complete sterilization
    - The effect of low temperature on microorganism depends on the particular microbe and the intensity of application, e.g. at ordinary freezing temperatures,

Career Avenues

($0\text{-}7^0C$), mostly all microbes cannot reproduces or produces toxins but the psychographs survive in these conditions.

- Psychographs do grow slowly i.e., they will alter the appearance of and taste of food after a time, e.g. a single migrate reproducing three times a dry would reach a population of 1 million by a week.
- Some bacterial can grow at temperature several degrees below freezing; most foods remain unfrozen until -2^0C or lower

- Rapidly attained subfreezing temperatures tend to render microbes dormant but not necessary kills them.
- Slow freezing is more harmful to bacteria because the ice crystal that forms and grow disrupt the cellular and molecular structure of the bacteria.
- Throwing (cooling after heating to normal temp) being slower is actually the more damaging part of freeze thro cycle.

* Once frozen only one third of the population of some vegetative bacteria may survive a year.

* Many eukaryotic parasites such as the round worms that cruse trichinosis are killed by several days of freezing temperature.

High Pressure

- High pressure applied to liquid suspension is transferred instantly and evenly throughout the sample.
- If the pressure is high enough, the molecular structure of proteins and carbohydrates are altered, resulting in rapid inactivation of vegetative bacterial cells.
- Endospores are relatively resistant to pressure
- They can however, be killed by other techniques, such as i) combining high pressure with elevated temperatures
- Alternating pressure cycles that cause spare germination followed by pressure caused death of resulting vegetative cells.
    - Advantage is that these treatments preserve the flavours, colors and nutrient values of the products.

* Fruit juices preserved by high pressure treatments have been marketed in Japan and United States.

DESSICATION

It is the condition of absence of water.

Under these conditions microbes cannot grow or reproduce but can remain viable for years.

Career Avenues

* This ability is used in laboratory when microbes are preserved by lyophilisation or freeze drying.

Certain foods are also freeze dried e.g. coffee and some fruit additives for dry cereals.

The resistance of vegetative cells to desiccation varies with species and the organisms environment, e.g. the gonorrhoea bacterium can withstand dryness only for an hour while tuberculosis bacterium can last for months.

Viruses are generally resistant to desiccation but they are not as resistant as bacterial endospores, some of which have survived for centuries.

* This ability is used in hospital setting dust, clothing bedding, and dressing might contain infections microbes in dried mucus, urine pus and faeces.

OSMOTIC PRESSURE

- The use of high concentration of salts and sugars to preserve food is based on effect of osmotic pressure.
- High concentration of these substances create a hypertonic environment that cruse water to leave the microbial cells and creates desiccate conditions.
- This principle is used in preoccupation of food (COMMERCIAL STERILIZATION) e.g. concentrated salt solutions are used to cure meets.

Thick sugar 80^4 are used to preserve fruits

- Moulds and yeasts are much more capable that bacteria of growing in materials with low moisture and high O.P.

* This property and ability to grow under acidic condition of molds is the reason why fruits and grains are spoiled by moulds and not bacteria.

* It is also the part of the reason moulds is able to form mildew on a damp wall or a shower certain.

RADIATION

Radiation has various effects on cells depending on its wavelength intensity and duration. Radiations that kill microorganisms are of two types

- Ionizing
- Non – Ionizing

Ionizing Radiation – gamma rays x – rays

A high energy e^- beam

- Has a wavelength shorter than that of no ionizing radiations ie. < inm.

Career Avenues

- Therefore it carries more energy.
- Production
- Gamma rays are emitted by certain radioactive elements such as cobalt.
- Electron beams are produced by accelerating electrons to high energy by speed machines.
- X- rays are produced in machines in a similar manner to e^- beam.

* X-rays are similar in nature to x- rays.

- Gamma rays penetrate deeply but may require hours to sterilize large masses.
- High energy electrons have much less penetration power but usually require.
- Principle effect is ionization of water which forms highly reactive hydroxyl radicals.
- These radicals react with organic cellular components especially, DNA.
- The RO called theory of damage by radiation supposes that ionizing particles, or packets or energy pass through or close to vital portions of the cell, these constitute "hits".

* One a few hits may only cruse nonlethal mutation some of them conceivably useful.

* More hits are likely to cruse sufficient mutations to kill the microbes.

- Low level ionizing radiation, used for years in many countries, has been approved in US for processing spices and certain meats and vegetables.
- Ionizing radiations, specially high energy electron beams is used for sterilization of pharmaceuticals and disposable dental and medical supplies, such as plastic syringes, surgical gloves, suturing materials and catheters.

* As a protection against bioterrorism, the postal service often uses electron from radiations to sterilize certain classes of mail.

GATE BT 2011

Q.7 Substrate consumption in lag phase of microbial growth is primarily used for
P. turn over of the cell material
Q. maintenance of intracellular pH
R. motility
S. increase in cell number
(A) P, Q and S only (B) Q, R and S only
(C) P, Q and R only (D) S only

Q.8 Determine the correctness or otherwise of the following Assertion (a) and the Reason (r)
Assertion: In synchronous culture, majority of the cells move to next phase of the cell cycle simultaneously

Career Avenues

Reason: Synchronous culture could be obtained by starving cells for essential nutrient components

(A) Both (a) and (r) are true and (r) is the correct reason for (a)
(B) Both (a) and (r) are true but (r) is not the correct reason for (a)
(C) (a) is true but (r) is false
(D) (a) is false but (r) is true

Answers:

- **Q.7 – C.**
- **Q.8 – A.**

Career Avenues

MICROBIAL METABOLISM

HETEROTROPHIC MICROBIAL METABOLISM

Most microbes are heterotrophic (more precisely chemoorganoheterotrophic), using organic compounds as both carbon and energy sources. Heterotrophic microbes live off of nutrients that they scavenge from living hosts (as commensals or parasites) or find in dead organic matter of all kind (saprophages). Microbial metabolism is the main contribution for the bodily decay of all organisms after death. Many eukaryotic microorganisms are heterotrophic by predation or parasitism, properties also found in some bacteria such as *Bdellovibrio* (an intracellular parasite of other bacteria, causing death of its victims) and Myxobacteria such as *Myxococcus* (predators of other bacteria which are killed and lysed by cooperating swarms of many single cells of Myxobacteria). Most pathogenic bacteria can be viewed as heterotrophic parasites of humans or the other eukaryotic species they affect. Heterotrophic microbes are extremely abundant in nature and are responsible for the breakdown of large organic polymers such as cellulose, chitin or lignin which are generally indigestible to larger animals. Generally, the breakdown of large polymers to carbon dioxide (mineralization) requires several different organisms, with one breaking down the polymer into its constituent monomers, one able to use the monomers and excreting simpler waste compounds as by-products, and one able to use the excreted wastes. There are many variations on this theme, as different organisms are able to degrade different polymers and secrete different waste products. Some organisms are even able to degrade more recalcitrant compounds such as petroleum compounds or pesticides, making them useful in bioremediation.

Biochemically, prokaryotic heterotrophic metabolism is much more versatile than that of eukaryotic organisms, although many prokaryotes share the most basic metabolic models with eukaryotes, e. g. using glycolysis (also called EMP pathway) for sugar metabolism and the citric acid cycle to degrade acetate, producing energy in the form of ATP and reducing power in the form of NADH or quinols. These basic pathways are well conserved because they are also involved in biosynthesis of many conserved building blocks needed for cell growth (sometimes in reverse direction). However, many bacteria and archaea utilize alternative metabolic pathways other than glycolysis and the citric acid cycle. A well-studied example is sugar metabolism via the keto-deoxy-phosphogluconate pathway (also called ED pathway) in *Pseudomonas*. Moreover, there is a third alternative sugar-catabolic pathway used by some bacteria, the pentose phosphate pathway. The metabolic diversity and ability of prokaryotes to use a large variety of organic compounds arises from the much deeper evolutionary history and diversity of prokaryotes, as compared to eukaryotes. It is also noteworthy that the mitochondrion, the small membrane-bound intracellular organelle that is the site of eukaryotic energy metabolism, arose from the endosymbiosis of a bacterium related to obligate intracellular *Rickettsia*, and also to plant-associated *Rhizobium* or *Agrobacterium*. Therefore it is not surprising that all mitrochondriate eukaryotes share metabolic properties with these Proteobacteria. Most microbes respire (use an electron transport chain), although oxygen is not the only terminal electron acceptor that may be used. As discussed below, the use of terminal electron acceptors other than oxygen has important biogeochemical consequences.

Career Avenues

FERMENTATION

Fermentation is a specific type of heterotrophic metabolism that uses organic carbon instead of oxygen as a terminal electron acceptor. This means that these organisms do not use an electron transport chain to oxidize NADH to NAD^+ and therefore must have an alternative method of using this reducing power and maintaining a supply of NAD^+ for the proper functioning of normal metabolic pathways (e.g. glycolysis). As oxygen is not required, fermentative organisms are anaerobic. Many organisms can use fermentation under anaerobic conditions and aerobic respiration when oxygen is present. These organisms are facultative anaerobes. To avoid the overproduction of NADH, obligately fermentative organisms usually do not have a complete citric acid cycle. Instead of using an ATP synthase as in respiration, ATP in fermentative organisms is produced by substrate-level phosphorylation where a phosphate group is transferred from a high-energy organic compound to ADP to form ATP. As a result of the need to produce high energy phosphate-containing organic compounds (generally in the form of CoA-esters) fermentative organisms use NADH and other cofactors to produce many different reduced metabolic by-products, often including hydrogen gas (H_2). These reduced organic compounds are generally small organic acids and alcohols derived from pyruvate, the end product of glycolysis. Examples include ethanol, acetate, lactate, and butyrate. Fermentative organisms are very important industrially and are used to make many different types of food products. The different metabolic end products produced by each specific bacterial species are responsible for the different tastes and properties of each food.

Not all fermentative organisms use substrate-level phosphorylation. Instead, some organisms are able to couple the oxidation of low-energy organic compounds directly to the formation of a proton (or sodium) motive force and therefore ATP synthesis. Examples of these unusual forms of fermentation include succinate fermentation by *Propionigenium modestum* and oxalate fermentation by *Oxalobacter formigenes*. These reactions are extremely low-energy yielding. Humans and other higher animals also use fermentation to produce lactate from excess NADH, although this is not the major form of metabolism as it is in fermentative microorganisms.

ANAEROBIC RESPIRATION

While aerobic organisms during respiration use oxygen as a terminal electron acceptor, anaerobic organisms use other electron acceptors. These inorganic compounds have a lower reduction potential than oxygen, meaning that respiration is less efficient in these organisms and leads to slower growth rates than aerobes. Many facultative anaerobes can use either oxygen or alternative terminal electron acceptors for respiration depending on the environmental conditions.

Most respiring anaerobes are heterotrophs, although some do live autotrophically. All of the processes described below are dissimilative, meaning that they are used during energy production and not to provide nutrients for the cell (assimilative). Assimilative pathways for many forms of anaerobic respiration are also known.

Career Avenues

DENITRIFICATION - NITRATE AS ELECTRON ACCEPTOR

Denitrification is the utilization of nitrate (NO_3^-) as a terminal electron acceptor. It is a widespread process that is used by many members of the Proteobacteria. Many facultative anaerobes use denitrification because nitrate, like oxygen, has a high reduction potential. Many denitrifying bacteria can also use ferric iron (Fe^{3+}) and some organic electron acceptors. Denitrification involves the stepwise reduction of nitrate to nitrite (NO_2^-), nitric oxide (NO), nitrous oxide (N_2O), and dinitrogen (N_2) by the enzymes nitrate reductase, nitrite reductase, nitric oxide reductase, and nitrous oxide reductase, respectively. Protons are transported across the membrane by the initial NADH reductase, quinones, and nitrous oxide reductase to produce the electrochemical gradient critical for respiration. Some organisms (e.g. *E. coli*) only produce nitrate reductase and therefore can accomplish only the first reduction leading to the accumulation of nitrite. Others (e.g. *Paracoccus denitrificans* or *Pseudomonas stutzeri*) reduce nitrate completely. Complete denitrification is an environmentally significant process because some intermediates of denitrification (nitric oxide and nitrous oxide) are important greenhouse gases that react with sunlight and ozone to produce nitric acid, a component of acid rain. Denitrification is also important in biological wastewater treatment where it is used to reduce the amount of nitrogen released into the environment thereby reducing eutrophication.

SULFATE REDUCTION - SULFATE AS ELECTRON ACCEPTOR

Sulfate reduction is a relatively energetically poor process used by many Gram negative bacteria found within the δ-Proteobacteria, Gram-positive organisms relating to *Desulfotomaculum* or the archaeon *Archaeoglobus*. Hydrogen sulfide (H_2S) is produced as a metabolic end product. For sulfate reduction electron donors and energy are needed.

ACETOGENESIS - CARBON DIOXIDE AS ELECTRON ACCEPTOR

Acetogenesis is a type of microbial metabolism that uses hydrogen (H_2) as an electron donor and carbon dioxide (CO_2) as an electron acceptor to produce acetate, the same electron donors and acceptors used in methanogenesis (see above). Bacteria that can autotrophically synthesize acetate are called homoacetogens. Carbon dioxide reduction in all homoacetogens occurs by the acetyl-CoA pathway. This pathway is also used for carbon fixation by autotrophic sulfate-reducing bacteria and hydrogenotrophic methanogens. Often homoacetogens can also be fermentative, using the hydrogen and carbon dioxide produced as a result of fermentation to produce acetate, which is secreted as an end product.

CHEMOLITHOTROPHY

Chemolithotrophy is a type of metabolism where energy is obtained from the oxidation of inorganic compounds. Most chemolithotrophic organisms are also autotrophic. There are two major objectives to chemolithotrophy: the generation of energy (ATP) and the generation of reducing power (NADH).

Career Avenues

HYDROGEN OXIDATION

Many organisms are capable of using hydrogen (H_2) as a source of energy. While several mechanisms of anaerobic hydrogen oxidation have been mentioned previously (e.g. sulfate reducing- and acetogenic bacteria), hydrogen can also be used as an energy source aerobically. In these organisms, hydrogen is oxidized by a membrane-bound hydrogenase causing proton pumping via electron transfer to various quinones and cytochromes. In many organisms, a second cytoplasmic hydrogenase is used to generate reducing power in the form of NADH, which is subsequently used to fix carbon dioxide via the Calvin cycle. Hydrogen-oxidizing organisms, such as *Cupriavidus necator* (formerly *Ralstonia eutropha*), often inhabit oxic-anoxic interfaces in nature to take advantage of the hydrogen produced by anaerobic fermentative organisms while still maintaining a supply of oxygen.

SULFUR OXIDATION

Sulfur oxidation involves the oxidation of reduced sulfur compounds (such as sulfide (H_2S), inorganic sulfur (S^0), and thiosulfate ($S_2O_3^{2-}$) to form sulfuric acid (H_2SO_4). A classic example of a sulfur-oxidizing bacterium is *Beggiatoa*, a microbe originally described by Sergei Winogradsky, one of the founders of environmental microbiology. Another example is *Paracoccus*. Generally, the oxidation of sulfide occurs in stages, with inorganic sulfur being stored either inside or outside of the cell until needed. This two step process occurs because energetically sulfide is a better electron donor than inorganic sulfur or thiosulfate, allowing for a greater number of protons to be translocated across the membrane. Sulfur-oxidizing organisms generate reducing power for carbon dioxide fixation via the Calvin cycle using reverse electron flow, an energy-requiring process that pushes the electrons against their thermodynamic gradient to produce NADH. Biochemically, reduced sulfur compounds are converted to sulfite (SO_3^{2-}) and subsequently converted to sulfate (SO_4^{2-}) by the enzyme sulfite oxidase. Some organisms, however, accomplish the same oxidation using a reversal of the APS reductase system used by sulfate-reducing bacteria. In all cases the energy liberated is transferred to the electron transport chain for ATP and NADH production. In addition to aerobic sulfur oxidation, some organisms (e.g. *Thiobacillus denitrificans*) use nitrate (NO_3^-) as a terminal electron acceptor and therefore grow anaerobically.

FERROUS IRON (FE2+) OXIDATION

Ferrous iron is a soluble form of iron that is stable at extremely low pHs or under anaerobic conditions. Under aerobic, moderate pH conditions ferrous iron is oxidized spontaneously to the ferric (Fe^{3+}) form and is hydrolyzed abiotically to insoluble ferric hydroxide ($Fe(OH)_3$). There are three distinct types of ferrous iron-oxidizing microbes. The first are acidophiles, such as the bacteria *Acidithiobacillus ferrooxidans* and *Leptospirillum ferrooxidans*, as well as the archaeon *Ferroplasma*. These microbes oxidize iron in environments that have a very low pH and are important in acid mine drainage. The second type of microbes oxidize ferrous iron at cirum-neutral pH. These micro-organisms (for example *Gallionella ferruginea* or *Leptothrix ochracea*) live at the oxic-anoxic interfaces and are microaerophiles. The third type of iron-oxidizing microbes are anaerobic photosynthetic bacteria such as Rhodopseudomonas, which use ferrous iron to produce NADH for autotrophic carbon

Career Avenues

dioxide fixation. Biochemically, aerobic iron oxidation is a very energetically poor process which therefore requires large amounts of iron to be oxidized by the enzyme rusticyanin to facilitate the formation of proton motive force. Like sulfur oxidation, reverse electron flow must be used to form the NADH used for carbon dioxide fixation via the Calvin cycle.

NITRIFICATION

Nitrification is the process by which ammonia (NH_3) is converted to nitrate (NO_3^-). Nitrification is actually the net result of two distinct processes: oxidation of ammonia to nitrite (NO_2^-) by nitrosifying bacteria (e.g. *Nitrosomonas*) and oxidation of nitrite to nitrate by the nitrite-oxidizing bacteria (e.g. *Nitrobacter*). Both of these processes are extremely energetically poor leading to very slow growth rates for both types of organisms. Biochemically, ammonia oxidation occurs by the stepwise oxidation of ammonia to hydroxylamine (NH_2OH) by the enzyme ammonia monooxygenase in the cytoplasm, followed by the oxidation of hydroxylamine to nitrite by the enzyme hydroxylamine oxidoreductase in the periplasm.

Electron and proton cycling are very complex but as a net result only one proton is translocated across the membrane per molecule of ammonia oxidized. Nitrite reduction is much simpler, with nitrite being oxidized by the enzyme nitrite oxidoreductase coupled to proton translocation by a very short electron transport chain, again leading to very low growth rates for these organisms. Oxygen is required in both ammonia and nitrite oxidation, meaning that both nitrosifying and nitrite-oxidizing bacteria are aerobes. As in sulfur and iron oxidation, NADH for carbon dioxide fixation using the Calvin cycle is generated by reverse electron flow, thereby placing a further metabolic burden on an already energy-poor process.

PHOTOTROPHY

Many microbes (phototrophs) are capable of using light as a source of energy to produce ATP and organic compounds such as carbohydrates, lipids, and proteins. Of these, algae are particularly significant because they are oxygenic, using water as an electron donor for electron transfer during photosynthesis. Phototrophic bacteria are found in the phyla Cyanobacteria, Chlorobi, Proteobacteria, Chloroflexi, and Firmicutes. Along with plants these microbes are responsible for all biological generation of oxygen gas on Earth. Because chloroplasts were derived from a lineage of the Cyanobacteria, the general principles of metabolism in these endosymbionts can also be applied to chloroplasts. In addition to oxygenic photosynthesis, many bacteria can also photosynthesize anaerobically, typically using sulfide (H_2S) as an electron donor to produce sulfate. Inorganic sulfur (S^0), thiosulfate ($S_2O_3^{2-}$) and ferrous iron (Fe^{2+}) can also be used by some organisms. Phylogenetically, all oxygenic photosynthetic bacteria are Cyanobacteria, while anoxygenic photosynthetic bacteria belong to the purple bacteria (Proteobacteria), Green sulfur bacteria (e.g. Chlorobium), Green non-sulfur bacteria (e.g. Chloroflexus), or the heliobacteria (Low %G+C Gram positives). In addition to these organisms, some microbes (e.g. the Archaeon *Halobacterium* or the bacterium *Roseobacter*, among others) can utilize light to produce energy using the enzyme bacteriorhodopsin, a light-driven proton pump. However, there are no known Archaea that carry out photosynthesis.

Career Avenues

As befits the large diversity of photosynthetic bacteria, there are many different mechanisms by which light is converted into energy for metabolism. All photosynthetic organisms locate their photosynthetic reaction centers within a membrane, which may be invaginations of the cytoplasmic membrane (Proteobacteria), thylakoid membranes (Cyanobacteria), specialized antenna structures called chlorosomes (Green sulfur and non-sulfur bacteria), or the cytoplasmic membrane itself (heliobacteria). Different photosynthetic bacteria also contain different photosynthetic pigments, such as chlorophylls and carotenoids, allowing them to take advantage of different portions of the electromagnetic spectrum and thereby inhabit different niches. Some groups of organisms contain more specialized light-harvesting structures (e.g. phycobilisomes in Cyanobacteria and chlorosomes in Green sulfur and non-sulfur bacteria), allowing for increased efficiency in light utilization.

Biochemically, anoxygenic photosynthesis is very different from oxygenic photosynthesis. Cyanobacteria (and by extension, chloroplasts) use the Z scheme of electron flow in which electrons eventually are used to form NADH. Two different reaction centers (photosystems) are used and proton motive force is generated both by using cyclic electron flow and the quinone pool. In anoxygenic photosynthetic bacteria, electron flow is cyclic, with all electrons used in photosynthesis eventually being transferred back to the single reaction center. A proton motive force is generated using only the quinone pool. In heliobacteria, Green sulfur, and Green non-sulfur bacteria, NADH is formed using the protein ferredoxin, an energetically favorable reaction. In purple bacteria, NADH is formed by reverse electron flow due to the lower chemical potential of this reaction center. In all cases, however, a proton motive force is generated and used to drive ATP production via an ATPase.

Most photosynthetic microbes are autotrophic, fixing carbon dioxide via the Calvin cycle. Some photosynthetic bacteria (e.g. Chloroflexus) are photoheterotrophs, meaning that they use organic carbon compounds as a carbon source for growth. Some photosynthetic organisms also fix nitrogen.

PHOTOSYNTHESIS

Microorganisms cannot only derive energy from the oxidation of inorganic and organic compounds, but many can capture light energy and use it to synthesize ATP and NADH or NADPH .This process in which light energy is trapped and converted to chemical energy is called photosynthesis. Usually a photosynthetic organism reduces and incorporates CO2, reactions that are also considered part of this process. Photosynthesis is one of the most significant metabolic processes on earth because almost all our energy is ultimately derived from solar energy. It provides photosynthetic organisms with the ATP and NADPH necessary to synthesize the organic material required for growth. In turn these organisms serve as the base of most food chains in the biosphere. Photosynthesis is also responsible for replenishing our supply of O2, a remarkable process carried out by a variety of organisms, both eukaryotic and prokaryotic. Although most people associate photosynthesis with the more obvious higher plants, over half the photosynthesis on earth is carried out by microorganisms. Photosynthesis as a whole is divided into two parts. In the light reactions light energy is trapped and converted to chemical energy. This energy is then used to reduce or fix CO2 and synthesize cell constituents in the dark reactions.

Career Avenues

Diversity of Photosynthetic Organisms:

Eukaryotic Organisms	Prokaryotic Organisms
Higher plants	Cyanobacteria (blue-green algae)
Multicellular green,brown, and red algae	Green sulfur bacteria
Unicellular algae (e.g., euglenoids, dinoflagellates, diatoms)	Green nonsulfur bacteria
	Purple sulfur bacteria
	Purple nonsulfur bacteria
	Prochloron

THE LIGHT REACTION IN EUKARYOTES AND CYANOBACTERIA

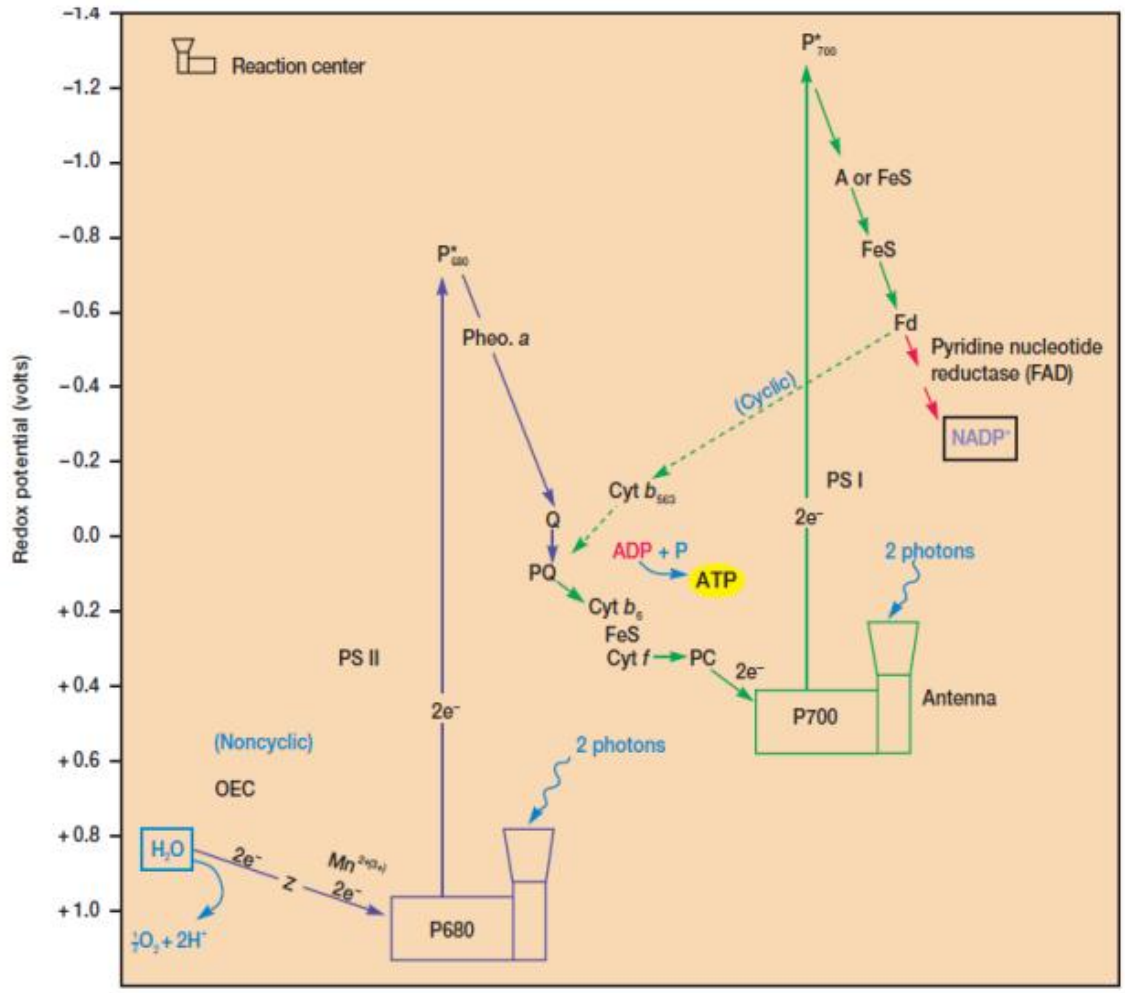
All photosynthetic organisms have pigments for the absorption of light. The most important of these pigments are the chlorophylls. Chlorophylls are large planar rings composed of four substituted pyrrole rings with a magnesium atom coordinated to the four central nitrogen atoms. Several chlorophylls are found in eukaryotes, the two most important of which are chlorophyll *a* and chlorophyll *b*. These two molecules differ slightly in their structure and spectral properties. When dissolved in acetone, chlorophyll *a* has a light absorption peak at 665 nm; the corresponding peak for chlorophyll *b* is at 645 nm. In addition to absorbing red light, chlorophylls also absorb blue light strongly (the second absorption peak for chlorophyll *a* is at 430 nm). Because chlorophylls absorb primarily in the red and blue ranges, green light is transmitted. Consequently many photosynthetic organisms are green in color. The long hydrophobic tail attached to the chlorophyll ring aids in its attachment to membranes, the site of the light reactions. Other photosynthetic pigments also trap light energy. The most widespread of these are the carotenoids, long molecules, usually yellowish in color, that possess an extensive conjugated double bond system. β-Carotene is present in *Prochloron* and most divisions of algae; fucoxanthin is found in diatoms, dinoflagellates, and brown algae (*Phaeophyta*). Red algae and cyanobacteria have photosynthetic pigments called phycobiliproteins, consisting of a protein with a tetrapyrrole attached. Phycoerythrin is a red pigment with a maximum absorption around 550 nm, and phycocyanin is blue (maximum absorption at 620 to 640 nm). Carotenoids and phycobiliproteins are often called accessory pigments because of their role in photosynthesis. Although chlorophylls cannot absorb light energy effectively in the blue green through yellow range (about 470 to 630 nm), accessory pigments do absorb light in this region and transfer the trapped energy to chlorophyll. In this way they make photosynthesis more efficient over a broader range of wavelengths. Accessory pigments also protect microorganisms from intense sunlight, which could\ oxidize and damage the photosynthetic apparatus in their absence. Chlorophylls and accessory pigments are assembled in highly organized arrays called antennas, whose purpose is to create a large surface area to trap as many photons as possible. An antenna has about 300 chlorophyll molecules. Light energy is captured in an antenna and transferred from chlorophyll to chlorophyll until it reaches a special reaction-center chlorophyll directly

Career Avenues

involved in photosynthetic electron transport (figure 9.28). In eukaryotic cells and cyanobacteria, there are two kinds of antennas associated with two different photosystems. Photosystem I absorbs longer wavelength light (>=680 nm) and funnels the energy to a special chlorophyll *a* molecule called P700. The term P700 signifies that this molecule most effectively absorbs light at a wavelength of 700 nm.

Photosystem II traps light at shorter wavelengths (<=680 nm) and transfers its energy to the special chlorophyll P680. When the photosystem I antenna transfers light energy to the reaction-center P700 chlorophyll, P700 absorbs the energy and is excited; its reduction potential becomes very negative. It then donates its excited or high-energy electron to a specific acceptor, probably a special chlorophyll *a* molecule (A) or an iron-sulfur protein . The electron is eventually transferred to ferredoxin and can then travel in either of two directions. In the cyclic pathway (the dashed lines in figure 9.29), the electron moves in a cyclic route through a series of electron carriers and back to the oxidized P700. The pathway is termed cyclic because the electron from P700 returns to P700 after traveling through the photosynthetic electron transport chain. PMF is formed during cyclic electron transport in the region of cytochrome *b*6 and used to synthesize ATP. This process is called cyclic photophosphorylation because electrons travel in a cyclic pathway and ATP is formed. Only photosystem I participates. Electrons also can travel in a noncyclic pathway involving both photosystems. P700 is excited and donates electrons to ferredoxin as before. In the noncyclic route, however, reduced ferredoxin reduces NADP+ to NADPH . Because the electrons contributed to NADP+ cannot be used to reduce oxidized P700, photosystem II participation is required. It donates electrons to oxidized P700 and generates ATP in the process. The photosystem II antenna absorbs light energy and excites P680, which then reduces pheophytin *a.* Pheophytin *a* is chlorophyll *a* in which two hydrogen atoms have replaced the central magnesium. Electrons subsequently travel to Q (probably a plastoquinone) and down the electron transport chain to P700. Oxidized P680 then obtains an electron from the oxidation of water to O2. Thus electrons flow from water all the way to NADP+ with the aid of energy from two photosystems, and ATP is synthesized by noncyclic photophosphorylation. It appears that one ATP and one NADPH are formed when two electrons travel through the noncyclic pathway.

Career Avenues



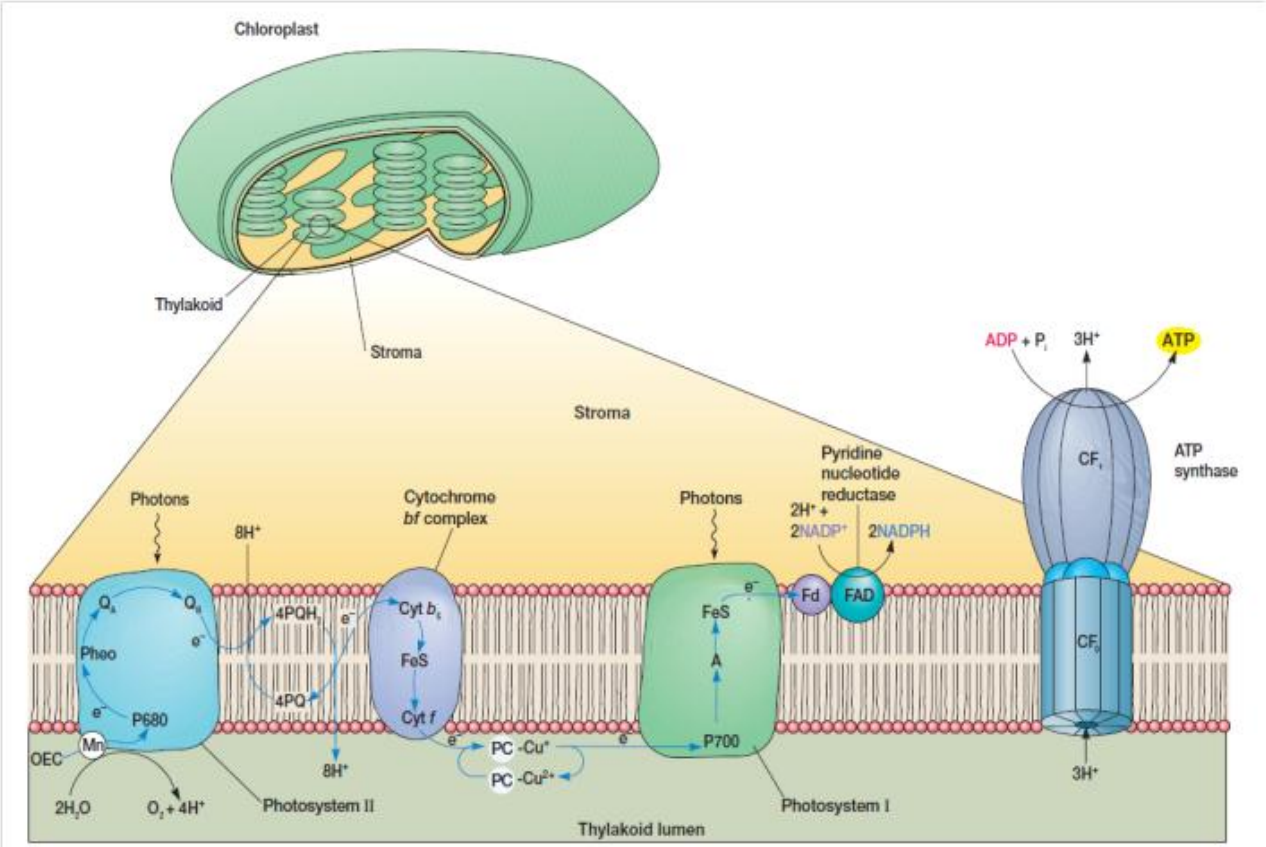
Green Plant Photosynthesis. Electron flow during photosynthesis in higher plants. Cyanobacteria and eucaryotic algae are similar in having two photosystems, although they may differ in some details. The carriers involved in electron transport are ferredoxin (Fd) and other FeS proteins; cytochromes b_6, b_{563}, and *f;* plastoquinone (PQ); copper containing plastocyanin (PC); pheophytin *a* (Pheo. *a*); possibly chlorophyll *a* (A); and the unknown quinone Q, which is probably a plastoquinone. Both photosystem I (PS I) and photosystem II (PS II) are involved in noncyclic photophosphorylation; only PS I participates in cyclic photophosphorylation. The oxygen evolving complex (OEC) that extracts electrons from water contains manganese ions and the substance Z, which transfers electrons to the PS II reaction center. See the text for further details.

Just as is true of mitochondrial electron transport, photosynthetic electron transport takes place within a membrane. Chloroplast granal membranes contain both photosystems and their antennas. Protons move to the thylakoid interior during photosynthetic electron transport and return to the stroma when ATP is formed. It is believed that stromal lamellae possess only photosystem I and are involved in cyclic photophosphorylation alone. In cyanobacteria, photosynthetic light reactions are also located in membranes. The dark reactions require three ATPs and two NADPHs to reduce one CO2 and use it to synthesize carbohydrate (CH_2O).

$$\mathbf{CO_2 + 3ATP + 2NADPH + 2H^+ + H_2O \rightarrow (CH_2O) + 3ADP + 3P_i + 2\ NADP^+}$$

The noncyclic system generates one NADPH and one ATP per pair of electrons; therefore four electrons passing through the system will produce two NADPHs and two ATPs. A total

of 8 quanta of light energy (4 quanta for each photosystem) is needed to propel the four electrons from water to NADP+. Because the ratio of ATP to NADPH required for CO2 fixation is 3:2, at least one more ATP must be supplied. Cyclic photophosphorylation probably operates independently to generate the extra ATP. This requires absorption of another 2 to 4 quanta. It follows that around 10 to 12 quanta of light energy are needed to reduce and incorporate one molecule of CO2 during photosynthesis.



The Mechanism of Photosynthesis. An illustration of the chloroplast thylakoid membrane showing photosynthetic electron transport chain function and noncyclic photophosphorylation. The chain is composed of three complexes: PS I, the cytochrome *bf* complex, and PS II. Two diffusible electron carriers connect the three complexes. Plastoquinone (PQ) connects PS I with the cytochrome *bf* complex, and plastocyanin (PC) connects the cytochrome *bf* complex with PS II. The light-driven electron flow pumps protons across the thylakoid membrane and generates an electrochemical gradient, which can then be used to make ATP. Water is the source of electrons and the oxygen-evolving complex (OEC) produces oxygen.

The Light Reaction in Green and Purple Bacteria Green and purple photosynthetic bacteria differ from cyanobacteria and eukaryotic photosynthesizers in several fundamental ways. In particular, green and purple bacteria do not use water as an electron source or produce O_2 photosynthetically—that is, they are anoxygenic. In contrast, cyanobacteria and eucaryoticphoto synthesizers are almost always oxygenic. NADPH is not directly produced in the photosynthetic light reaction of purple bacteria. Green bacteria can reduce NAD+ directly during the light reaction. To synthesize NADH and NADPH, green and purple bacteria must use electron donors like hydrogen, hydrogen sulfide, elemental sulfur, and organic compounds that have more negative reduction potentials than water and are therefore easier to oxidize (better electron donors). Finally, green and purple bacteria possess slightly

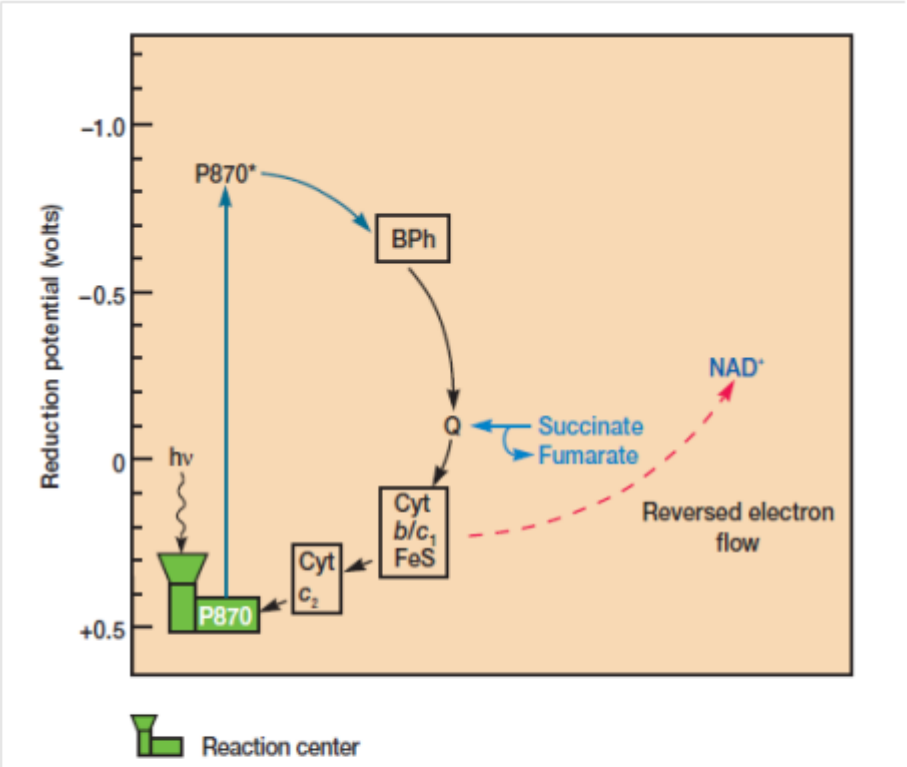
different photosynthetic pigments, bacteriochlorophylls many with absorption maxima at longer wavelengths.

Bacteriochlorophylls *a* and *b* have maxima in ether at 775 and 790 nm, respectively. In vivo maxima are about 830 to 890 nm (bacteriochlorophyll *a*) and 1,020 to 1,040 nm (Bchl *b*). This shift of absorption maxima into the infrared region better adapts these bacteria to their ecological niches . There are four groups of green and purple photosynthetic bacteria, each containing several genera: green sulfur bacteria (*Chlorobium*), green nonsulfur bacteria (*Chloroflexus*), purple sulfur bacteria (*Chromatium*), and purple nonsulfur bacteria (*Rhodospirillum, Rhodopseudomonas*). The biology of thesegroups will be discussed later. Many differences found in green and purple bacteria are due to their lack of photosystem II; they cannot use water as an electron donor in noncyclic electron transport. Without photosystem II they cannot produce O2 from H2O photosynthetically and are restricted to cyclic photophosphorylation. Indeed, almost all purple and green sulfur bacteria are strict anaerobes. When the special reaction-center chlorophyll P870 is excited, it donates an electron to bacteriopheophytin. Electrons then flow to quinones and through an electron transport chain back to P870 while driving ATP synthesis. Note that although both green and purple bacteria lack two photosystems, the purple bacteria have a photosynthetic apparatus similar to photosystem II, whereas the green sulfur bacteria have a system similar to photosystem I. Green and purple bacteria face a further problem because they also require NADH or NADPH for CO2 incorporation. They may synthesize NADH in at least three ways. If they are growing in the presence of hydrogen gas, which has a reduction potential more negative than that of NAD+, the hydrogen can be used directly to produce NADH. Like chemolithotrophs, many photosynthetic purple bacteria use proton motive force to reverse the flow of electrons in an electron transport chain and move them from inorganic or organic donors to NAD+. Green sulfur bacteria such as *Chlorobium* appear to carry out a simple form of noncyclic photosynthetic electron flow to reduce NAD+.

Properties of Microbial Photosynthetic Systems:

Property	Eukaryotes	Cyanobacteria	Green and Purple Bacteria
Photosynthetic pigment	Chlorophyll *a*	Chlorophyll *a*	Bacteriochlorophyll
Photosystem II	Present	Present	Absent
Photosynthetic electron donors	H_2O	H_2O	H_2, H_2S, S, organic matter
O_2 production pattern	Oxygenic	Oxygenic	Anoxygenic
Primary products of energy conversion	ATP + NADPH	ATP + NADPH	ATP
Carbon source	CO_2	CO_2	Organic and/or CO_2

Career Avenues



Purple Nonsulfur Bacterial Photosynthesis. The photosynthetic electron transport system in the purple nonsulfur bacterium, *Rhodobacter sphaeroides*. This scheme is incomplete and tentative. Ubiquinone (Q) is very similar to coenzyme Q. BPh stands for bacteriopheophytin. NAD^+ and the electron source succinate are in color.

GATE BT 2010

Q.9 During lactic acid fermentation, net yield of ATP and NADH per mole of glucose is

(A) 2 ATP and 2 NADH
(B) 2 ATP and 0 NADH
(C) 4 ATP and 2 NADH
(D) 4 ATP and 0 NADH

GATE BT 2013

Q.10 Which one of the following aminoacids in proteins does **NOT** undergo phosphorylation?

(A) Ser (B) Thr (C) Pro (D) Tyr

Answers:

- **Q.9 – B.**
- **Q.10 – C.**

Career Avenues

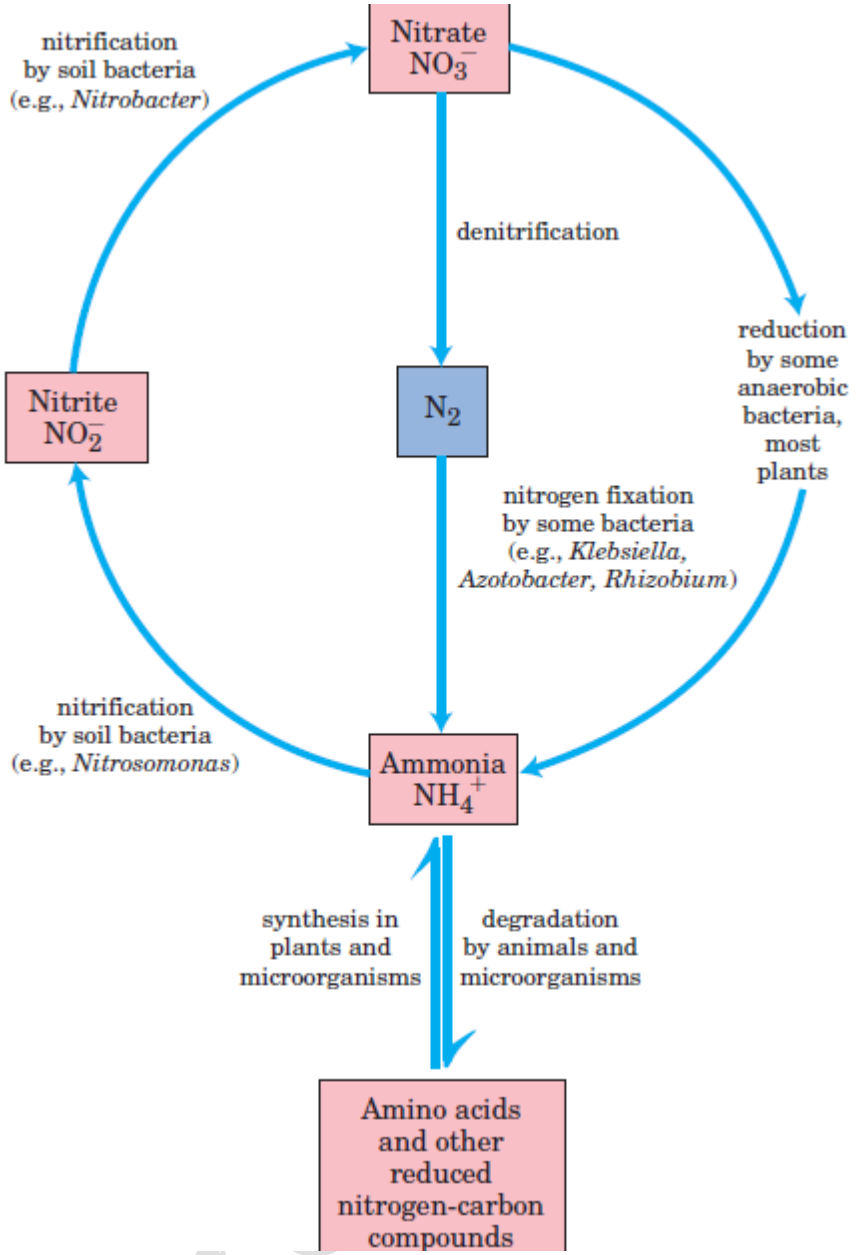
NITROGEN FIXATION

NITROGEN METABOLISM

The biosynthetic pathways leading to amino acids and nucleotides share a requirement for nitrogen.

The most important source of nitrogen is air, which is four-fifths molecular nitrogen (N_2). However, relatively few species can convert atmospheric nitrogen into forms useful to living organisms. In the biosphere, the metabolic processes of different species function interdependently to salvage and reuse biologically available nitrogen in a vast **nitrogen cycle**. The first step in the cycle is **fixation** (reduction) of atmospheric nitrogen by nitrogen-fixing bacteria to yield ammonia (NH_3 or NH_4^+). Although ammonia can be used by most living organisms, soil bacteria that derive their energy by oxidizing ammonia to nitrite (NO_2^-) and ultimately nitrate (NO_3^-) are so abundant and active that nearly all ammonia reaching the soil is oxidized to nitrate. This process is known as **nitrification.** Plants and many bacteria can take up and readily reduce nitrate and nitrite through the action of nitrate and nitrite reductases. The ammonia so formed is incorporated into amino acids by plants. Animals then use plants as a source of amino acids, both nonessential and essential, to build their proteins. When organisms die, microbial degradation of their proteins returns ammonia to the soil, where nitrifying bacteria again convert it to nitrite and nitrate. A balance is maintained between fixed nitrogen and atmospheric nitrogen by bacteria that convert nitrate to N_2 under anaerobic conditions, a process called **denitrification**. These soil bacteria use NO_3^- rather than O2 as the ultimate electron acceptor in a series of reactions that (like oxidative phosphorylation) generates a transmembrane proton gradient, which is used to synthesize ATP.

Career Avenues



NITROGEN FIXATION

Only certain prokaryotes can fix atmospheric nitrogen. These include the cyanobacteria of soils and fresh and salt waters, other kinds of free-living soil bacteria such as *Azotobacter* species, and the nitrogen-fixing bacteria that live as **symbionts** in the root nodules of leguminous plants. The first important product of nitrogen fixation is ammonia, which can be used by all organisms either directly or after its conversion to other soluble compounds such as nitrites, nitrates, or amino acids.

Biological nitrogen fixation is carried out by a highly conserved complex of proteins called the **nitrogenise complex**, the crucial components of which are **dinitrogenase reductase** and **dinitrogenase**. Dinitrogenase reductase (*M*r 60,000) is a dimer of two identical subunits. It

contains a single 4Fe- 4S redox center, bound between the subunits, and can be oxidized and reduced by one electron. It also has two binding sites for ATP/ADP (one site on each subunit). Dinitrogenase (*M*r 240,000), a tetramer with two copies of two different subunits, contains both iron and molybdenum; its redox centers have a total of 2 Mo, 32 Fe, and 30 S per tetramer.

Another form of nitrogenase that contains vanadium rather than molybdenum has been discovered, and some bacterial species can produce both types of nitrogenase systems. The vanadium-containing enzyme may be the primary nitrogen-fixing system under some environmental conditions, but it is not yet as well characterized as the molybdenum-dependent enzyme.

Nitrogen fixation is carried out by a highly reduced form of dinitrogenase and requires eight electrons: six for the reduction of N_2 and two to produce one molecule of H2 as an obligate part of the reaction mechanism. Dinitrogenase is reduced by the transfer of electrons from dinitrogenase reductase. The dinitrogenase tetramer has two binding sites for the reductase. The required eight electrons are transferred from reductase to dinitrogenase one at a time: a reduced reductase molecule binds to the dinitrogenase and transfers a single electron, then the oxidized reductase dissociates from dinitrogenase, in a repeating cycle. Each turn of the cycle requires the hydrolysis of two ATP molecules by the dimeric reductase. one species, the ultimate source of electrons to reduce ferredoxin is pyruvate.

Career Avenues

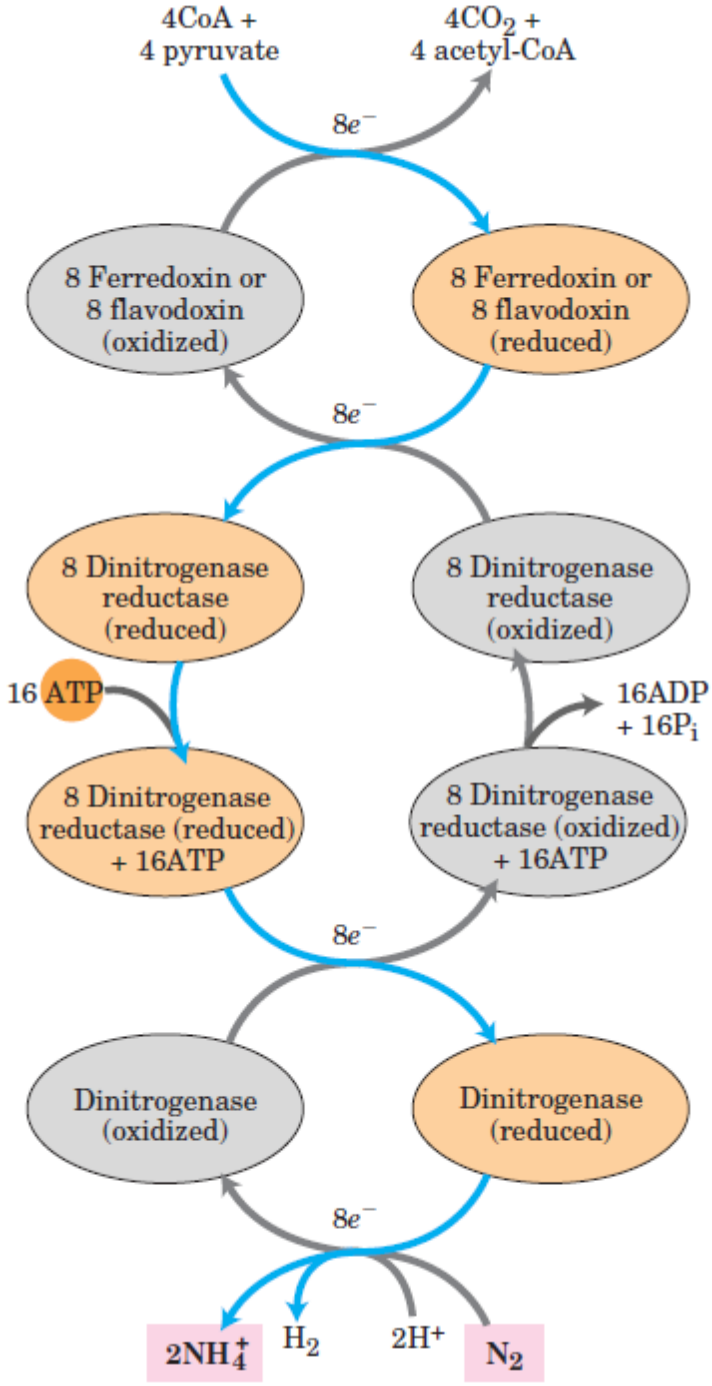


Figure: Nitrogen fixation by the nitrogenase complex. Electrons are transferred from pyruvate to dinitrogenase via ferredoxin (or flavodoxin) and dinitrogenase reductase. Dinitrogenase reductase reduces dinitrogenase one electron at a time, with at least six electrons required to fix one molecule of N_2. An additional two electrons are used to reduce $2H^+$ to H_2 in a process that obligatorily accompanies nitrogen fixation in anaerobes, making a total of eight electrons required per N_2 molecule.

The role of ATP in this process is somewhat unusual. ATP can contribute not only chemical energy, through the hydrolysis of one or more of its phosphoanhydride bonds, but also binding energy , through noncovalent interactions that lower the activation energy. In the reaction carried out by dinitrogenase reductase, both ATP binding and ATP hydrolysis bring about protein conformational changes that help overcome the high activation energy of nitrogen fixation.

Another important characteristic of the nitrogenise complex is an extreme lability in the presence of oxygen.

The symbiotic relationship between leguminous plants and the nitrogen-fixing bacteria in their root nodules takes care of both the energy requirements and the oxygen lability of the nitrogenise complex. The energy required for nitrogen fixation was probably the evolutionary driving force for this plant-bacteria association. The bacteria in root nodules have access to a large reservoir of energy in the form of abundant carbohydrate and citric acid cycle intermediates made available by the plant. This may allow the bacteria to fix hundreds of times more nitrogen than their free-living cousins can fix under conditions generally encountered in soils. To solve the oxygen-toxicity problem, the bacteria in root nodules are bathed in a solution of the oxygen-binding heme protein **leghemoglobin,** produced by the plant (although the heme may be contributed by the bacteria). Leghemoglobin binds all available oxygen so that it cannot interfere with nitrogen fixation, and efficiently delivers the oxygen to the bacterial electron-transfer system.

- heterocyst formation (cyanobacteria, e.g. *Anabaena*) where one cell does not photosynthesize but instead fixes nitrogen for its neighbors which in turn provide it with energy
- root nodule symbioses (e.g. *Rhizobium*) with plants that supply oxygen to the bacteria bound to molecules of leghaemoglobin
- anaerobic lifestyle (e.g. *Clostridium pasteurianum*)
- very fast metabolism (e.g. *Azotobacter vinelandii*)

The production and activity of nitrogenases is very highly regulated, both because nitrogen fixation is an extremely energetically expensive process (16-24 ATP are used per N_2 fixed) and due to the extreme sensitivity of the nitrogenase to oxygen.

Career Avenues

MUTATIONS AND THEIR CHEMICAL BASIS

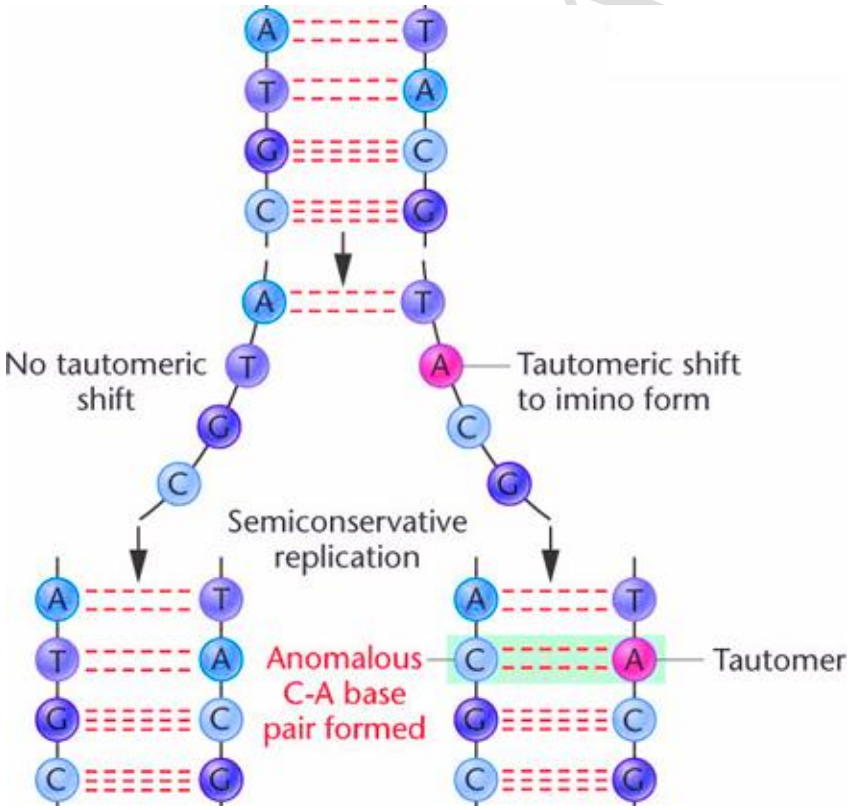
Mutations [Latin *mutare,* to change] were initially characterized as altered phenotypes or phenotypic expressions. Long before the existence of direct proof that a mutation is a stable, heritable change in the nucleotide sequence of DNA, geneticists predicted that several basic types of transmitted mutations could exist. They believed that mutations could arise from the alteration of single pairs of nucleotides and from the addition or deletion of one or two nucleotide pairs in the coding regions of a gene. Clearly, mutations may be characterized according to either the kind of genotypic change that has occurred or their phenotypic consequences.

Two classes of mutations are spontaneous mutations (molecular decay) and induced mutations caused by mutagens.

SPONTANEOUS MUTATION

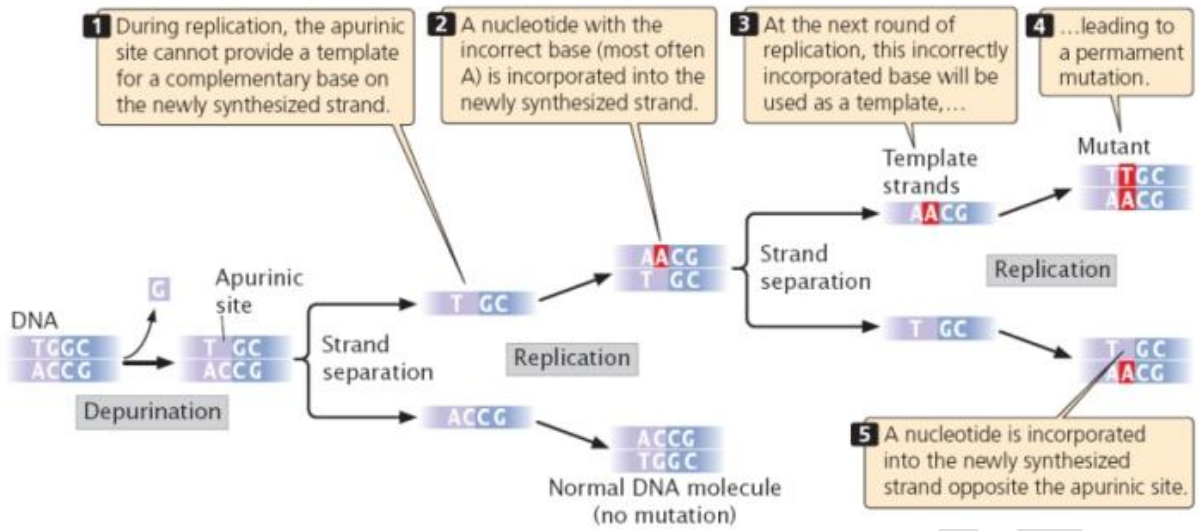
Spontaneous mutations on the molecular level can be caused by:

- **Tautomerism** – A base is changed by the repositioning of a hydrogen atom, altering the hydrogen bonding pattern of that base resulting in incorrect base pairing during replication.

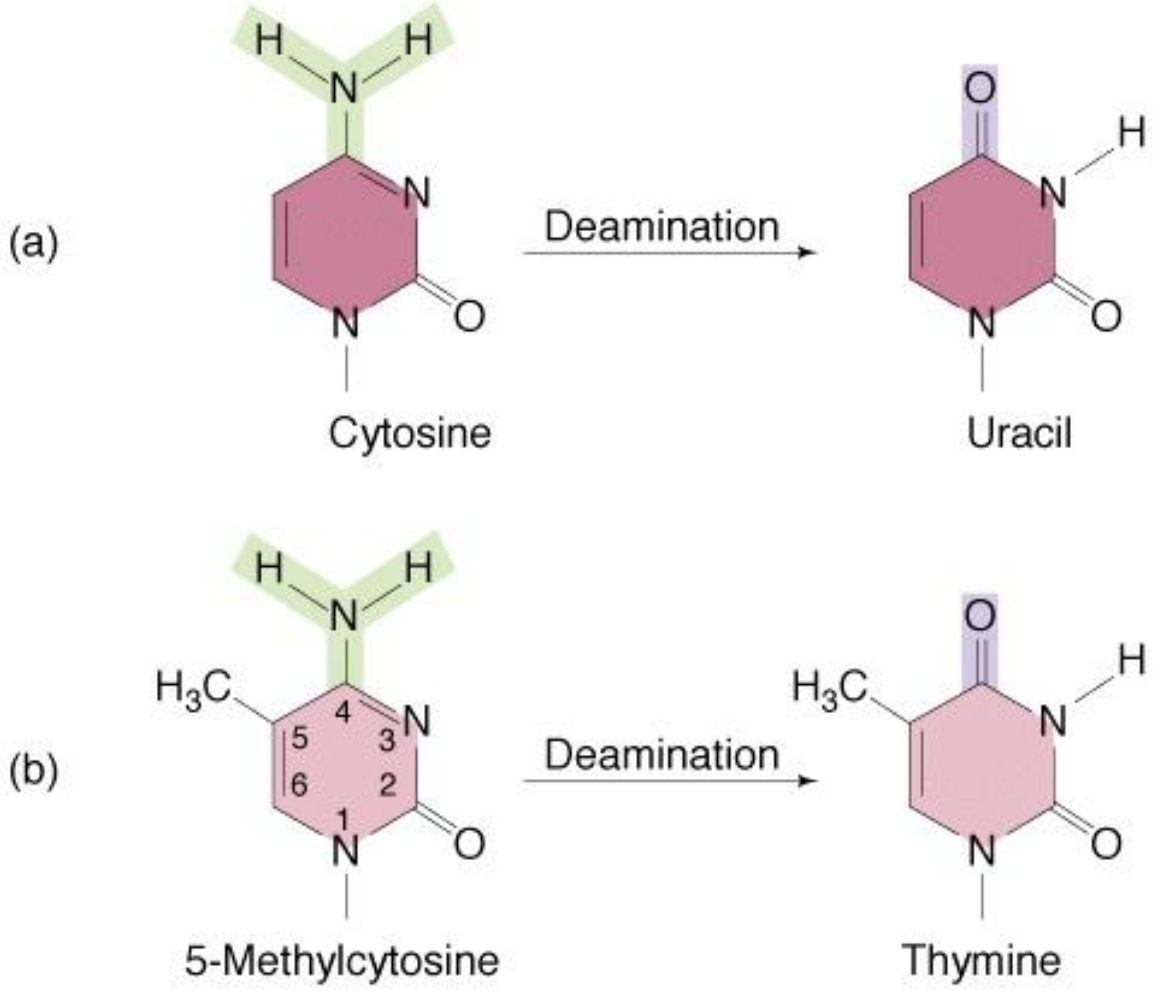


- **Depurination** – Loss of a purine base (A or G) to form an apurinic site (AP site).

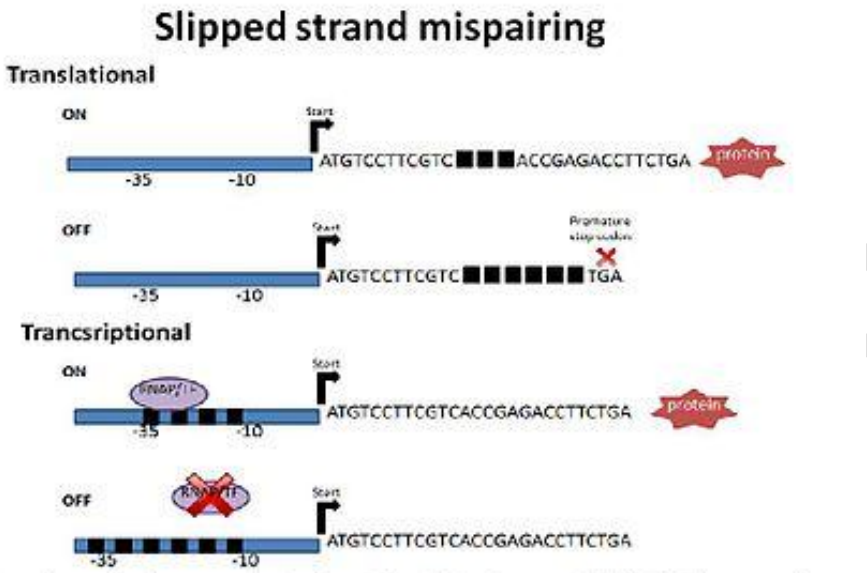
Career Avenues



- **Deamination** – Hydrolysis changes a normal base to an atypical base containing a keto group in place of the original amine group. Examples include C → U and A → HX (hypoxanthine), which can be corrected by DNA repair mechanisms; and 5MeC (5-methylcytosine) → T, which is less likely to be detected as a mutation because thymine is a normal DNA base.



- **Slipped strand mispairing** – Denaturation of the new strand from the template during replication, followed by renaturation in a different spot ("slipping"). This can lead to insertions or deletions.



Purple oval can either be a transcription factor (TF) or RNA polymerase (RNAP). Black boxes are short sequence repeats, Start (ATG) is the start codon in which the ribosome initiates translation of nucleotide sequence into amino acids and -10 -35 is the promoter which is the binding site for the RNAP to initiate transcription of DNA into RNA

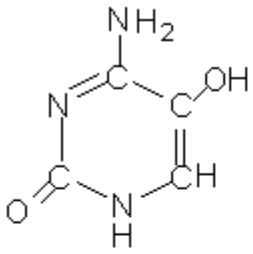
INDUCED MUTATION

Induced mutations on the molecular level can be caused by:

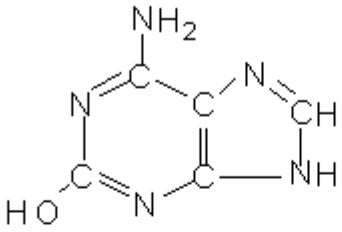
- Chemicals
  - Hydroxylamine N H_2OH

Career Avenues

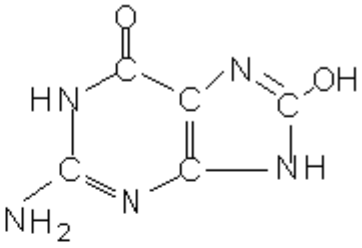
(a)



5-Hydroxycytosine

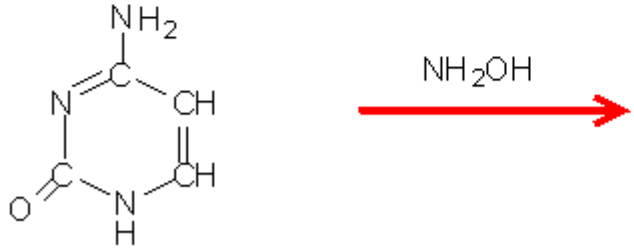


2-Hydroxyadenine



8-Hydroxyguanine
(8OHG)

(b)



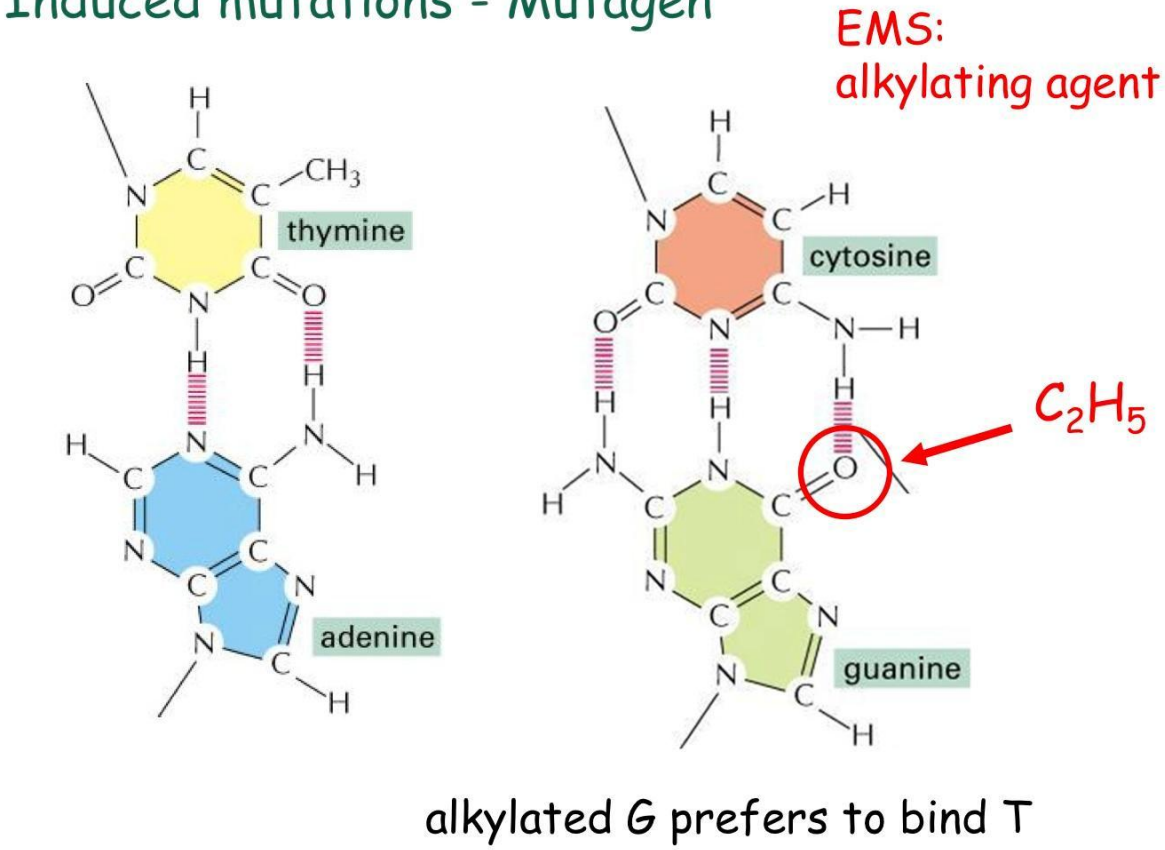
Cytosine

Hydroxylaminocytosine

- Base analogs (e.g. BrdU)
- Alkylating agents (e.g. *N*-ethyl-*N*-nitrosourea): These agents can mutate both replicating and non-replicating DNA. In contrast, a base analog can only mutate the DNA when the analog is incorporated in replicating the DNA. Each of these classes of chemical mutagens has certain effects that then lead to transitions, transversions, or deletions.

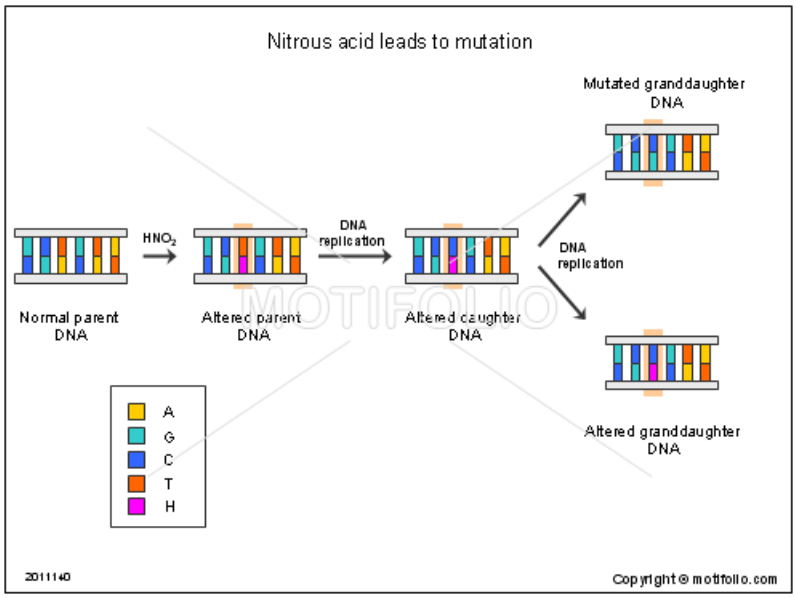
Career Avenues

Induced mutations - Mutagen



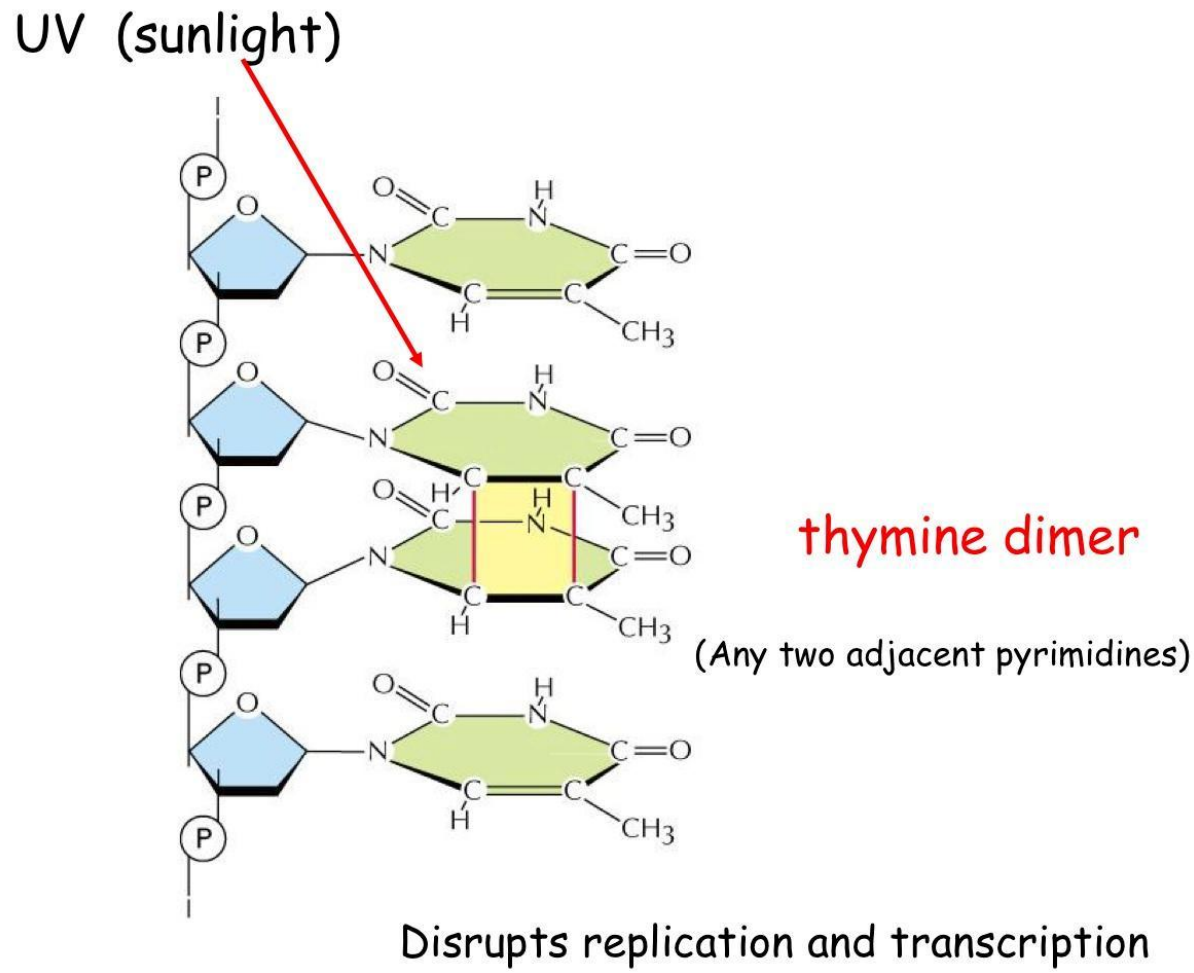
- Agents that form DNA adducts (e.g. ochratoxin A metabolites)
- DNA intercalating agents (e.g. ethidium bromide)
- DNA crosslinkers
- Oxidative damage
- Nitrous acid converts amine groups on A and C to diazo groups, altering their hydrogen bonding patterns which leads to incorrect base pairing during replication.

Career



- Radiation
- **Ultraviolet radiation (nonionizing radiation):** Two nucleotide bases in DNA – cytosine and thymine – are most vulnerable to radiation that can change their properties. UV light can induce adjacent pyrimidine bases in a DNA strand to become covalently joined as a pyrimidine dimer. UV radiation, particularly longer-wave UVA, can also cause oxidative damage to DNA. Mutation rates also vary across species.

Career Avenues



CLASSIFICATION OF MUTATION TYPES

BY EFFECT ON STRUCTURE

The sequence of a gene can be altered in a number of ways. Gene mutations have varying effects on health depending on where they occur and whether they alter the function of essential proteins. Mutations in the structure of genes can be classified as:

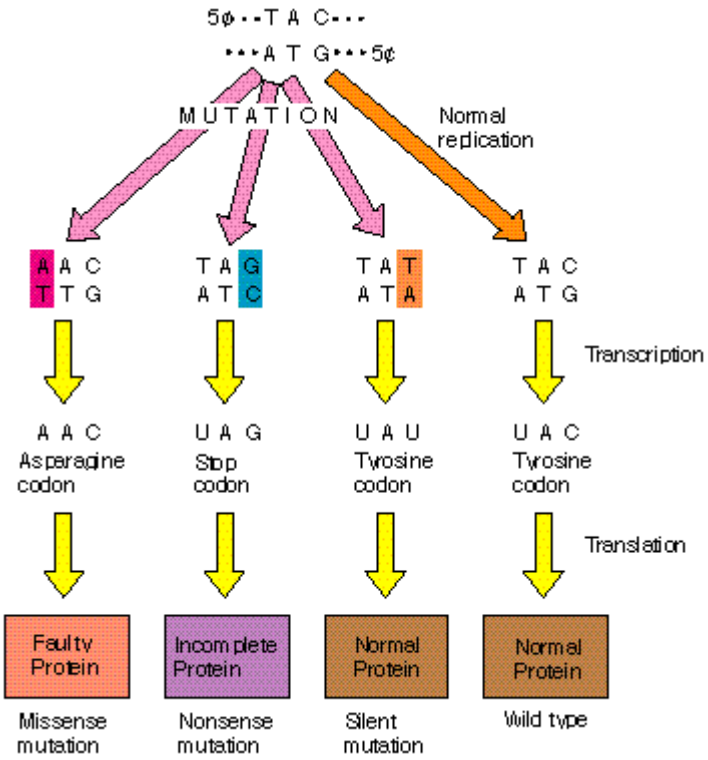
Small-scale mutations, such as those affecting a small gene in one or a few nucleotides, including:

Point mutations, often caused by chemicals or malfunction of DNA replication, exchange a single nucleotide for another. These changes are classified as transitions or transversions. Most common is the transition that exchanges a purine for a purine (A ↔ G) or a pyrimidine for a pyrimidine, (C ↔ T). A transition can be caused by nitrous acid, base mis-pairing, or mutagenic base analogs such as 5-bromo-2-deoxyuridine (BrdU).

Less common is a transversion, which exchanges a purine for a pyrimidine or a pyrimidine for a purine (C/T ↔ A/G). An example of a transversion is adenine (A) being converted into

a cytosine (C). A point mutation can be reversed by another point mutation, in which the nucleotide is changed back to its original state (true reversion) or by second-site reversion (a complementary mutation elsewhere that results in regained gene functionality). Point mutations that occur within the protein coding region of a gene may be classified into three kinds, depending upon what the erroneous codon codes for:

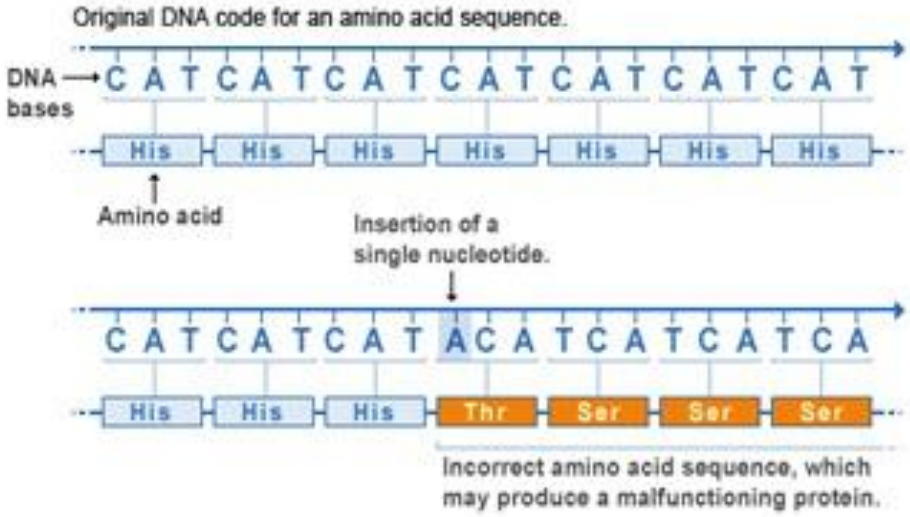
- Silent mutations: which code for the same amino acid.
- Missense mutations: which code for a different amino acid.
- Nonsense mutations: which code for a stop and can truncate the protein.



Insertions add one or more extra nucleotides into the DNA. They are usually caused by transposable elements, or errors during replication of repeating elements. Insertions in the coding region of a gene may alter splicing of the mRNA (splice site mutation), or cause a shift in the reading frame (frameshift), both of which can significantly alter the gene product. Insertions can be reverted by excision of the transposable element.

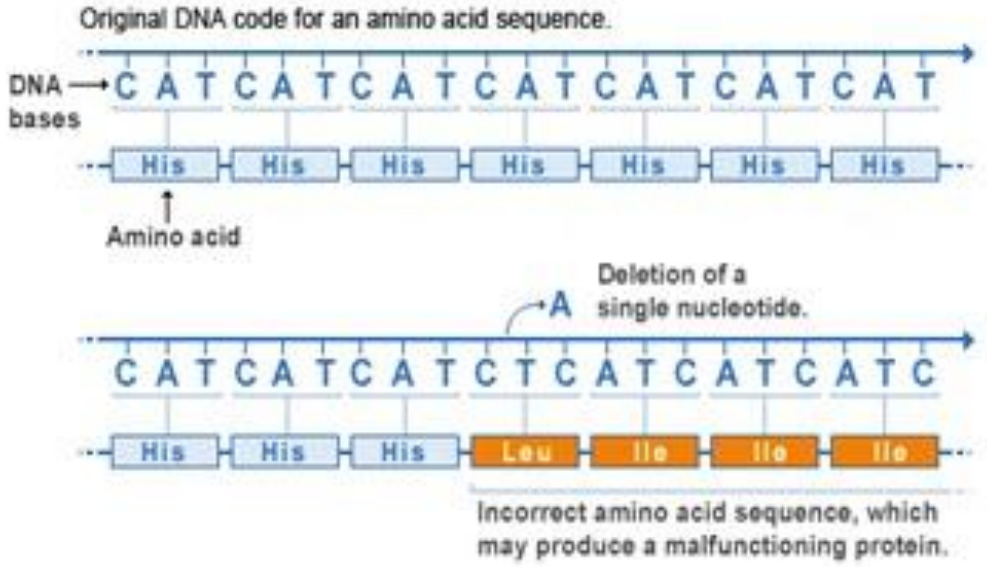
Career Avenues

Insertion mutation



Deletions remove one or more nucleotides from the DNA. Like insertions, these mutations can alter the reading frame of the gene. They are generally irreversible: though exactly the same sequence might theoretically be restored by an insertion, transposable elements able to revert a very short deletion (say 1–2 bases) in *any* location are either highly unlikely to exist or do not exist at all. Note that a deletion is not the exact opposite of an insertion: the former is quite random while the latter consists of a specific sequence inserting at locations that are not entirely random or even quite narrowly defined.

Deletion mutation

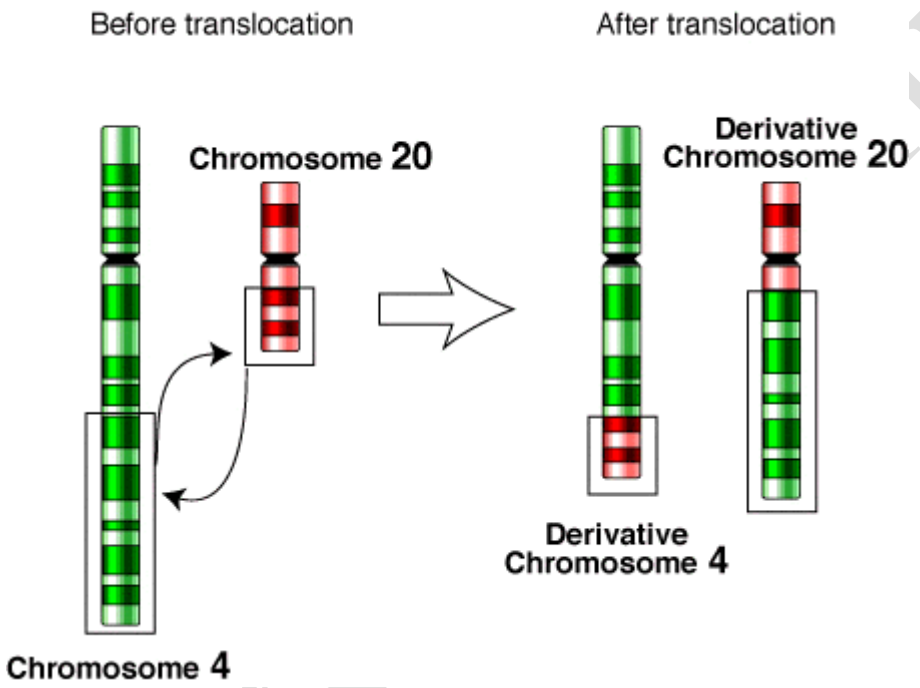


Large-scale mutations in chromosomal structure, including:

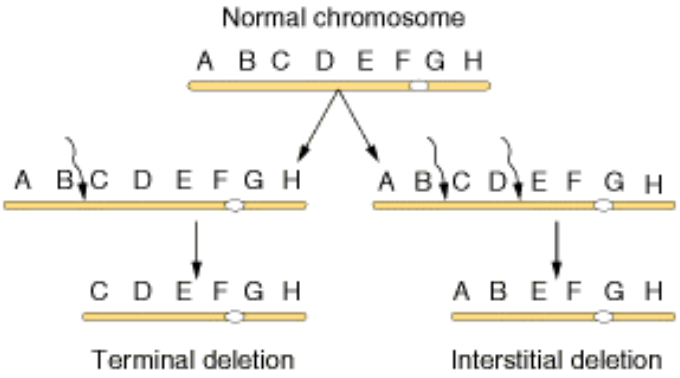
- **Amplifications** (or gene duplications) leading to multiple copies of all chromosomal regions, increasing the dosage of the genes located within them.
- **Deletions** of large chromosomal regions, leading to loss of the genes within those regions.

Mutations whose effect is to juxtapose previously separate pieces of DNA, potentially bringing together separate genes to form functionally distinct fusion genes (e.g. bcr-abl). These include:

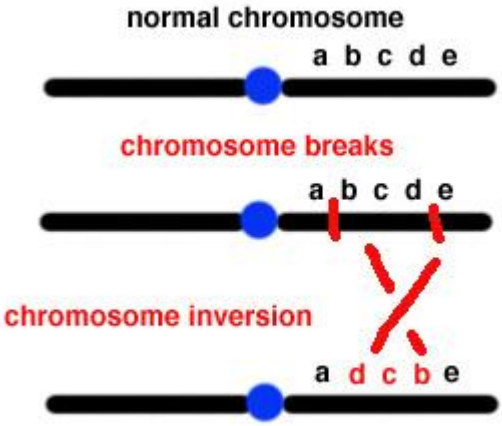
Chromosomal translocations: Interchange of genetic parts from nonhomologous chromosomes.



Interstitial deletions: An intra-chromosomal deletion that removes a segment of DNA from a single chromosome, thereby apposing previously distant genes. For example, cells isolated from a human astrocytoma, a type of brain tumor, were found to have a chromosomal deletion removing sequences between the "fused in glioblastoma" (fig) gene and the receptor tyrosine kinase "ros", producing a fusion protein (FIG-ROS). The abnormal FIG-ROS fusion protein has constitutively active kinase activity that causes oncogenic transformation (a transformation from normal cells to cancer cells).

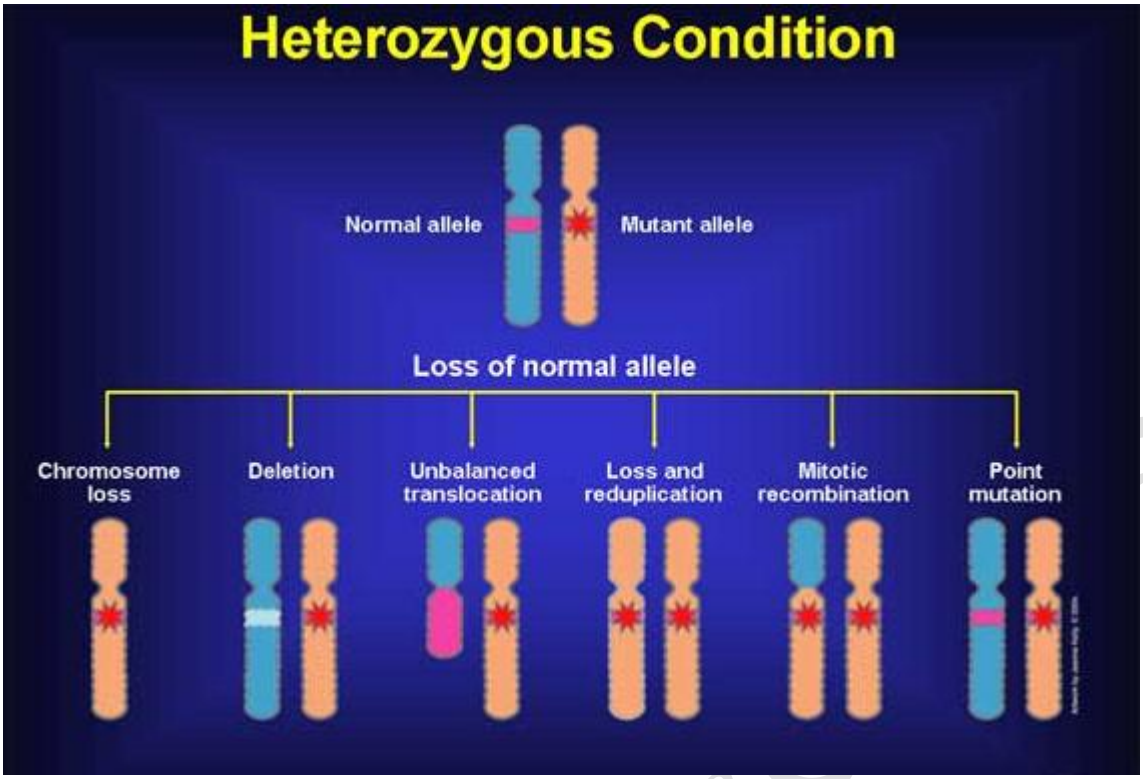


Chromosomal inversions: reversing the orientation of a chromosomal segment.



- **Loss of heterozygosity**: loss of one allele, either by a deletion or recombination event, in an organism that previously had two different alleles.

Career Avenues



By Effect On Function

- **Loss-of-function mutations** are the result of gene product having less or no function. When the allele has a complete loss of function (null allele) it is often called an **amorphic mutation**. Phenotypes associated with such mutations are most often recessive. Exceptions are when the organism is haploid, or when the reduced dosage of a normal gene product is not enough for a normal phenotype (this is called haploinsufficiency).
- **Gain-of-function mutations** change the gene product such that it gains a new and abnormal function. These mutations usually have dominant phenotypes. Often called a neomorphic mutation.
- **Dominant negative mutations** (also called **antimorphic mutations**) have an altered gene product that acts antagonistically to the wild-type allele. These mutations usually result in an altered molecular function (often inactive) and are characterised by a dominant or semi-dominant phenotype.
- **Lethal mutations** are mutations that lead to the death of the organisms which carry the mutations.
- A **back mutation** or **reversion** is a point mutation that restores the original sequence and hence the original phenotype.

By Effect On Fitness

In applied genetics it is usual to speak of mutations as either harmful or beneficial.

- A **harmful mutation** is a mutation that decreases the fitness of the organism.

Career Avenues

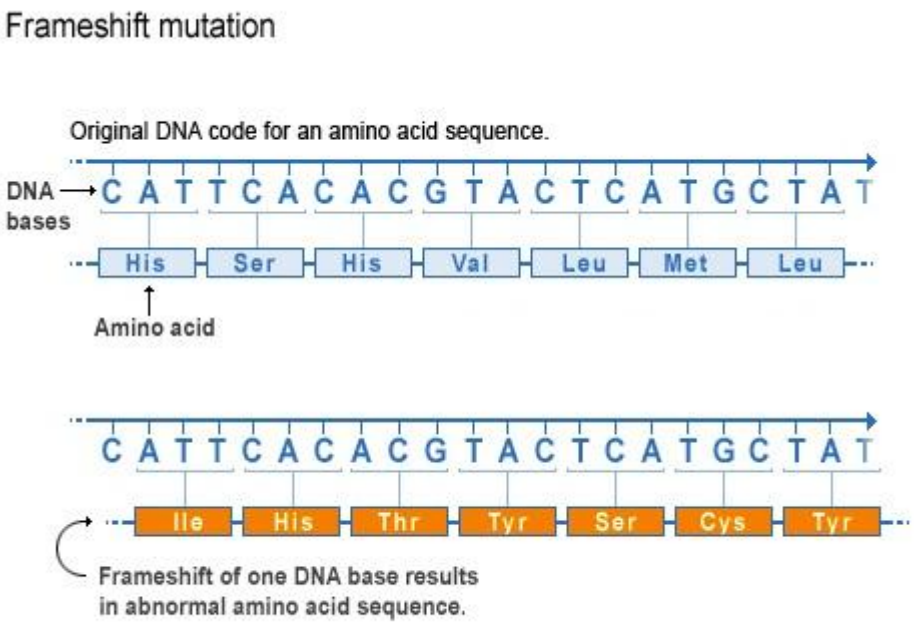
- A **beneficial mutation** is a mutation that increases fitness of the organism, or which promotes traits that are desirable.

In theoretical population genetics, it is more usual to speak of such mutations as deleterious or advantageous. In the neutral theory of molecular evolution, genetic drift is the basis for most variation at the molecular level.

- A **neutral mutation** has no harmful or beneficial effect on the organism. Such mutations occur at a steady rate, forming the basis for the molecular clock.
- A **deleterious mutation** has a negative effect on the phenotype, and thus decreases the fitness of the organism.
- An **advantageous mutation** has a positive effect on the phenotype, and thus increases the fitness of the organism.
- A **nearly neutral mutation** is a mutation that may be slightly deleterious or advantageous, although most nearly neutral mutations are slightly deleterious.

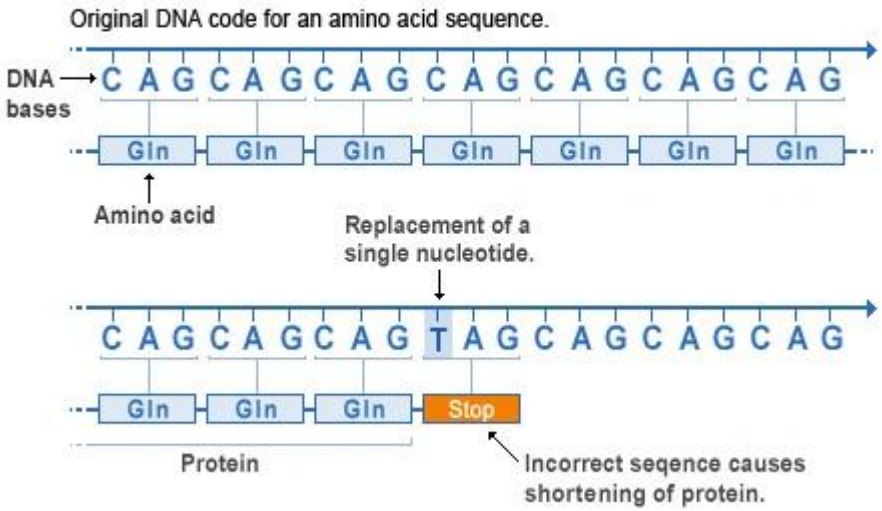
BY IMPACT ON PROTEIN SEQUENCE

- A **frameshift mutation** is a mutation caused by insertion or deletion of a number of nucleotides that is not evenly divisible by three from a DNA sequence. Due to the triplet nature of gene expression by codons, the insertion or deletion can disrupt the reading frame, or the grouping of the codons, resulting in a completely different translation from the original. The earlier in the sequence the deletion or insertion occurs, the more altered the protein produced is.



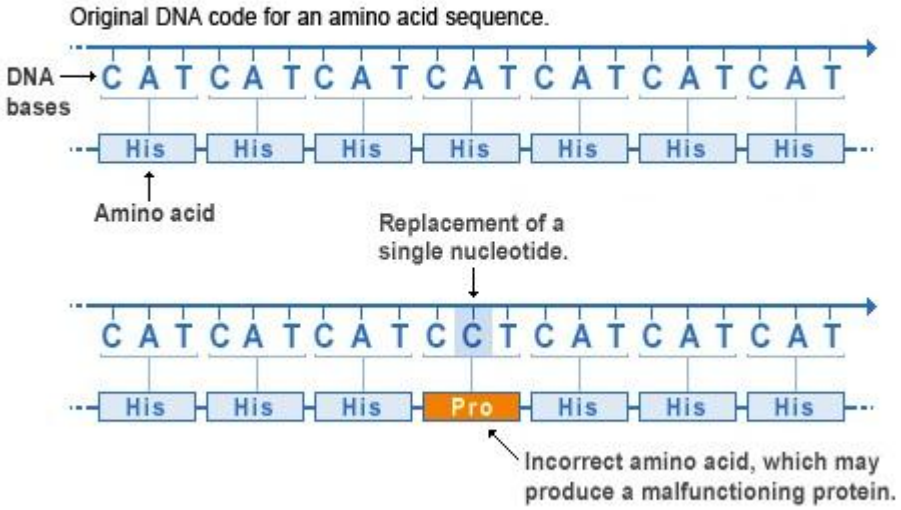
- A **nonsense mutation** is a point mutation in a sequence of DNA that results in a premature stop codon, or a *nonsense codon* in the transcribed mRNA, and possibly a truncated, and often nonfunctional protein product.

Nonsense mutation



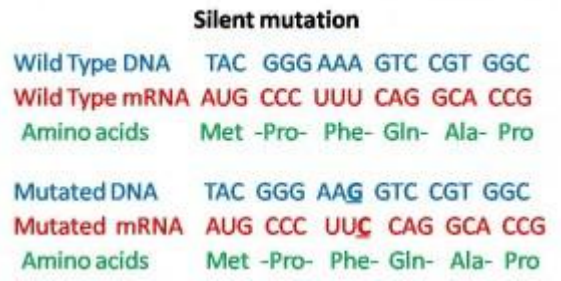
- **Missense mutations** or *nonsynonymous mutations* are types of point mutations where a single nucleotide is changed to cause substitution of a different amino acid. This in turn can render the resulting protein nonfunctional.

Missense mutation



- A **neutral mutation** is a mutation that occurs in an amino acid codon which results in the use of a different, but chemically similar, amino acid. The similarity between the two is enough that little or no change is often rendered in the protein. For example, a change from AAA to AGA will encode arginine, a chemically similar molecule to the intended lysine.

- **Silent mutations** are mutations that do not result in a change to the amino acid sequence of a protein. They may occur in a region that does not code for a protein, or they may occur within a codon in a manner that does not alter the final amino acid sequence.



BY INHERITANCE

In multicellular organisms with dedicated reproductive cells, mutations can be subdivided into germ line mutations, which can be passed on to descendants through their reproductive cells, and somatic mutations (also called acquired mutations), which involve cells outside the dedicated reproductive group and which are not usually transmitted to descendants.

A germline mutation gives rise to a *constitutional mutation* in the offspring, that is, a mutation that is present in every cell. A constitutional mutation can also occur very soon after fertilisation, or continue from a previous constitutional mutation in a parent.

The distinction between germline and somatic mutations is important in animals that have a dedicated germ line to produce reproductive cells. However, it is of little value in understanding the effects of mutations in plants, which lack dedicated germ line. The distinction is also blurred in those animals that reproduce asexually through mechanisms such as budding, because the cells that give rise to the daughter organisms also give rise to that organisms germ line. A new mutation that was not inherited from either parent is called a *de novo* mutation.

Diploid organisms (e.g. human) contain two copies of each gene – a paternal and a maternal allele. Based on the occurrence of mutation on each chromosome, we may classify mutations into three types.

- A **heterozygous mutation** is a mutation of only one allele.
- A **homozygous mutation** is an identical mutation of both the paternal and maternal alleles.
- **Compound heterozygous** mutations or a **genetic compound** comprises two different mutations in the paternal and maternal alleles.

A **wild type** or **homozygous non-mutated** organism is one in which neither allele is mutated.

Career Avenues

SPECIAL CLASSES

- **Conditional mutation** is a mutation that has wild-type (or less severe) phenotype under certain "permissive" environmental conditions and a mutant phenotype under certain "restrictive" conditions. For example, a temperature-sensitive mutation can cause cell death at high temperature (restrictive condition), but might have no deleterious consequences at a lower temperature (permissive condition).

MUTAGENESIS

DNA may be modified, either naturally or artificially, by a number of physical, chemical and biological agents, resulting in mutations.

MECHANISMS

Mutagenesis may occur endogenously, for example through spontaneous hydrolysis, or through normal cellular processes that can generate reactive oxygen species and DNA adducts, or through error in replication and repair. Mutagenesis may also arise as a result of the presence of environmental mutagens that induces changes to the DNA. The mechanism by which mutation arises varies according to the causative agent, the mutagen, involved. Most mutagens act either directly, or indirectly via mutagenic metabolites, on the DNA producing lesions. Some however may affect the replication or chromosomal partition mechanism, and other cellular processes.

Many chemical mutagens require **biological activation** to become mutagenic. An important group of enzymes involved in the generation of mutagenic metabolites is cytochrome P450. Other enzymes that may also produce mutagenic metabolites include glutathione S-transferase and microsomal epoxide hydrolase. Mutagens that are not mutagenic by themselves but require biological activation are called promutagens.

Many mutations arise as a result of problems caused by the DNA lesions during replication, resulting in **errors in replication**. In bacteria, extensive damage to the DNA due to mutagens results in single-stranded DNA gaps during replication. This induces the SOS response, an emergency repair process that is also error-prone, thereby generating mutations. In mammalian cells, stalling of replication at a damaged sites induces a number of rescue mechanisms that help bypass DNA lesions, but which also may result in errors. The Y family of DNA polymerases specialize in DNA lesion bypass in a process termed translesion synthesis (TLS) whereby these lesion-bypass polymerases replace the stalled high-fidelity replicative DNA polymrase, transits the lesion and extend the DNA until the lesion has been passed so that normal replication can resume. These processes may be error-prone or error-free.

SPONTANEOUS HYDROLYSIS

DNA is not entirely stable in aqueous solution. Under physiological conditions the glycosidic bond may be hydrolyzed spontaneously and 10,000 purine sites in DNA are estimated to be

Career Avenues

depurinated each day in a cell. Numerous DNA repair pathway exist for the DNA, however, if the apurinic site failed to be repaired, misincorporation of nucleotide may occur during replication. Adenine is preferentially incorporated by DNA polymerases in an apurinic site.

Cytidine may also become deaminated to uridine at one five-hundredth of the rate of depurination and can result in G to A transition. Eukaryotic cells also contains 5-methylcytosine, thought to be involved in the control of gene transcription, which can become deaminated into thymine.

MODIFICATION OF BASES

Bases may be modified endogenously by normal cellular molecules. For example DNA may be methylated by S-adenosylmethionine, and glycosylated by reducing sugars.

Many compounds, such as PAHs, aromatic amines, aflatoxin and pyrrolizidine alkaloids, may form reactive oxygen species catalyzed by cytochrome P450. These metabolites form adducts with the DNA, which can cause errors in replication, and the bulky aromatic adducts may form stable intercalation between bases and block replication. The adducts may also induce conformational changes in the DNA. Some adducts may also result in the depurination of the DNA, it is however uncertain how significant such depurination as caused by the adducts is in generating mutation.

Some alkylating agents such as N-Nitrosamines may also require the catalytic reaction of cytochrome-P450 for the formation of a reactive alkyl cation. Alkylation and arylation of bases can cause errors in replication. N^7 and O^6 of guanine and the N^3 and N^7 of adenine are most susceptible to attack; while N^7-guanine adducts, which form the bulk of DNA adducts, appear to be non-mutagenic, alkylation at O^6 of guanine is harmful because excision repair of O^6-adduct of guanine may be poor in some tissues such as the brain. The O^6 methylation of guanine can result in G to A transition, while O^4-methylthymine can be mispaired with guanine. The type of the mutation generated however may be dependent on the size and type of the adduct as well as the DNA sequence.

Ionizing radiations and reactive oxygen species often oxidize guanine to produce 8-oxoguanine.

CROSSLINKING

Some alkylating agents may produce crosslinking of DNA. Some natural occurring chemicals may also promote crosslinking, such as psoralens after activation by UV radiation, and nitrous acid. Interstrand cross-linking is more damaging as it blocks replication and transcription and can cause chromosomal breakages and rearrangements. Some crosslinkers such as cyclophosphamide, mitomycin C and cisplatin are used as anticancer chemotherapeutic because their high degree of toxicity to proliferating cells.

Career Avenues

DIMERIZATION

UV radiation promotes the formation of a cyclobutyl ring between adjacent thymines, resulting in the formation of pyrimidine dimers. In human skin cells, thousands of dimers may be formed in a day due to normal exposure to sunlight. DNA polymerase η may help bypass these lesions in an error-free manner; however, individuals with defective DNA repair function, such as sufferers of Xeroderma pigmentosum, are sensitive to sunlight and may be prone to skin cancer.

INTERCALATION BETWEEN BASES

The planar structure of chemicals such as ethidium bromide and proflavine allows them to insert between bases in DNA, and cause frameshift mutation. The intercalation into DNA of anthracyclines such as daunorubicin and doxorubicin interferes with the functioning of the enzyme topoisomerase II, blocking replication as well as causing mitotic homologous recombination.

BACKBONE DAMAGE

Ionizing radiations may produce highly reactive free radicals that can break the bonds in the DNA. Double-stranded breakages are especially damaging and hard to repair, producing translocation and deletion of part of a chromosomes. Alkylating agents like mustard gas may also cause breakages in the DNA backbone. Oxidative stress may also generate highly reactive oxygen species that can damage the DNA. Incorrect repair of other damages induced by the highly reactive species can also lead to mutations.

INSERTIONAL MUTAGENESIS

Transposon and virus may insert DNA sequence into coding region or functional elements of a gene and result in inactivation of the gene.

ERROR IN REPLICATION

While most mutagens produce effects that ultimately result in error in replication, some mutagens may affect directly the replication process. Base analog such as 5-bromouracil may substitute for thymine in replication. Some metals such as cadmium, chromium, and nickel may alter the fidelity of DNA replication.

SITE-DIRECTED MUTAGENESIS

It is desirable that specific changes can be introduced to the DNA. Analogs of nucleotides and other chemicals were first used to generate localized point mutations. Such chemicals may be aminopurine which induces AT to GC transition, while nitrosoguanidine,bisulfite, and N^4-hydroxycytidine may induce GC to AT transition. These technique allows specific

Career Avenues

mutations to be engineered into a protein, however, they are not flexible in the kinds of mutants generated.

Current techniques for site-specific mutation commonly involve using mutagenic oligonucleotides in a primer extension reaction with DNA polymerase. This methods allows for point mutation, or deletion or insertion of small stretches of DNA to be introduced at specific sites. Advances in methodology have made such mutagenesis now a relatively simple and efficient process.

The site-directed approach may be done systematically in such technique as **alanine scanning mutagenesis** whereby residues are systematically mutated to alanine in order to identify residues important to the structure or function of a protein.

COMBINATORIAL MUTAGENESIS

Combinatorial mutagenesis is a technique whereby large number of mutants may be screened for a particular characteristic. In this technique, a few selected positions or a short stretch of DNA may be exhaustively modified to obtain a comprehensive library of mutant proteins. One approach of this technique is to excise a portion of DNA and replaced with a library of sequences containing all possible combinations at the desired mutation sites. The segment may be at an enzyme active site, or sequences that have structural significance or immunogenic property. A segment however may also be inserted randomly into the gene in order to assess the structural or functional significance of particular part of protein.

GATE BT 2010

Q.11 Ames test is used to determine

(A) The mutagenicity of a chemical
(B) Carcinogenicity of a chemical
(C) Both mutagenicity and carcinogenicity of a chemical
(D) Toxicity of a chemical

GATE BT 2011

Q.12 Determine the correctness or otherwise of the following Assertion (a) and the reason (r)

Assertion: N – methyl – N' – nitro – N – nitrosoguanidine (NTG) is an effective Chemical mutagen

Reason: Mutations induced by NTG mainly are the GC $\rightarrow$ At transitions

(A) Both (a) and (r) are true and (r) is the correct reason for (a)
(B) Both (a) and (r) are true and (r) is not the correct reason for (a)
(C) (a) is true but (r) is false
(D) (a) is false but (r) is true

Career Avenues

GATE BT 2012

Q.13 A truncated polypeptide is synthesized due to a nonsense mutation. Where would you introduce another mutation to obtain a full – length polypeptide?

(A) Ribosomal protein gene
(B) Transfer RNA gene
(C) DNA repair gene
(D) Ribosomal RNA gene

GATE BT 2013

Q.14 The total number of fragments generated by the complete and sequential cleavage of the polypeptide given below by Trypsin followed by CNBr is _____

Phe-Trp-Met-Gly-Ala-Lys-Leu-Pro-Met-Asp-Gly-Arg-Cys-Ala-Gln

Answer:

- **Q.11 - C.**
- **Q.12 – A.**
- **Q.13 – B.**
- **Q.14 – 5.**

Career Avenues

MICROBIAL GENETICS

- Plasmids
- Transformation
- Transduction
- Conjugation

PLASMIDS

Many bacteria possess plasmids in addition to their chromosome. These are double-stranded DNA molecules, usually circular, that can exist and replicate independently of the chromosome or may be integrated with it; in either case they normally are inherited or passed on to the progeny. However, plasmids are not usually attached to the plasma membrane and sometimes are lost to one of the progeny cells during division. Plasmids are not required for host growth and reproduction, although they may carry genes that give their bacterial host a selective advantage. Plasmid genes can render bacteria drug-resistant, give them new metabolic abilities, make them pathogenic, or endow them with a number of other properties. Because plasmids often move between bacteria, properties such as drug resistance can spread throughout a population.

BACTERIAL PLASMIDS

Conjugation, the transfer of DNA between bacteria involving direct contact, depends on the presence of an "extra" piece of circular DNA known as a plasmid. Plasmids play many important roles in the lives of bacteria. They also have proved invaluable to microbiologists and molecular geneticists in constructing and transferring new genetic combinations and in cloning genes. In this section the different types of bacterial plasmids are discussed.

Plasmids are small double-stranded DNA molecules, usually circular, that can exist independently of host chromosomes and are present in many bacteria (they are also present in some yeasts and other fungi). They have their own replication origins and are autonomously replicating and stably inherited. A replicon is a DNA molecule or sequence that has a replication origin and is capable of being replicated. Plasmids and bacterial chromosomes are separate replicons. Plasmids have relatively few genes, generally less than 30. Their genetic information is not essential to the host, and bacteria that lack them usually function normally. Single-copy plasmids produce only one copy per host cell. Multi-copy plasmids may be present at concentrations of 40 or more per cell.

Characteristically, plasmids can be eliminated from host cells in a process known as curing. Curing may occur spontaneously or be induced by treatments that inhibit plasmid replication while not affecting host cell reproduction. The inhibited plasmids are slowly diluted out of the growing bacterial population. Some commonly used curing treatments are acridine

Career Avenues

mutagens, UV and ionizing radiation, thymine starvation, and growth above optimal temperatures.

Plasmids may be classified in terms of their mode of existence and spread. An episome is a plasmid that can exist either with or without being integrated into the host's chromosome. Some plasmids, conjugative plasmids, have genes for pili and can transfer copies of themselves to other bacteria during conjugation.

BACTERIAL CONJUGATION

The initial evidence for bacterial conjugation, the transfer of genetic information by direct cell to cell contact, came from an elegant experiment performed by Joshua Lederberg and Edward L. Tatum in 1946. They mixed two auxotrophic strains, incubated the culture for several hours in nutrient medium, and then plated it on minimal medium. To reduce the chance that their results were due to simple reversion, they used double and triple auxotrophs on the assumption that two or three reversions would not often occur simultaneously. For example, one strain required biotin (Bio_), phenylalanine (Phe_), and cysteine (Cys_) for growth, and another needed threonine (Thr_), leucine (Leu_), and thiamine (Thi_). Recombinant prototrophic colonies appeared on the minimal medium after incubation (figure 13.12). Thus the chromosomes of the two auxotrophs were able to associate and undergo recombination.

Lederberg and Tatum did not directly prove that physical contact of the cells was necessary for gene transfer. This evidence was provided by Bernard Davis (1950), who constructed a U tube consisting of two pieces of curved glass tubing fused at the base to form a U shape with a fritted glass filter between the halves. The filter allows the passage of media but not bacteria. The U tube was filled with nutrient medium and each side inoculated with a different auxotrophic strain of E. coli (figure 13.13). During incubation, the medium was pumped back andforth through the filter to ensure medium exchange between the halves. After a 4-hour incubation, the bacteria were plated on minimal medium. Davis discovered that when the two auxotrophic strains were separated from each other by the fine filter, gene transfer could not take place. Therefore direct contact was required for the recombination that Lederberg and Tatum had observed.

$F_+ \times F_-$ MATING

In 1952 William Hayes demonstrated that the gene transfer observedby Lederberg and Tatum was polar. That is, there were definitedonor (F+) and recipient (F-) strains, and gene transfer was nonreciprocal. He also found that in F+×F- mating the progeny were only rarely changed with regard to auxotrophy (that is, bacterial genes were not often transferred), but F-strains frequently became F+.

These results are readily explained in terms of the F factor previously described (figure 13.5). The F_ strain contains an extra-chromosomal F factor carrying the genes for pilus formation and plasmid transfer. During F+× F- mating or conjugation, the F factor replicates by the rolling-circle mechanism, and a copy moves to the recipient. The entering strand is copied to produce double-stranded DNA. Because bacterial chromosome genes are rarely transferred

Career Avenues

with the independent F factor, the recombination frequency is low. It is still not completely clear how the plasmid moves between bacteria. The sex pilus or F pilus joins the donor and recipient and may contract to draw them together. The channel for DNA transfer could be either the hollow F pilus or a special conjugation bridge formed upon contact.

THE U-TUBE EXPERIMENT

The U-tube experiment used to show that genetic recombination by conjugation requires direct physical contact between bacteria.

HFR CONJUGATION

Because certain donor strains transfer bacterial genes with great efficiency and do not usually change recipient bacteria to donors, a second type of conjugation must exist. The F factor is an episome and can integrate into the bacterial chromosome at several different locations by recombination between homologous insertion sequences present on both the plasmid and host chromosomes. When integrated, the F plasmid's tra operon is still functional; the plasmid can direct the synthesis of pili, carry out rolling-circle replication, and transfer genetic material to an F- recipient cell. Such a donor is called an Hfr strain (for high frequency of recombination) because it exhibits a very high efficiency of chromosomal gene transfer in comparison with F+ cells. DNA transfer begins when the integrated F factor is nicked at its site of transfer origin . As it is replicated, the chromosome moves through the pilus or conjugation bridge connecting the donor and recipient. Because only part of the F factor is transferred at the start (the initial break is within the F plasmid), the F_ recipient does not become F_ unless the whole chromosome is transferred. Transfer is standardized at 100 minutes in E. coli, and the connection usually breaks before this process is finished. Thus a complete F factor usually is not transferred, and the recipient remains F_.As mentioned earlier, when an Hfr strain participates in conjugation, bacterial genes are frequently transferred to the recipient.

Gene transfer can be in either a clockwise or counterclockwise direction around the circular chromosome, depending on the orientation of the integrated F factor. After the replicated donor chromosome enters the recipient cell, it may be degraded or incorporated into the F- genome by recombination.

F' CONJUGATION

Because the F plasmid is an episome, it can leave the bacterial chromosome. Sometimes during this process the plasmid makes an error in excision and picks up a portion of the chromosomal material to form an F' plasmid. It is not unusual to observe the inclusion of one or more genes in excised F plasmids.

The F' cell retains all of its genes, although some of them are on the plasmid, and still mates only with an F_ recipient. F'× F- conjugation is virtually identical with F+ × F- mating.

Career Avenues

Once again, the plasmid is transferred, but usually bacterial genes on the chromosome are not. Bacterial genes on the F′ plasmid are transferred with it and need not be incorporated into the recipient chromosome to be expressed. The recipient becomes F′ and it is a partially diploid merozygote since it has two sets of the genes carried by the plasmid. In this way specific bacterial genes may spread rapidly throughout a bacterial population. Such transfer of bacterial genes is often called sexduction. F′ conjugation is very important to the microbial geneticist. A partial diploid's behavior shows whether the allele carried by an F′ plasmid is dominant or recessive to the chromosomal gene. The formation of F′ plasmids also is useful in mapping the chromosome since if two genes are picked up by an F factor they must be neighbors.

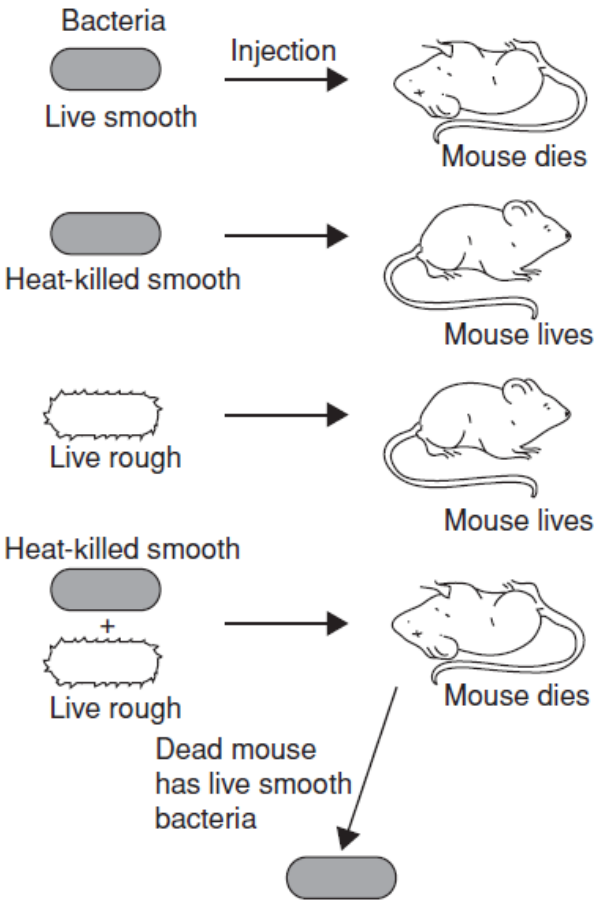
Career Avenues

GENETIC TRANSFER IN MICROORGANISMS

The various mechanisms of mutation described above result in an alteration in the genetic make-up of an organism. This can also occur by *recombination*, in which genetic material from two cells combines to produce a variant different to either parent cell. In bacteria, this involves the transfer of DNA from one cell (*donor*) to another (*recipient*). Because transfer occurs between cells of the same generation (unlike the genetic variation brought about by sexual reproduction in eucaryotes, where it is passed to the next generation), it is sometimes referred to as *horizontal transfer*. There are three ways in which gene transfer can occur in bacteria, which we shall explore in the following sections.

TRANSFORMATION

Transformation is the simplest of these, and also the first to have been described. We have already referred to the classic experiment of Fred Griffith in 1928, the first demonstration.



That genetic transfer can occur in bacteria. Griffith had previously demonstrated the existence of two strains of the bacterium *Streptococcus pneumoniae*, which is one of the causative agents of pneumonia in humans, and is also extremely virulent in mice.

The S (smooth)-form produced a polysaccharide capsule, whilst the R (rough)-form did not. These formed recognisably different colonies when grown on a solid medium, but more importantly, differed in their ability to bring about disease in experimental animals. The R-form, lacking the protective capsule, was easily destroyed by the defence system of the host. Griffith observed the effects of injecting mice with bacterial cells of both forms; these are outlined below and in Figure 11.24. The results of experiments

- 1–3 were predictable, those of experiment 4 startlingly unexpected:
- 1 Mice injected with live cells of the R-form were unaffected.
- 2 Mice injected with live cells of the S-form died, and large numbers of S-form bacteria were recovered from their blood.
- 3 Mice injected with cells of the S-form that had been killed by heating at 60 °C were unharmed and no bacteria were recovered from their blood. 4 Mice injected with a mixture of living R-form and heat-killed S-form cells died, and living S-form bacteria were isolated from their blood.

The S-form bacteria recovered from the mice in the crucial fourth experiment possessed a polysaccharide capsule like other S-forms, and, critically, were able to pass on this characteristic to subsequent generations. This finding went against the prevailing view that bacteria simply underwent binary fission, a completely asexual process involving no genetic transfer. Griffith deduced that some as yet unknown substance had passed from the heat-killed S-form cells to some of the living R-forms and conferred on them the ability to make capsules. Not long afterwards it was shown that this process of *transformation* could happen in the test tube, without the involvement of a host animal, and, as we have seen, it was eventually shown that Griffith's 'transforming principle' was DNA.

How Does Transformation Occur?

The uptake of foreign DNA from the environment is known to occur naturally in a number of bacterial types, both Gram-positive and Gram-negative, by taking up fragments of naked DNA released from dead cells in the vicinity. Being linear and very fragile, the DNA is easily broken into fragments, each carrying on average around 10 genes.

Transformation will only happen at a specific stage in the bacterial life cycle, when cells are in a physiological state known as competence. This occurs at different times in different bacteria, but is commonly during late log phase. One of the reasons why only a low percentage of recipient cells become transformed is that only a small proportion of them are at any one time in a state of competence. The expression of proteins essential to the transformation process is dependent on the secretion of a competence factor.

The exact mechanism of transformation varies somewhat according to species; the process for Bacillus subtilis is shown in Figure 11.25. Mere uptake of exogenous DNA is not enough to cause transformation; it must also be integrated into the host genome, displacing a single strand, which is subsequently degraded. Upon DNA replication and cell division, one

Career Avenues

daughter cell will inherit the parent genotype, and the other the recombinant genotype. In the latter, there is a stable alteration of the cell's genetic composition, and a new phenotype is expressed in subsequent generations. Uptake of donor DNA from an unrelated species will not result in transformation; this is due to a failure to locate a homologous sequence and integrate into the host's chromosome, rather than an inability to gain entry to the cell.

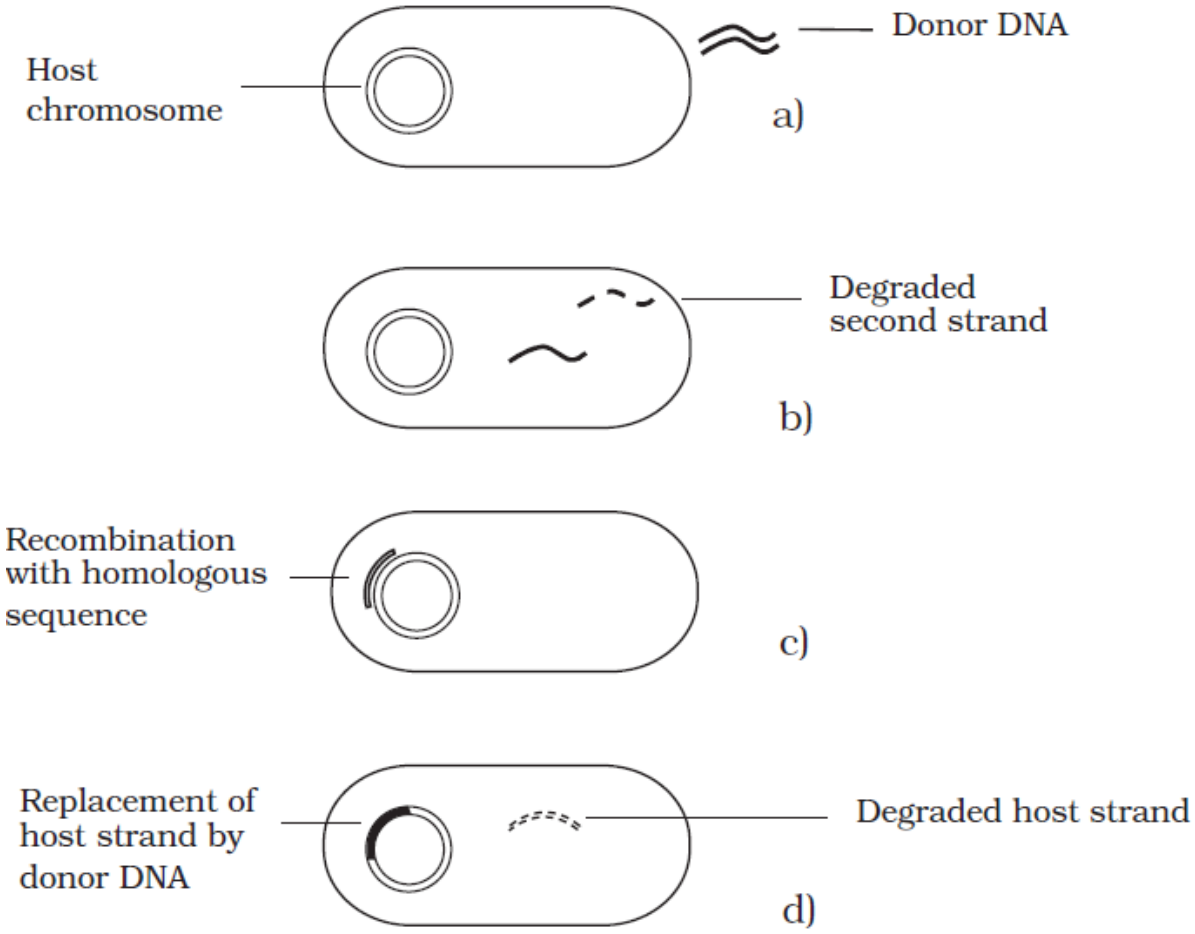


Figure 11.25 : Transformation in Bacillus subtilis. (a) A fragment of donor DNA become bound to the recipient cell surface by means of a DNA-binding protein. (b) After binding, a nuclease contained at the cell surface degrades one strand of the donor DNA, leaving the other strand to be ferried, by other transformation-specific proteins, to the interior of the cell. (c) A fragment of single-stranded DNA aligns with a homologous stretch on the recipient chromosome. (d) The donor fragment becomes integrated by a process of non-reciprocal recombination. At the next cell division, one daughter cell is a transformant, whilst the other retains the parental genotype.

CONJUGATION

In 1946, Edward Tatum and Joshua Lederberg (the latter aged only 21 and still a student!) demonstrated a second form of genetic transfer between bacteria. These experiments involved the use of auxotrophic mutants which have lost the ability to make a particular enzyme involved in the biosynthesis of an essential nutrient.

These can be recognised experimentally by their inability to grow on a medium lacking the nutrient in question (Figure 11.26). The results of Tatum and Lederberg's experiments suggested that a process was taking place in bacteria akin to sex in eucaryotes, in which there was a recombination of the cells' genetic material.

Transformation, as demonstrated by Griffith, could not explain their results, since the addition of a DNA-containing extract of one strain to whole cells of the other did not result in prototroph formation. This was confirmed in an ingenious experiment in which Bernard Davis showed that Tatum and Lederberg's results were only obtained if direct cell-to cell contact was allowed (Figure 11.28).

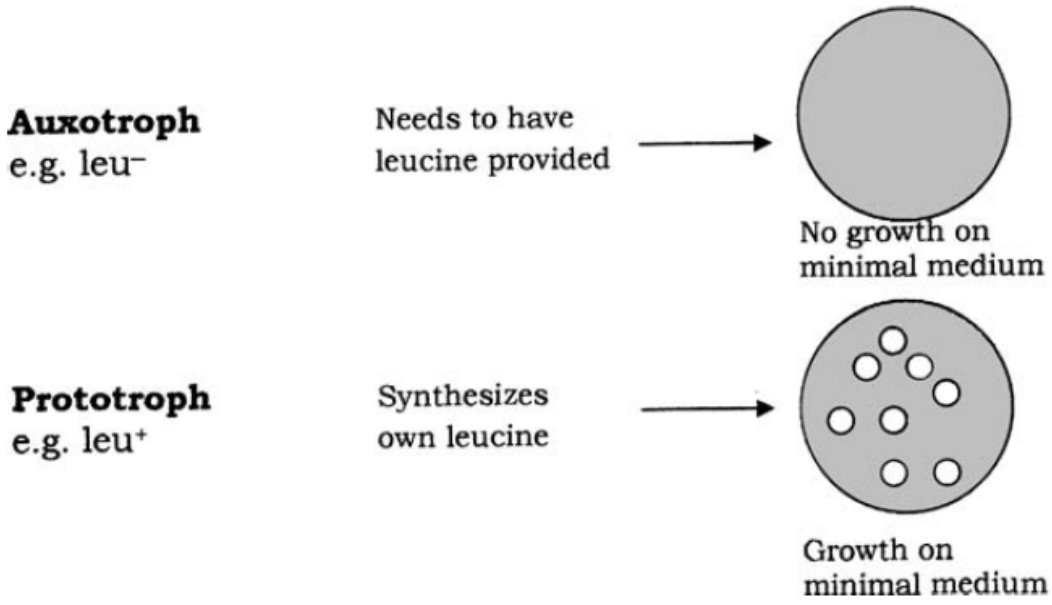


Figure 11.26 Auxotrophic mutants provide useful genetic markers. They are unable to synthesise a particular nutrient, and can be detected by their inability to grow on a minimal medium (one containing only inorganic salts and a carbon source such as glucose).

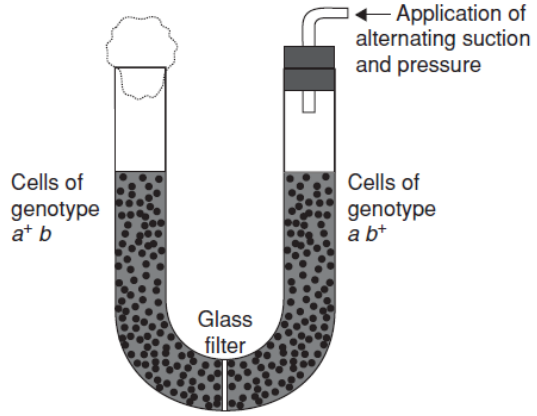


Figure 11.28 The Davis U-tube experiment: Two bacterial strains were placed in two arms of a U-tube, separated by a filter which did not permit the passage of cells. Although suction was used to transfer medium between compartments, no recombinants resulted. The results provided direct evidence that cell-to-cell contact is essential for conjugation to occur.

Gene transfer in conjugation is one way only

The process of conjugation was initially envisaged as a fusion of the two partner cells to give a diploid zygote, which subsequently underwent meiosis to give haploid offspring with modified genotypes. The work of William Hayes, however, showed that the development of colonies of recombinant cells was dependant on the survival of only one of the participating strains, the other strain being required only as a donor of DNA.

It became apparent that in E. Coli, there are two distinct mating types, which became known as F^+ and F^-, depending on whether or not they possessed a plasmid called the F (fertility) plasmid. This contains some 30 or 40 genes responsible for its own replication and for the synthesis of a thread-like structure expressed on the cell surface called a sex pilus. In a mixture of F^+ and F^- cells, the sex pilus contacts an F^-, contracts to pull the two cells together. A single strand of plasmid DNA is then passed across a channel made between the two cells, and enters the F− cell (Figure 11.29). This then serves as a template for the production of the complementary second strand, and similarly the single strand left behind in the donor cell is replicated, leaving us with a double-stranded copy of the F plasmid in both cells. By acquiring the F plasmid, the recipient F− cell has been converted to F+.

When conjugation involves a form of donor cell called Hfr (high frequency of recombination), genes from the main bacterial chromosome may be transferred. In these, the F plasmid has become integrated into the main bacterial chromosome; and thus loses its ability to replicate independently (Figure 11.30). It behaves just like any other part of the chromosome, although of course it still carries the genes for conjugation and pilus formation. When conjugation occurs, a single strand of Hfr DNA is broken within the F sequence, and is transferred in a linear fashion, carrying behind it chromosomal DNA.

Recombination in the recipient cell results in the replacement of a homologous segment of DNA by the transferred material, which is then faithfully replicated in subsequent generations. If transfer is uninterrupted, a process which takes around 100 minutes in E. coli,

Career Avenues

eventually the remaining stretch of F plasmid will enter the F− cell, bringing up the rear. The fragile nature of the pilus means, however, that transfer is rarely complete, and only a limited portion of the bacterial genome is transferred. This means that those F− cells receiving DNA from Hfr cells usually remain F−, unlike those in a cross with F+ cells, because the remainder of the F sequence is not transferred. It soon became clear that this phenomenon afforded a great opportunity to determine the relative positions of genes on the bacterial chromosome. This was done by interrupted mating experiments, in which the time allowed for conjugation is deliberately limited by mechanical breakage of the sex pili, and correlated with the phenotypic traits transferred to the recipient cells (Figure 11.31). By these means a genetic map of the bacterial chromosome could be developed.

The integration of the F plasmid into the bacterial chromosome is reversible; thus Hfr cells can revert to F+. Excision of the integrated plasmid is not always precise, and sometimes a little chromosomal DNA is removed too. When this happens, the plasmid, and the cell containing it, are called F_ ('F prime'); transfer of the plasmid to an F− cell takes with it the extra DNA from the host chromosome. The recipient genome thus becomes partially diploid (merodiploid), because it has its own copy, plus the 'guest' copy of certain genes.

Career Avenues

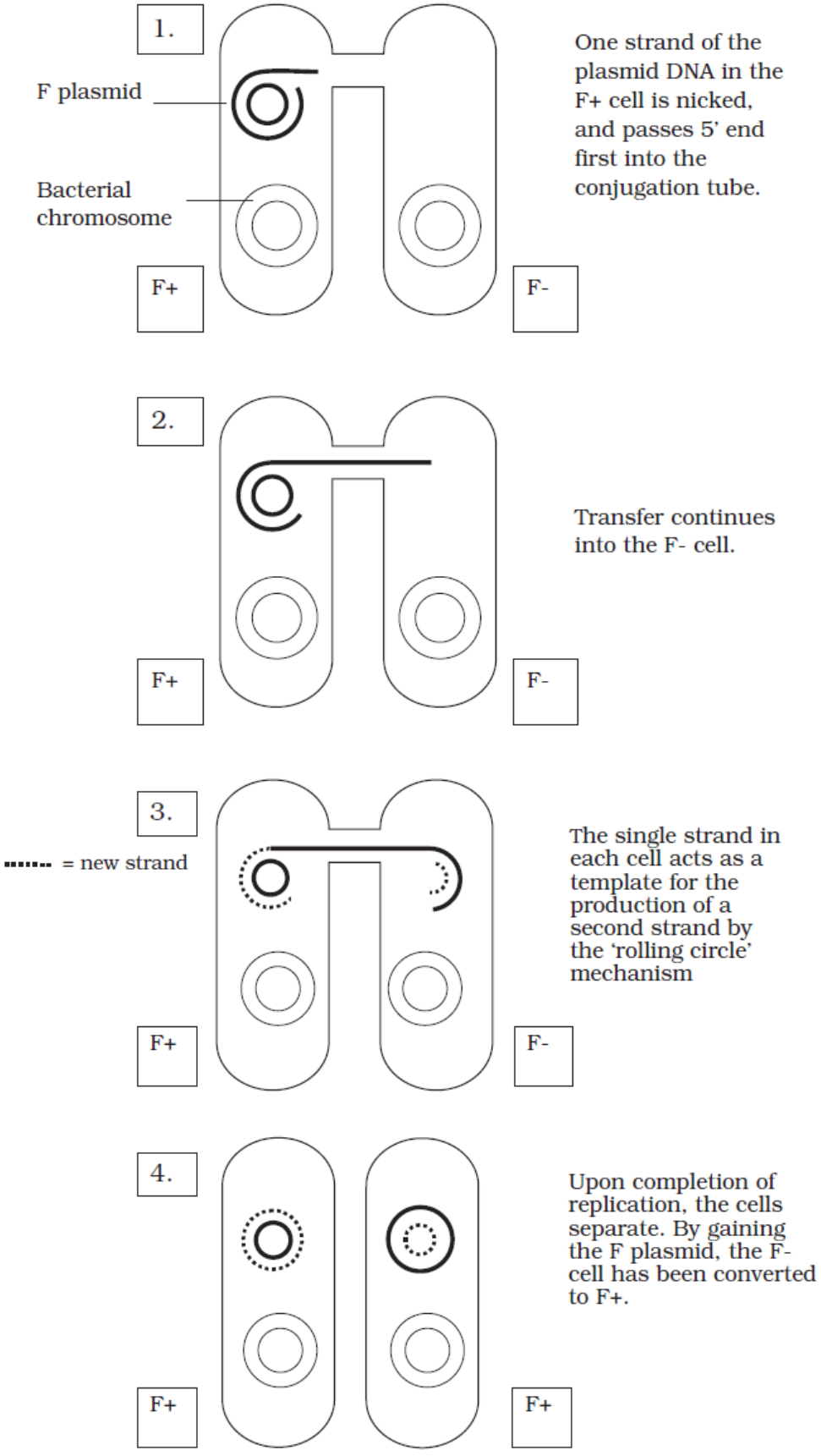


Figure 11.29 Conjugation in bacteria.

A single-stranded copy of the F plasmid passes across a conjugation tube from an F+ to an F − cell. Both this and the copy left behind act as templates for their own replication, leaving both cells with a complete F plasmid. This means that the recipient cell is converted from F− to F+.

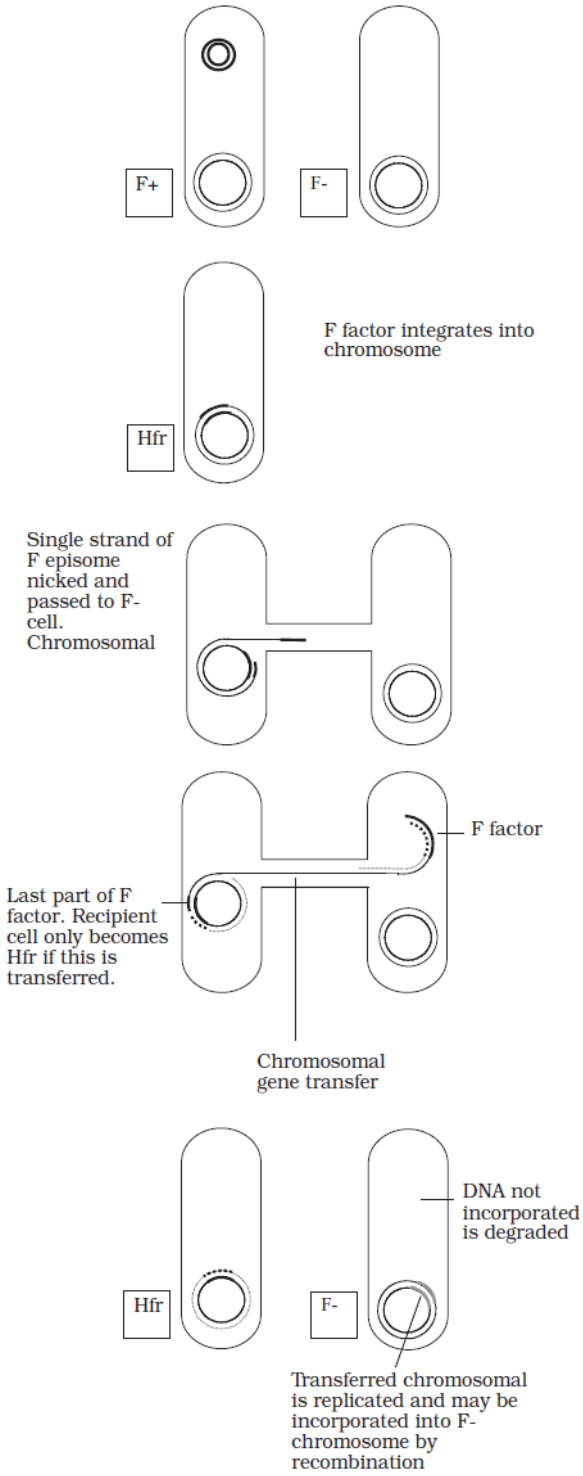


Figure 11.30 Conjugation with an Hfr cell results in transfer of chromosomal genes.

enues

Hfr cells are formed by the integration of the F factor into the bacterial chromosome. During conjugation, transfer begins part of the way along the F episome, and continues with chromosomal DNA. The amount of chromosomal material transferred depends on how long conjugation is able to proceed. Conjugation may be followed by recombination of transferred chromosomal material with its homologous sequence on the recipient cell's chromosome.

TRANSDUCTION

In the third form of genetic transfer in bacteria, bacteriophages act as carriers of DNA from one cell to another. In order to appreciate the way in which this is done, it is necessary to recall the sequence of events in phage replication cycles discussed in the previous chapter (see Figure 10.11).

Generalised transduction occurs in virulent phages, that is, those with a lytic life cycle. Sometimes, the enzymes responsible for packaging phage DNA into its protein coat package instead similarly sized fragments of degraded chromosomal DNA (Figure 11.32). Despite containing the wrong DNA, this transducing phage particle is still infective, since this is dependent on its protein element. Thus following infection of another bacterial cell, the DNA can be incorporated by recombining with the homologous segment in the recipient cell. Since any chromosomal fragment can be mistakenly packaged in this way (as long as it finds an area of homology), all genes are transferred at a similar (low) frequency.

Specialised transduction results in a much higher efficiency of transfer for specific genes, however it is limited to genes having a particular chromosomal location. Recall from Chapter 10 that in lysogenic life cycles, the phage DNA is integrated into the host chromosome, and later, perhaps after many rounds of cell division, excised again before re-entering a lytic cycle. If this excision does not happen precisely, some of the adjoining chromosomal DNA, carrying a gene or two, may be incorporated into the phage particle (we saw a similar mechanism in the case of F_ plasmid formation). Upon infecting another cell, the transduced genes would undergo recombination and become incorporated into the recipient's chromosome (Figure 11.33). Although limited to genes in the vicinity of the lysogenic phage's integration, this is a highly efficient form of transfer, since the genes become stably integrated into the host cell. Transduction experiments, like those involving conjugation, can be used to determine the relative positions of genes on a bacterial chromosome.

Career Avenues

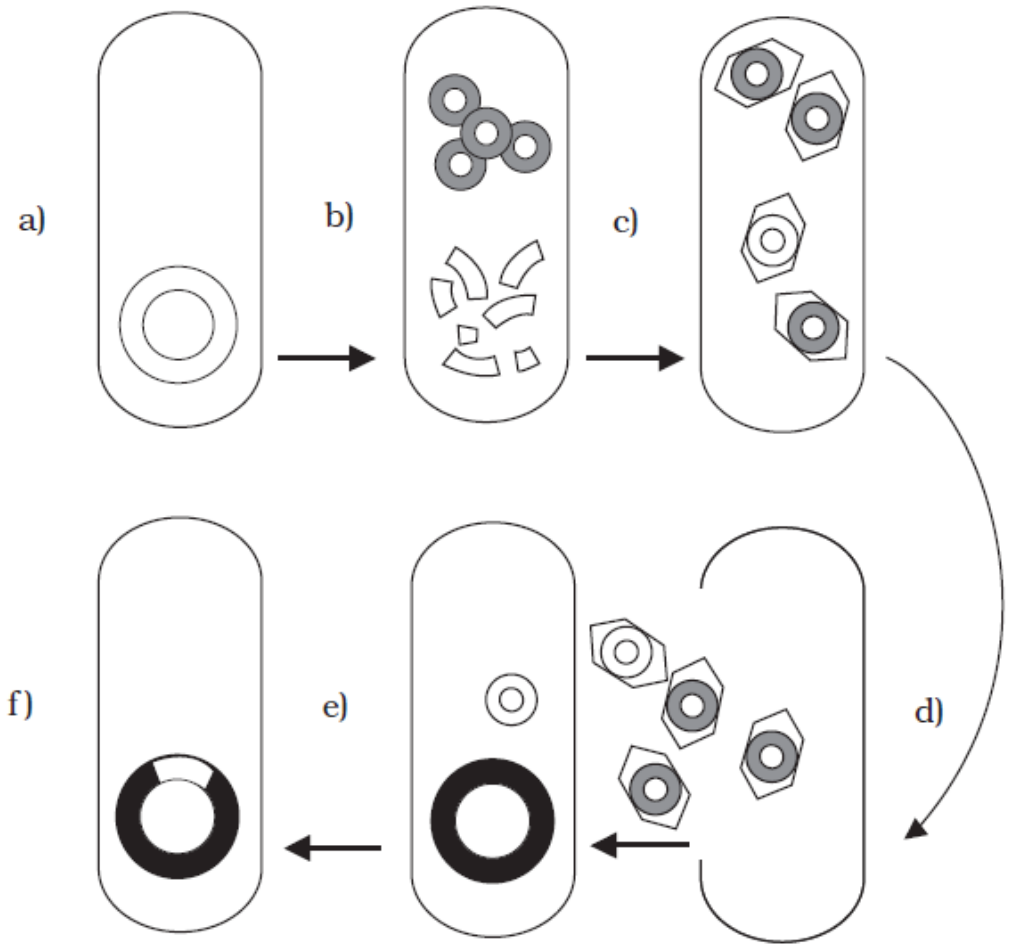


Figure 11.32: Generalised transduction.

During the lytic cycle of a phage, the host DNA is degraded (b), and a fragment may be mistakenly packaged into a newly synthesised phage particle (c). Upon infecting a new host cell, the transducing phage releases its DNA (e); although unable to replicate, this can undergo recombination with a homologous sequence on the host chromosome (f).

Career

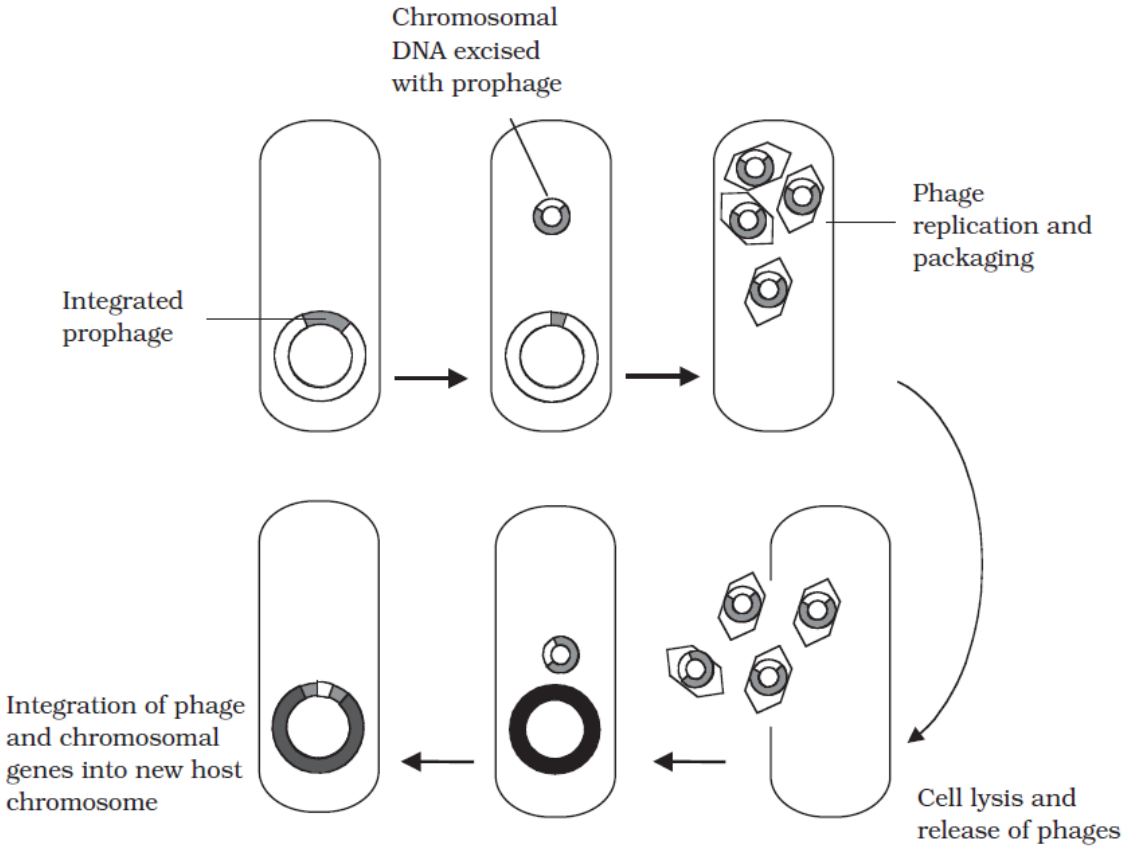


Figure 11.33: Specialised transduction.

During the replication cycle of a lysogenic bacteriophage, phage DNA is incorporated into the host chromosome (see Figure 10.11). When a lytic cycle resumes and the phage DNA is excised, it may take with it an amount of surrounding chromosomal DNA. This is packaged into phage particles and infects new host cells, where it is integrated into the bacterial chromosome. Only genes surrounding the site of phage integration may be transduced in this way.

GATE BT 2010

Q.15 Match the following antibiotics in Group I with their mode of action in Group II

Group I	Group II
P. Chloramphenicol	1.Binds to DNA gyrase
Q. Norfloxacin	2.Binds to RNA polymerase
R. Puromycin	3. Inhibits peptidyl transferase
S. Rifampicin	4. Mimics aminoacyl tRNA

(A) P-1, Q-3, R-2, S-4 (B) P-3, Q-1, R-2, S-4

(C) P-3, Q-1, R-4, S-2 (D) P-4, Q-2, R-3, S-1

Career

GATE BT 2011

Q.16 Diphtheria toxin, tetracycline and streptomycin inhibit
(A) DNA repair
(B) DNA replication
(C) Transcription
(D) Translation

GATE BT 2013

Q.17 In nature, the horizontal gene transfer across bacteria is mediated by
(A) gene cloning followed by transformation
(B)conjugation and transformation
(C) conjugation only
(D) transformation only

Q.18 The RNA primer synthesized during the replication process in bacteria is removed by
(A) DNA gyrase (B)primase
(C) DNA polymeraseI (D) DNA polymerase II

Answers:

- **Q.15 – C.**
- **Q.16 – D.**
- **Q.17 – B.**
- **Q.18 – C.**

Career Avenues

MICROBIAL DIVERSITY

BRIEF DESCRIPTION OF SOME SPECIFIC MICROORGANISM

A) Spirilla

Collected together under this heading are several genera of aerobic (mostly microaerophilic) spiral-shaped bacteria with polar flagella. These include free-living, symbiotic and parasitic types.

Spirilla such as *Aquaspirillum* and *Magnetospirillum* contain *magnetosomes*, intracellular particles of iron oxide (magnetite, Fe3O4). Such magnetotactic bacteria have the remarkable ability to orientate themselves with respect to the earth's magnetic field (*magnetotaxis*).

Two important pathogens of humans are included in the spirilla; *Campylobacter jejuni* is responsible for foodborne gastroenteritis, while *Helicobacter pylori* has in recent times been identified as the cause of many cases of peptic ulcers.

Representative genera: *Magnetospirillum, Campylobacter*

B) Rickettsia

This group comprises arthropod-borne intracellular parasites of vertebrates, and includes the causative agents of human diseases such as *typhus* and *Rocky Mountain spotted fever*. The bacteria are taken up by host phagocytic cells, where they multiply and eventually cause lysis.

The Rickettsia are aerobic organotrophs, but some possess an unusual mode of energy metabolism, only being able to oxidise intermediate metabolites such as glutamate and succinate, which they obtain from their host. *Rickettsia* and *Coxiella*, the two main genera, are not closely related phylogenetically and are placed in the α- and γ - Proteobacteria, respectively.

Representative genera: *Rickettsia, Coxiella*

C) Neisseria and related Proteobacteria

All members of this loose collection of bacteria are aerobic non-motile cocci, typically seen as pairs, with flattened sides where they join. Some however only assume this morphology during stationary growth phase. Many are found in warm-blooded animals, and some species are pathogenic. The genus *Neisseria* includes species responsible for gonorrhoea and meningitis in humans.

Career Avenues

Characterstics of Whittaker's five kingdoms :

	Monera (Procaryotae)	Protista	Fungi	Plantae	Animalia
Cell Type	Procaryotic	Eucaryotic	Eucaryotic	Eucaryotic	Eucaryotic
Cell Organization	Unicellular; occasionally grouped	Unicellular; occasionally multicellular	Unicellular or multicellular	Multicellualr	Multicellular
Cell Wall	Present in most	Present in some; absent in others	Present	Present	Absent
Nutrition	Absorption; some photosynthetic, some chemosynthetic	Ingestion or absorption, some photosynthetic	Absorption	Absorptive, photosynthetic	Ingestion; occasionally in some parasites by absorption
Reproduction	Asexual, usually by binary fission	Mostly asexual, occasionally both sexual and asexual	Both asexual and sexual, often involving a complex life cycle	Both asexual and sexual	Primarily sexual

Career Avenues

D) Cyanobacteria: the blue-green bacteria

Members of the Cyanobacteria were once known as blue-green algae because they carry out the same kind of *oxygenic photosynthesis* as algae and green plants They are the only group of procaryotes capable of carrying out this form of photosynthesis; all the other groups of photosynthetic bacteria to be discussed in this chapter carry out an anoxygenic form. When it became possible to examine cell structure in more detail with the electron microscope, it became clear that the cyanobacteria were in fact procaryotic, and hence quite distinct from the true algae. Old habits die hard, however, and the term 'blue-green algae' is still encountered, particularly in the popular press. Being procaryotic, cyanobacteria do not possess chloroplasts; however they contain lamellar membranes called *thylakoids*, which serve as the site of photosynthetic pigments and as the location for both light-gathering and electron transfer processes.

Early members of the Cyanobacteria evolved when the oxygen content of the earth's atmosphere was much lower than it is now, and these organisms are thought to have been

responsible for its gradual increase, since photosynthetic eucaryotes did not arise until many millions of years later.

Cyanobacteria are Gram-negative bacteria which may be unicellular or filamentous; in spite of the name by which they were formerly known, they may also appear variously as red, black or purple, according to the pigments they possess. A characteristic of many cyanobacteria is the ability to fix atmospheric nitrogen, that is, to reduce it to ammonium ions (NH4 +) for incorporation into cellular constituents. In filamentous forms, this activity is associated with specialised, enlarged cells called *heterocysts*.

The tiny unicellular cyanobacterium *Prochlorococcus* is found in oceans throughout the tropical and temperate regions and is thought to be the most abundant photosynthetic organism on our planet. It has several strains adapted to different light conditions. Some cyanobacteria are responsible for the production of unsightly (and smelly!) 'algal' blooms in waters rich in nutrients such as phosphate. When they die, their decomposition by other bacteria leads to oxygen depletion and the death of other aquatic life forms. Bloom-forming species contain gas vacuoles to aid their buoyancy.

Representative genera: *Oscillatoria, Anabaena*

Note:

Control measures for microorganisms include capitalizing on our knowledge of:						
	Growth on artificial media	Division by binary fission	Whether they have both DNA and RNA	Whether they have ribosomes	Whether they have muramic acid	Their sensitivity to antibiotics
Bacteria	Yes	Yes	Yes	Yes	Yes	Yes
Mycoplasma	Yes	Yes	Yes	Yes	No	Yes
Rickettsia	No	Yes	Yes	Yes	Yes	Yes
Chlamydia	No	Yes	Yes	Yes	No	Yes
Viruses	No	No	No	No *	No	No
* The arenavirus family (an RNA virus family) appears to package ribosomes 'accidentally'. The packaged ribosomes appear to play no role in viral protein synthesis.						

GATE BT 2010

Q.19 Interferon - β is produced by

(A) Bacteria infected cells
(B) Virus infected cells
(C) Both virus and bacteria infected cells

Career Avenues

(D) Fungi infected cells

Q.20 Match the products in Group I with the microbial cultures in Group II used for their industrial production

Group I	Group II
P. Gluconic acid	1. Leuconostoc mesenteroids
Q. L-Lysine	2. Aspergillus niger
R. Dextran	3. Brevibacterium flavum
S. Cellulase	4. Trichoderma reesei

(A) P-2, Q-1, R-3, S-4 (B) P-1, Q-3, R-4, S-2
(C) P-2, Q-3, R-1, S-4 (D) P-3, Q-2, R-4, S-1

GATE BT 2013

Q.21 Match the cell structures in **Group I** with the organismsin **Group II.**

Group I	**Group II**
P. Endospores	1. *Methanobacterium*
Q. Bipolar flagella	2. *Treponema*
R. Pseudomurine in cell wall	3. *Spirillum*
S. Periplasmic flagella	4. *Clostridium*

(A) P-4, Q-3, R-1, S-2 (B) P-4, Q-3, R-2, S-1
(C) P-3, Q-4, R-1, S-2 (D) P-4, Q-1, R-3, S-2

Q.22 Match the commercial microbial sources in **Group I** with the products in **Group II.**

Group I	**Group II**
P. *Corynebacteriumlilium*	1. 2,3-Butane di-ol
Q. *Klebsiellaoxytoca*	2. Poly-β-hydroxybutyric acid
R. *Aspergillusniger*	3. Glutamic acid
S. *Alcaligeneseutrophus*	4. Citric acid

(A) P-3,Q-1,R-2,S-4 (B) P-3,Q-1,R-4,S-2
(C) P-1,Q-3,R-2,S-4 (D) P-1,Q-3,R-4,S-2

Q.23 Phylum proteobacteria is subdivided into α - , β -, γ -, δ - and ε -proteobacteria based on

(A) G+C content (B) 23S rRNA sequences
(C)tRNA sequences (D)16S rRNA sequences

Career Avenues

Answers:

- **Q.19 – B.**
- **Q.20 – C.**
- **Q.21 – A.**
- **Q.22 – B.**
- **Q.23 – D.**

Career Avenues

VIRUS

WHAT ARE VIRUSES?

All viruses are obligate intracellular parasites; they inhabit a no-man's-land between the living and the non-living worlds, and possess characteristics of both. They are now known to differ radically from the simplest true organisms, bacteria, in a number of respects:

- they cannot be observed using a light microscope
- they have no internal cellular structure
- they contain either DNA or RNA, but not both
- they are incapable of replication unless occupying an appropriate living host cell
- they are incapable of metabolism
- individuals show no increase in size

When inside a host cell, viruses show some of the features of a living organism, such as the ability to replicate themselves, but outside the cell they are just inert chemical structures, thus fuelling the debate as to whether they can be considered to be life forms. A particular virus has a limited host range, that is, it is only able to infect certain cell types.

GENERAL PROPERTIES OF VIRUSES

Viruses are a unique group of infectious agents whose distinctiveness resides in their simple, acellular organization and pattern of reproduction. A complete virus particle or **virion** consists of one or more molecules of DNA or RNA enclosed in a coat of protein, and sometimes also in other layers. These additional layers may be very complex and contain carbohydrates, lipids, and additional proteins. Viruses can exist in two phases: extracellular and intracellular. Virions, the extracellular phase, possess few if any enzymes and cannot reproduce independent of living cells. In the intracellular phase, viruses exist primarily as replicating nucleic acids that induce host metabolism to synthesize virion components; eventually complete virus particles or virions are released.

In summary, viruses differ from living cells in at least three ways: (1) their simple, acellular organization; (2) the presence of either DNA or RNA, but not both, in almost all virions (human cytomegalovirus has a DNA genome and four mRNAs); and (3) their inability to reproduce independent of cells and carry out cell division as procaryotes and eucaryotes do. Although bacteria such as Chlamydia and rickettsia are obligately intracellular parasites like viruses, they do not meet the first two criteria.

Career Avenues

VIRAL STRUCTURE

The demonstration by Wendel Stanley in 1935 that a preparation of tobacco mosaic virus could be crystallised was an indication of the relative chemical homogeneity of viruses, and meant that they could not be thought of in the same terms as other living things. Compared to even the most primitive cellular organism, viruses have a very simple structure (see following figure). An intact viral particle, or *virion*, has in essence just two components: a core of nucleic acid, surrounded and protected by a protein coat or *capsid*, the combination of the two being known as the *nucleocapsid*. In certain virus types, the nucleocapsid is further surrounded by a membranous *envelope*, partly derived from host cell material. Most viruses are smaller than even the smallest bacterial cells.

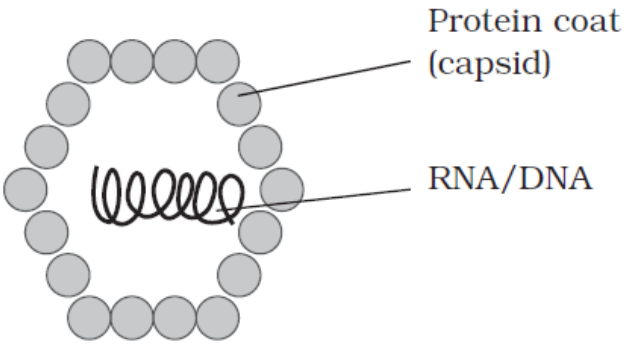


Figure: Viral structure. Viruses comprise a nucleic acid genome surrounded by a protein coat (capsid). Both naked and enveloped forms are shown.

Viroids

Viroids contain RNA only. They are small (less than 400 nucleotides), single stranded, circular RNAs. The RNAs are not packaged, do not appear to code for any proteins, and so far have only been shown to be associated with plant disease. However, there are some suggestions that somewhat similar agents may possibly be involved in some human diseases.

Hepatitis delta virus

At present, the only known human disease agent to resemble viroids is hepatitis delta virus (HDV). In some ways HDV (also called hepatitis delta agent) appears to be intermediate between 'classical viruses' and viroids. HDV has a very small RNA genome (~1700 nucleotides) compared to most viruses, although it is somewhat larger than viroids. However, features of HDV's nucleic acid sequence and structure are similar to some viroids. HDV differs from viroids in that it codes for a protein (various forms of the hepatitis delta antigen). Unlike the viroids, it is packaged. However, it differs from true viruses in that it does not code for its own attachment protein. The RNA is encapsidated by the hepatitis delta antigen, and HDV acts as a parasite on the unrelated hepatitis B virus (HBV), using HBV envelopes containing the hepatitis B attachment protein (HBsAg).

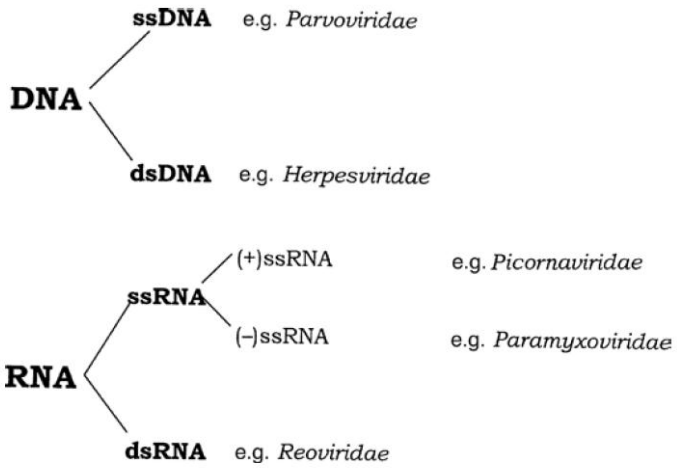
Career Avenues

Prions

Prions contain protein only (although this is somewhat controversial). They are small, proteinaceous particles and there is controversy as to whether they contain any nucleic acid, but if there is any, there is very little, and almost certainly not enough to code for protein: Examples of prion-caused human diseases are Kuru, Creutzfeldt-Jakob disease and Gerstmann-Straussler syndrome. Prions also cause scrapie in sheep.

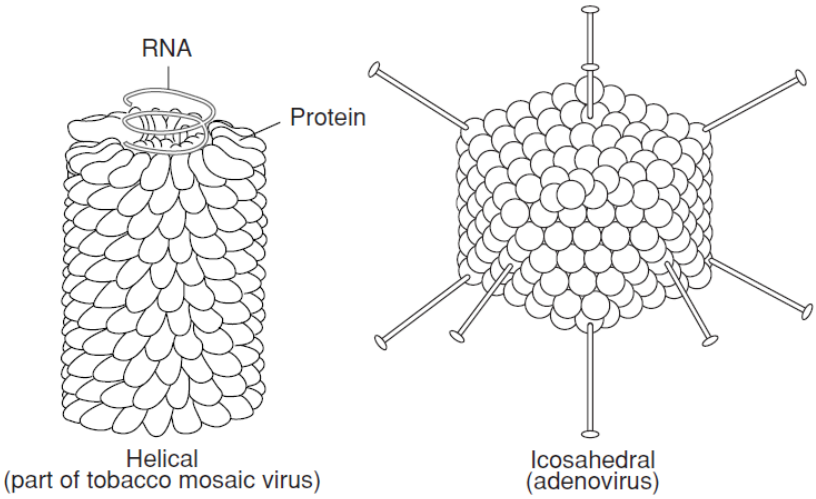
The Viral Genome

The genetic material of a virus may be either RNA or DNA, and either of these may be single-stranded or double-stranded (Figure 10.3). As shown in Figure, the genome may furthermore be circular or linear. An additional variation in the viral genome is seen in certain RNA viruses, such as the influenza virus; here, instead of existing as a single molecule, it is segmented, existing as several pieces, each of which may encode a separate protein. In some plant viruses, the segments may be present in separate particles, so in order for replication to occur, a number of virions need to co-infect a cell, thereby complementing each other (multipartite genomes)! Double-stranded RNA is always present in the segmented form.



The size of the genome varies greatly; it may contain as few as four genes or as many as over 200 (see Box 10.2). These genes may code for both structural and non-structural proteins; the latter include enzymes such as RNA/DNA polymerases required for viral replication.

Single-stranded RNA viral genomes can be divided into two types, known as *(+) sense* and *(−) sense RNA*. The former is able to act as mRNA, attach to ribosomes and become translated into the relevant proteins within the host cell. As such, it is infectious in its own right. Minus (−) sense RNA, on the other hand, is only infectious in the presence of a capsid protein possessing RNA polymerase activity. This is needed to convert the (−) RNA into its complementary (+) strand, which then acts as a template for protein production, as described above. When DNA forms the genome of viruses, it is usually double-stranded (dsDNA), although some of the smaller ones such as the parvoviruses have ssDNA.



CAPSID STRUCTURE

The characteristic shape of a virus particle is determined by its protein coat or capsid. In the non-enveloped viruses, the capsid represents the outermost layer, and plays a role in attaching the virus to the surface of a host cell. It also acts to protect the nucleic acid against harmful environmental factors such as UV light and desiccation, as well as the acid and degradative enzymes encountered in the gastrointestinal tract. The capsid is made up of a number of subunits called *capsomers* and may comprise a few different protein types or just one. The number of capsomers is constant for a particular viral type. This repetitive subunit construction is necessitated by the small amount of protein-encoding RNA/DNA in the viral genome.

The capsomers have the ability to interact with each other spontaneously to form the completed capsid by a process of self-assembly. This would be less easily achieved if there were large numbers of different protein types. Capsomers are arranged symmetrically, giving rise to two principal capsid shapes, *icosahedral* and *helical*. Both shapes can be found in either enveloped or non-enveloped viruses. Complex viruses, such as certain bacteriophages, contain elements of both helical and icosahedral symmetry.

HELICAL CAPSIDS

A number of plant viruses, including the well-studied tobacco mosaic virus, have a rod like structure when viewed under the electron microscope. This is caused by a helical arrangement of capsomers, resulting in a tube or cylinder, with room in the centre for the nucleic acid element, which fits into a groove on the inside. The diameter of the helix is determined by the nature of the protein(s) making up the capsomers; its length depends on the size of the nucleic acid core.

ICOSAHEDRAL CAPSIDS

An icosahedron is a regular three-dimensional shape with 20 triangular faces, and 12 points or corners. The overall effect is of a roughly spherical structure.

Career Avenues

THE VIRAL ENVELOPE

Envelopes are much more common in animal viruses than in those of plants. The lipid bilayer covering an enveloped virus is derived from the nuclear or cytoplasmic membrane of a previous host. Embedded in this, however, are proteins (usually glycoproteins) encoded by the virus's own genome. These may project from the surface of the virion as spikes, which may be instrumental in allowing the virus to bind to or penetrate its host cell. The envelope is more susceptible than the capsid to environmental pressures, and the virus needs to remain moist in order to survive. Consequently, such viruses are transmitted by means of body fluids such as blood (e.g. hepatitis B virus) or respiratory secretions (e.g. influenza virus).

PRINCIPAL EVENTS INVOLVED IN VIRAL REPLICATION

ADSORPTION

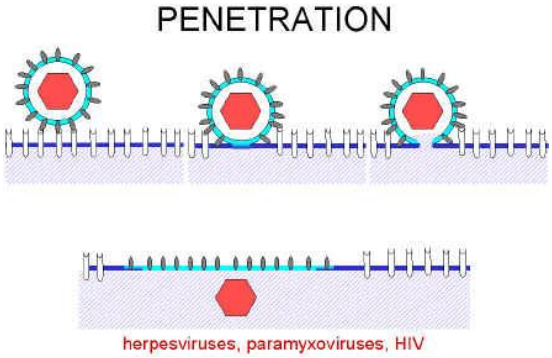
The first step in infection of a cell is attachment to the cell surface. Attachment is via ionic interactions which are temperature-independent. The viral attachment protein recognizes specific receptors, which may be protein, carbohydrate or lipid, on the outside of the cell. Cells without the appropriate receptors are not susceptible to the virus.

PENETRATION

The virus enters the cell in a variety of ways according to the nature of the virus.

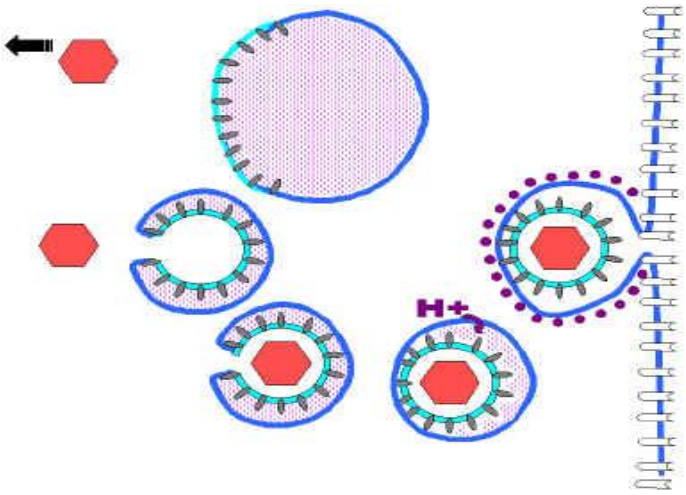
ENVELOPED VIRUSES

- *Entry by fusing with the plasma membrane*. Some enveloped viruses fuse directly with the plasma membrane. Thus, the internal components of the virion are immediately delivered to the cytoplasm of the cell.



Career Avenues

- Entry via endosomes at the cell surface.



Some enveloped viruses require an acid pH for fusion to occur and are unable to fuse directly with the plasma membrane. These viruses are taken up by invagination of the membrane into endosomes. As the endosomes become acidified, the latent fusion activity of the virus proteins becomes activated by the fall in pH and the virion membrane fuses with the endosome membrane. This results in delivery of the internal components of the virus to the cytoplasm of the cell

NON-ENVELOPED VIRUSES

Non-enveloped viruses may cross the plasma membrane directly or may be taken up into endosomes. They then cross (or destroy) the endosomal membrane.

UNCOATING

Nucleic acid has to be sufficiently uncoated that virus replication can begin at this stage. When the nucleic acid is uncoated, infectious virus particles cannot be recovered from the cell - this is the start of the ECLIPSE phase - which lasts until new infectious virions are made.

SYNTHESIS OF VIRAL NUCLEIC ACID AND PROTEIN

ASSEMBLY/MATURATION

New virus particles are assembled. There may be a maturation step that follows the initial assembly process.

RELEASE

Virus may be released due to cell lysis, or, if enveloped, may bud from the cell. Budding viruses do not necessarily kill the cell. Thus, some budding viruses may be able to set up

Career Avenues

persistent infections. Not all released viral particles are infectious. The ratio of non-infectious to infectious particles varies with the virus and the growth conditions.

BACTERIOPHAGES

Viruses that infect bacterial cells are called *bacteriophages* (phages for short), which means, literally, 'bacteria eaters'. Perhaps the best understood of all viral replication cycles are those of a class of bacteriophages which infect *E. coli*, known as the *T-even phages*. These are large, complex viruses, with a characteristic head and tail structure (Figure).

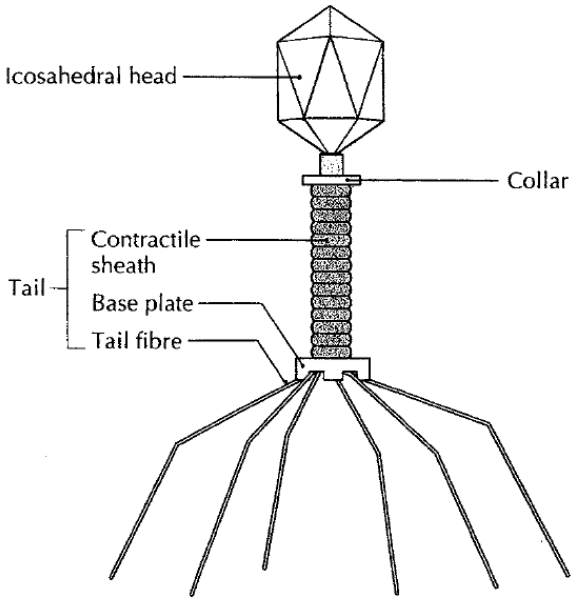


Figure: A T-even bacteriophage.

Note the characteristic 'head plus tail' structure. The tail fibres and base plate are involved in the attachment of the phage to its host cell's surface. The double-stranded, linear DNA genome contains over 100 genes, and is contained within the icosahedral head. The growth cycle is said to be *lytic*, because it culminates in the lysis (=bursting) of the host cell. Figure 10.9 shows the lytic cycle of phage T4, and the main stages are described below.

Career Avenues

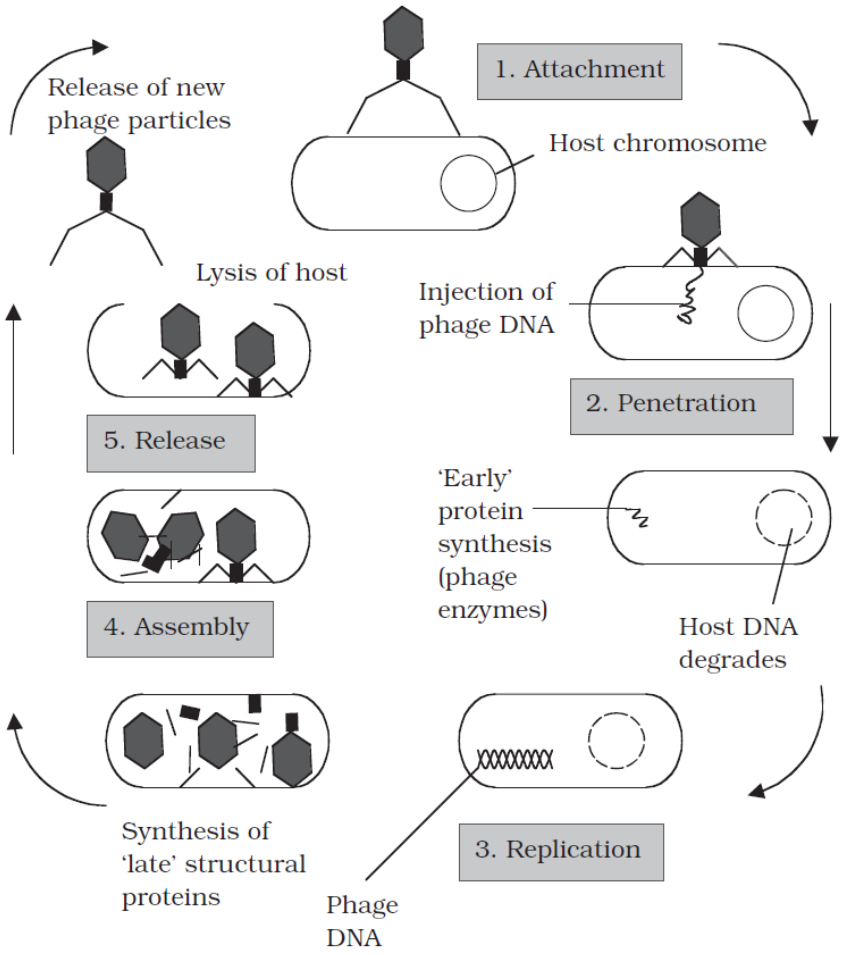


Figure: The lytic cycle of phage T4.

The cycle comprises the five main stages described in the text; from injection of phage of DNA to cell lysis takes 22 minutes. The number of phage particles released per cell is called the *burst size*, and for T4 it ranges from 50 to 200.

1. Adsorption (attachment)

T4 attaches by means of specific tail fibre proteins to complementary receptors on the host cell's surface. The nature of these receptors is one of the main factors in determining a virus's host specificity.

2. Penetration

The enzyme *lysozyme*, present in the tail of the phage, weakens the cell wall at the point of attachment, and a contraction of the tail sheath of the phage causes the core to be pushed down into the cell, releasing the viral DNA into the interior of the bacterium. The capsid remains entirely outside the cell.

Career Avenues

3. Replication

Phage genes cause host protein and nucleic acid synthesis to be switched off, so that all of the host's metabolic machinery becomes dedicated to the synthesis of phage DNA and proteins. Host nucleic acids are degraded by phage-encoded enzymes, thereby providing a supply of nucleotide building blocks. Host enzymes are employed to replicate phage DNA, which is then transcribed into mRNA and translated into protein.

4. Assembly

Once synthesised in sufficient quantities, capsid and DNA components assemble spontaneously into viral particles. The head and tail regions are synthesised separately, then the head is filled with the DNA genome, and joined onto the tail.

5. Release

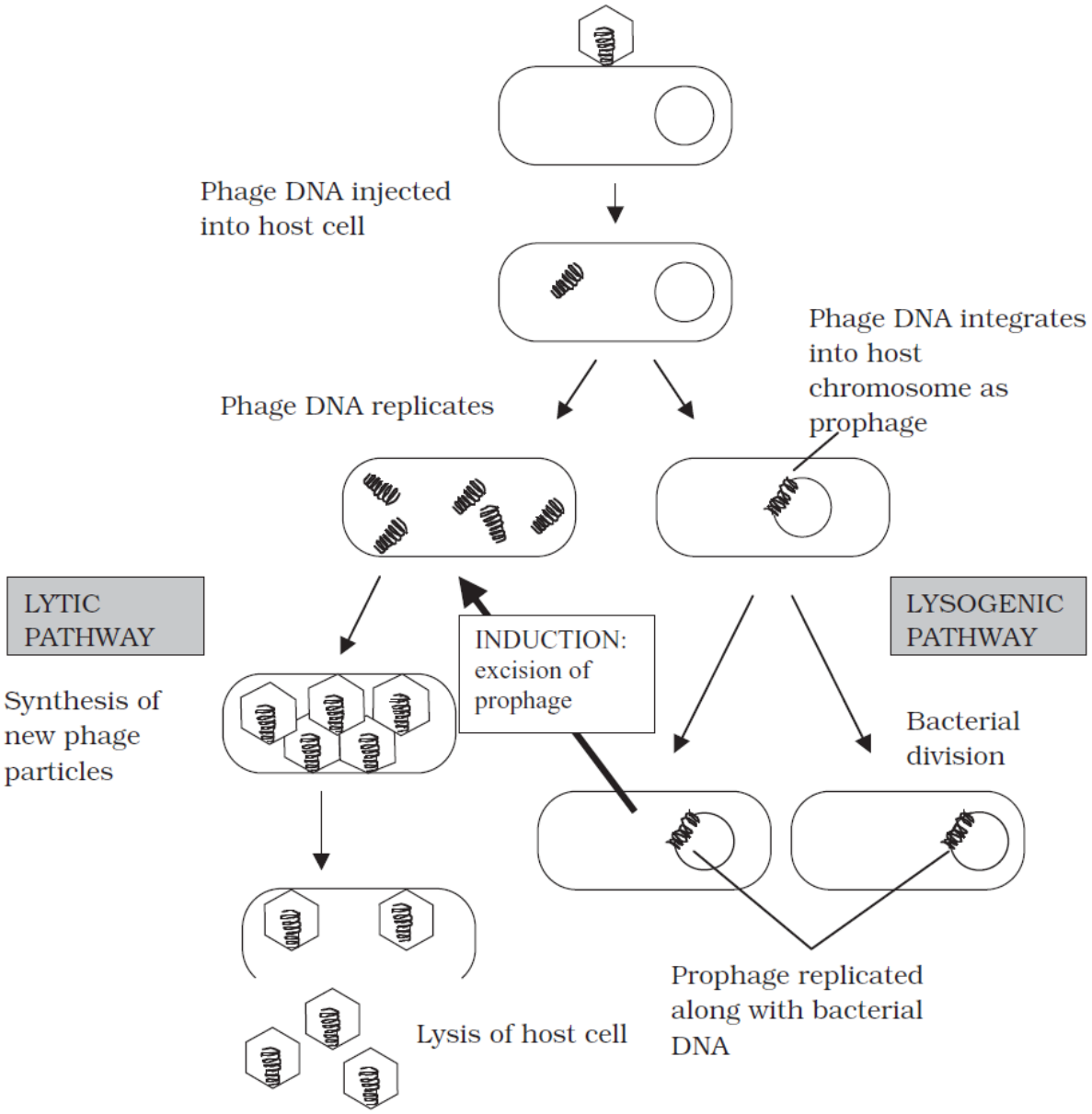
Phage-encoded lysozyme weakens the cell wall, and leads to lysis of the cell and release of viral particles; these are able to infect new host cells, and in so doing recommence the cycle. During the early phase of infection, the host cell contains components of phage, but no complete particles. This period is known as the *eclipse period*. The time which elapses between the attachment of a phage particle to the cell surface and the release of newly-synthesised phages is the *latent period* (sometimes known as the *burst time*); for T4 under optimal conditions, this is around 22 minutes.

THE LYSOGENIC CYCLE

LYSOGENIC REPLICATION CYCLE

Phages such as T4, which cause the lysis of their cells, are termed *virulent* phages. *Temperate* phages, in addition to following a lytic cycle as outlined above, are able to undergo an alternative form of growth cycle. Here, the phage DNA actually becomes incorporated into the host's genome as a *prophage* (Figure). In this condition of *lysogeny*, the host cell suffers no harm. This is because the action of repressor proteins, encoded by the phage, prevents most of the other phage genes being transcribed. These genes are, however, replicated along with the bacterial chromosome, so all the bacterial offspring contain the incorporated prophage. The lysogenic state is ended when the survival of the host cell is threatened, usually by an environmental factor such as UV light or a chemical mutagen. Inactivation of the repressor protein allows the phage DNA to be excised, and adopt a circular form in the cytoplasm. In this form, it initiates a lytic cycle, resulting in destruction of the host cell. An example of a temperate phage is bacteriophage λ (Lambda), which infects certain strains of *E. coli*. Bacterial strains that can incorporate phage DNA in this way are termed *lysogens*.

Career Avenues



Plaque count

The number of plaque forming units (pfu) per millilitre of suspension can be calculated by a plaque count by using the equation:

$$\text{pfu/ml} = \frac{\text{no. of plaques}}{\text{dilution factor} \times \text{volume (ml)}}$$

e.g. 100μl of a 1 in 10,000 dilution gives 53 colonies when plated out.
Thus the original suspension had:

$$\frac{53}{10^{-4} \times 0.1} = 5.3 \times 10^6 \text{ pfu/ml}$$

Virus vaccines

Smallpox, once the scourge of millions, was in 1979 the first infectious disease to be declared successfully eradicated. This followed a worldwide campaign of vaccination by the World Health Organisation over the previous decade, and was made feasible by the fact that humans are the only reservoir for the virus. *Vaccination* is a preventative strategy that aims to stimulate the host immune system, by exposing it to the infectious agent in question in an inactivated or incomplete form.

There are four main classes of virus vaccines:

- ***Attenuated*** **(='weakened') vaccines** contain 'live' viruses, but ones whose pathogenicity has been greatly reduced. The aim is to mimic an infection in order to stimulate an immune response, but without bringing about the disease itself. A famous example of this type of vaccine is the polio vaccine developed by Albert Sabin in the 1960s. The cowpox virus used by Edward Jenner in his pioneering vaccination work in the late 18th century was a naturally occurring attenuated version of the smallpox virus.
- ***Inactivated*** **vaccines** contain viruses which have been exposed to a denaturing agent such as formalin. This has the effect of rendering them non-infectious, while at the same time retaining their ability to stimulate an immune response. Vaccines directed against influenza are of this type.
- ***Subunit*** **vaccines** depend on the stimulation of an immune response by just a part of the virus. Since the complete virus is not introduced, there is no chance of infection, so vaccines of this type have the attraction of being very safe. Subunit vaccines are often made using recombinant DNA technology; the first example to be approved for human use was the hepatitis B vaccine, which consists of part of the protein coat of the virus produced in specially engineered yeast cells.
- ***DNA*** **vaccines** are also the product of modern molecular biology techniques. DNA coding for virus antigens is directly injected into the host, where it is expressed and triggers a response by the immune system. Vaccines of this type have not so far been approved for use in humans.

GATE BT 2011

Q.24 Match the viruses in Group I with their host cell receptors in Group II

Group I	Group II
P. Hepatitis A virus	1. Heparan sulphate
Q. Human immunodeficiency virus	2. Acetylcholine receptor
R. Rabies virus	3. CD4 protein
S. Herpes simplex virus type I	4. Alpha – 2 macroglobulin

(A) P-1, Q-3, R-2, S-4 (B) P-3, Q-4, R-1, S-2

(C) P-4, Q-3, R-2, S-1 (D) P-2, Q-3, R-1, S-4

Career Avenues

Q.25 Match items in Group I with Group II

Group I	Group II
P. Alzheimer's disease	1. H1 N1
Q. Mad cow disease	2.Hemoglobin
R. Sickle cell anaemia	3. Prions
S. Swine flu	4. Amyloid

(A) P-4, Q-3, R-2, S-1 (B) P-3, Q-4, R-1, S-2

(C) P-2 Q-1, R-4, S-3 (D) P-1 Q-2, R-3, S-4

Answers:

- **Q.24 – C.**
- **Q.25 – A.**

Career Avenues

PRACTICE QUESTIONS

1. The best descriptive term for the resident flora is _____.
2. Resident flora is commonly found in the _____.
3. Resident flora is absent from the _____.
4. Virulence factors include _____.
5. The specific action of hemolysins is to damage _____.
6. The _____ is the time that lapses between encounter with a pathogen and the first symptoms.
7. A short period early in a disease that manifests with general malaise and achiness is the _____.
8. The presence of a few bacteria in the blood is termed _____.
9. A _____ infection is acquired in a hospital.
10. A/an _____ is a passive animal transporter of pathogens.
11. An example of a non-communicable infection is _____.
12. A general term that refers to an increased white blood cell count is _____.
13. A positive antibody test for HIV would be a _____ of infection.
14. Which of the following would not be a portal of entry?
15. Which of the following is not a condition of Koch's postulates?
16. In toxocariasis _____.
17. Onchoceriasis is _____
18. Propiobacterium acnes is _____.
19. What is a microorganism?
20. What is a pathogenic microorganism?
21. Name the parts of this microorganism using the nomenclature system: Mycobacterium tuberculosis.
22. Why is a bacteria called a prokaryotic organism?
23. Why a fungus is called a eukaryotic microorganism?

Career Avenues

24. What is Archaea?

25. What is phagocytosis?

26. What is a compound microscope?

27. What is Germ Theory?

28. What is Edward Jenner's contribution to microbiology?

29. Viruses may contain _____.

30. Viruses are _____.

31. Which of the following statements are true?

    a] All viruses are sensitive to antiviral agents. Virus infected cells may be transformed.
    b] Viruses may have a lipid envelope.
    c] Viruses may produce cytopathic changes in cell culture. Some viruses are destroyed by lipid solvents.

32. The following are direct detection methods.

    a] Detection of rotavirus antigen in faecal specimens
    b] Single radial haemolysis (SRH) CMV DEAFF test
    c] Electron microscopy
    d] Polymerase chain reaction

33. The following methods may be used for serological diagnosis:

    a] Complement-fixation tests (CFT)
    b] Polymerase chain reaction (PCR)
    c] Single Radial Haemolysis (SRH)
    d] CMV DEAFF test
    e] Western blot

34. A serological diagnosis of a primary viral infection may be made

    a] Detection of viral-specific IgA
    b] Detection of viral-specific IgD
    c] Detection of viral-specific Ig
    d] Detection of viral-specific IgM
    e] Seroconversion

35. The following are examples of viral genome detection (molecular methods)

    a] Southern blot
    b] Western blot
    c] RIBA (Recombinant immunoblot assay)
    d] Branched DNA
    e] Polymerase chain reaction (PCR)

Career Avenues

36. The following statements are true

a] For cytomegalovirus (CMV), the cytopathic (CPE) effect usually appears within 24-48 hours
b] For some viruses, the CPE is so characteristic that so further identification is required.
c] Paramyxovirus causes syncytia formation in cell culture.
d] A given virus always produce identical CPE in different cell cultures.
e] Immmunofluroescence may be used to identify a virus in cell culture.

37. Poliovirus can be typed by

a] Single radial haemolysis (SRH)
b] Haemagglutination inhibition test (HAI)
c] DEAFF test
d] Neutralization test
e] Hybridization with specific nucleic acid probes

38. Immunofluorescence techniques can be used to detect the following directly from the specimen.

a] True Chlamydia
b] True CMV
c] True Respiratory Syncytial Virus (RSV)
d] True Influenza virus
e] True Rabies virus

39. The following statements are true for the haemagglutination-inhibition (HAI) test.

a] Not a quantitative test
b] Treatment of patient serum is necessary to remove non-specific inhibitors
c] Animal blood is necessary
d] Usually more specific than complement fixation tests (CFT)
e] May be used for the diagnosis of rubella infection

40. Regarding cell culture

a] Viruses can only be cultured using cell lines
b] The presence of cytopathic effect is the only way to detect a virus.
c] The neutralization test is the mainstay of identification of a poliovirus isolate.
d] The haemagglutination inhibition test is the mainstay of identification of a respiratory syncytial virus (RSV) isolate.
e] Whole blood is the specimen of choice for many common viruses

41. A standard Polymerase Chain Reaction (PCR) consists of

a] Denaturation, annealing, and ligation steps.
b] Denaturation, annealing, and extension steps.
c] dNTPs.
d] Mg++ ions.
e] Taq polymerase

Career Avenues

42. Modification of a standard PCR include

a] Nested PCR
b] branched DNA (bDNA)
c] RT-PCR (Reverse transcription PCR
d] Quantitative PCR
e] 3SR (Isothermal amplification)

43. Safety measures for preventing PCR contamination include.

a] The use of uracil-N-glycosylase (UNG).
b] Use of filtered pipette tips.
c] Separate areas for master mix, template, and PCR product operation.
d] Dedicated pipettes for master mix, template, and PCR products.
e] Ultraviolet irradiation

44. The main product of glycolysis under aerobic conditions is

a] Pyruvate b] Lactate c] None of these d] Both a and b

45. The protein moiety of an enzyme is known as

a] Holo enzyme b] Apo enzyme c] Co enzyme d] Enzyme

46. Yeast extract is an excellent source of

a] A Vitamin b] Proteins c] B Vitamin d] Carbohydrates

47. Example of anaerobic medium

a] Wilson blair medium
b] Mac conkey broth
c] Robertson's cooked meat medium
d] EMB agar

48. Biological Oxygen Demand (BOD) is a measure of:

a] Industrial wastes poured into water bodies
b] Extent to which water is polluted with organic compounds
c] Amount of carbon monoxide inseparably combined with haemoglobin
d] Amount of oxygen needed by green plants during night

49. An example of competitive inhibition of an enzyme is the inhibition of

a] Succinic dehydrogenase by malonic acid
b] Cytochrome oxidase by cyanide
c] Hexokinase by glucose-6-phosphate
d] Carbonic anhydrase by carbon dioxide

50. The following organisms have been proposed as sources of single cell protein

a] Bacteria b] Yeasts c] Algae d] All the three

51. Nitrites are oxidized to nitrates by a microorganism

a] Nitrosomonas b] Nitrosococcus c] Nitrobacter d] Azatobacter

52. The major constituents in agar are

a] Fats b] Aminoacids c] Polysaccharides d] Polypeptides

53. Match the following expressions with their respective bacteria A to E:

Career Avenues

1]	$K = \log (a/a - x) \times t^1$	A. Temperature effect
2]	$K = Cn\ t$	B. Watson's expression
3]	$K1/K2 = q(T2-T1)$	C. Concentration ofbactericide
4]	$x2 = 4D\ t\ In\ (mo/m)$	D. Film coefficient
		E. Fick's law

54. Multiple antibiotic resistance is mediated by

a] Episome b] Plasmid c] Colplasmid d] Both b and c

55. "Antagonism" is seen in

a] Lag phase b] Plasmids c] Log phase d] None of these

56. The first phase of a growth curve is

a] Log phase b] Lag phase c] gamma phase d] Both a and b

57. In gram positive and gram negative bacteria the electron transport contains

a] Naphthquinone b] Plastoquinone
c] Ubiquinone d] Both a and b

58. Growth in a closed system, affected by nutrient limitation and waste product accumulation is called

a] Batch culturing b] Ascus
c] Fruiting body d] Sporangiosphore

59. Cells are active and synthesizing new protoplasm. This stage of growth is called

a] Lag phase b] Stationary phase
c] Log phase d] All of these

60. Which one of the following tissues can metabolize glucose, fatty acids and ketone bodies for ATP production?

a] Liver b] Muscle c] Brain d] RBC

61. Which one of the following mineral elements play an important role in biological nitrogen fixation?

a] Copper b] Magnesium c] Zinc d] Molybdenum

62. Rapid bacterial growth phase is known as

a] Log b] Lag c] Lack d] None of these

Career Avenues

63. Clostridium welchii spore formation can be induced only on specified media such as

a] Wilson-Blair medium
b] Macconkey medium
c] Ellner medium
d] Thayee-Martion medium

64. Mycotoxins are formed during the end of

a] Lag phase b] Log phase
c] Death phase d] Stationary phase

65. Bacteria which need oxygen for growth are called

a] Thermophilic bacteria
b] Microaerophilic bacteria
c] Facultative anaerobic bacteria
d] Mycobacteria

66. pH required for the growth of bacteria is

a] 6.8 – 7.2 b] 5.6 – 8.2 c] 3.0 – 6.0 d] 8.0 – 14.0

67. Drug resistance in bacteria is mainly determined by factor:

a] F b] R c] Col d] Lysogenic factor

68. The ion that is required in trace amounts for the growth of bacteria is

a] Calcium b] Magnesium c] Cobalt d] Sodium

69. The most important vitamin for the growth of bacteria is

a] B-complex b] Vitamin A c] Vitamin D d] Vitamin C

70. The principle in microbiological assays is

a] At certain range the concentration of growth factor will bear a linear relationship to the amount of nutrients added
b] Concentration of growth factor have a linear relationship with the growth of the organism
c] Both a and b
d] None of the above

71. If the source of energy for bacteria is from chemical compounds they are said to be

a] Phototrophs b] Autotrophs c] Chemotrophs d] Chemolithotroph

72. In the synthesis of cell components the major element required is

a] Nitrogen b] Sulphur c] Carbon d] Oxygen

73. For the formation of cell-components the elements required are

a] Nitrogen b] Oxygen c] Sulphur d] All of these

Career Avenues

74. For the synthesis of amino acids cysteine, cystine and methionine the element required is

a] Sulphur b] Oxygen c] Nitrogen d] None of these

75. Sulphur can be utilized by bacteria in the form of

a] Organic compounds b] Inorganic compounds
c] Elemental compounds d] All of these

76. Phosphorous is an essential component of

a] Nucleotides
b] Nucleic acids
c] Phospholipids and Heichoic acids
d] All the above

77. Trace elements are

a] Zn+2, Cu+2, Mn+2 b] MO6+, Ni2+, B3+ and CO2+
c] Both a and b d] None of these

78. Most bacteria do not require the ion

a] Mg2+ b] Ca2+
c] Na+ d] Fe+2

79. The vitamin required for Lactobacillus species is

a] Riboflavin b] Niacin
c] Pyridoxine d] Folic acid

80. Vitamin K is necessary for the species

a] Lactobacillus spp. b] Bacillus anthracis
c] Bacteroides melaninogenicus d] All of these

81. The bacteria which are able to grow at 0°C but which grow at 20°C to 30°C, are known as

a] Psychrophiles b] Facultative psychrophiles
c] Average psychrophiles d] Mesophiles

82. Radical shifts can be prevented by adding

a] Acids b] Alkali
c] Buffer d] None of these

83. The orderly increase in the quantity of all the cellular components is known as

a] Reproduction b] Growth c] Binary fission d] None of these

84. The most common mode of cell division in bacteria is

a] Binary fission b] Transverse binary fission
c] Longitudinal binary fission d] None of these

Career Avenues

85. How much time a bacteria take for the complete duplication?

a] 30 min. b] 10 min.
c] 20 min. d] 25 min.

86. The generation time is

a] The time required for the cell to divide
b] The total division of the cell during its life time
c] The total no.of cells formed
d] None of these

87. In bacteria, the increase in population is in the manner

a] Geometric progression b] Multiplication
c] Doubling d] None of these

88. Physiologically the cells are active and are synthesizing new protoplasm in which stage of the growth in bacteria

a] Log phase b] Lag phase
c] Stationary phase d] None of these

89. The most active stage in the sigmoid curve of bacteria in which maximum growth is attained

a] Lag phase b] Stationary phase
c] Decline phase d] Log phase

90. Log-phase is also known as

a] Death phase b] Exponential phase
c] Lag-phase d] None

91. The number of generations per hour in a bacteria is

a] Growth rate b] Generation time
c] Sigmoid curve d] None of these

92. In the sigmoid curve (or) growth curve of bacteria how many stages are there

a] 3 b] 4 c] 2 d] 5

93. The reproduction rate is equal to death rate in which stage

a] Decline phase b] Stationary phase
c] Lag phase d] Log phase

94. Minimum growth temperature is

a] The growth of organisms at lowest temperature
b] The lowest temperature at which the microorganisms grow
c] The maximum temperature at which the growth is stable
d] None of these

Career Avenues

95. Optimum growth temperature is greater that 45°C is

a] Mesophiles b] Thermophiles
c] Psychrophiles d] None of these

96. The organisms which can grow both in presence and absence of oxygen

a] Aerobes b] Anaerobes
c] Faculative anaerobes d] Strict aerobes

97. The organisms which can grow best in the presence of a low concentration of oxygen

a] Aerophilic b] Microaerophilic
c] Aerobic d] Anaerobic

98. The compound that is added to the medium to absorb oxygen for the creation of anaerobic conditions

a] Sodium Thioglycollate b] Nitrous acid
c] Citrate d] None of these

99. The utilization of light energy to drive the synthesis of ATP is called as

a] Photolysis b] Photophosphorylation
c] Photosynthesis d] Respiration

100. During cyclic phosphorylation NADP is formed or not.

a] No NADP formation b] No NADP utilization
c] NADP is converted into NADPH d] All are correct

101. Cyclic phosphorylation is generally present in

a] Cyanobacteria b] Algae c] Bacteria d] Plants

102. Non-cyclic photophosphorylation is also known as

a] Oxygenic photosynthesis b] Photosynthesis
c] Anoxygenic photosynthesis d] Photophosphorylation

103. The number of ATP molecules formed during cyclic phosphorylation are

a] One b] Two c] Four d] Six

104. Artificial transformation in laboratory is carried out by treating the cells with

a] MgCl2 b] Cacl2 c] NaCl d] HCl

105. The process of formation of mesozygote is called

a] Meromixis b] Exozygote c] Mitosis d] Meiosis

106. Which of the following organisms requires tryptophan for growth?

a] H.influenza b] Vibrio c] Gonococci d] S.typhi

Career Avenues

107. Tubercular bacilli grow best in

a] Absence of O2 b] Presence of CO2
c] Presence of O2 d] None of these

108. Mycotoxins are formed during the end of

a] Lag phase b] Log phase
c] Death phase d] Stationary phase

109. Match the following growth characteristics with their respective temperature ranges A to E.

1.	Psychrotrophs	A] Grows between 55 to 65oC
2.	Mesophils	B] May survive above 60oC
3.	Thermophils	C] Grow well between 25 to 45oC
4.	vegetable bacteria	D] Grow below 25oC
		E] Multiply slowly at 0-4oC

110. Match the following microorganisms with their respective sources A to E:

1.	Achrommobacter	A] Bread spp
2.	Aspergillus flavus	B] Water supply
3.	Oscillatiria	C] Meat scytonema
4.	Clostridium	D] Salad nigereticans
		E] Milk and cheese products

111. Match the following microorganisms with their respective appearance of colonies on bismuth Sulphite agar from A to E:

1.	Salmonella typhi	A] Brown
2.	Salmonella	B] No growth choleraesuis
3.	Shigella flexneri	C] Green
4.	Escherichia coli	D] Yellow
		E] Black

112. The suitable temperature to transport viral culture is –

a] $30^{o}C$ b] $5^{o}C$ c] $25^{o}C$ d] $45^{o}C$
e] None of these

113. Growth curve does not include following phases of bacteria –

a] Decline phase b] Stationary phase
c] Lag phase d] Synchronous growth

114. Bacteria are more sensitive to antibiotics at which phase of growth curve?

a] Decline phase b] Stationary phase
c] Lag phase d] Log phase

ANSWER KEY

1. Commensals

Career Avenues

2. Urethra

3. Lungs

4. Toxins, enzymes, capsules and many more…

5. Damage red blood cells

6. Period of incubation

7. Prodromium

8. Bacteremia

9. Nosocomial

10. Mechanical vector

11. Tetanus

12. Leukocytosis

13. Sign

14. The meninges

15. Test the effects of pathogen on humans

16. The infestation is acquired from both dogs and cats

17. Ocular involvement results from lymphatic migration of the parasites

18. an important cause of acute post-operative endophthalmitis

19. A microorganism is a small organism that takes in and breaks downfood for energy and nutrients, excretes unused food as waste, and iscapable of reproduction.

20. A disease-causing microorganism

21. Mycobacterium is the genus and tuberculosis is the specific epithet

22. A bacterium is a one-cell organism that does not have a distinct nucleus.

23. Fungus has cells that have a nucleus, nuclear envelope, cytoplasm

24. Archaea is a classification of an organism that identifies prokaryotes that do not have peptidoglycan cell walls

25. The ability of a cell to engulf and digest solid materials by use of pseudopods, or “false feet.”

26. A microscope that has two sets of lenses: an ocular lens and an eyepiece.

Career Avenues

27. Germ Theory states that a disease-causing microorganism should be present in animals infected by the disease and not in healthy animals.

28. Edward Jenner discovered how to create vaccinations to trigger the body's immune system to develop antibodies that fight microorganisms.

29. Either DNA, or RNA. They may contain enzymes such as polymerases, and have glycoproteins in their envelope

30. Unlike bacteria, viruses do not have their own metabolism and do not divide by binary fission. They are intracellular parasites and may contain enzymes for their replication. Some viruses possess a lipid envelope.

31. To date, only a few viruses can be treated by antiviral agents. Some viruses such as oncornaviruses can transform cells. Some may have a lipid envelope which may be destroyed by lipid solvents.

32. SRH is a serological assay, and the CMV DEAFF test is a rapid culture test. The rest are used to detect virus particles, antigens, and nucleic acid directly from a specimen

33. PCR detects viral genomes directly. The CMV DEAFF test is a rapid culture assay. Western blot is mainly used as a serological test.

34. A diagnosis of a primary viral infection may be made by the detection of IgM and/or seroconversion.

35. Southern blot, branched DNA, and PCR are viral genome detection methods. Western blot and RIBA are serological assays

36. It takes 1-3 weeks for CMV-specific CPE to appear. For some viruses such as herpes simplex, the CPE is so characteristic that so further identification is required

37. Polioviruses can be typed by neutralization tests and hybridization with specific nucleic acid probes. SRH and HAI are not used. The DEAFF test is used for the isolation of CMV

38. All the above are true

39. HAI is a quantitative test. It is widely used in rubella serology. It is usually more specific than CFT. Treatment of patient serum is necessary to remove non-specific inhibitors.

40. Viruses may also be cultured in eggs and animals. The presence of virus may be detected by CPE as well as haemadsoption, and presence of viral particles and antigen.

41. Denaturation, annealing, and ligation steps occur in LCR. PCR consists of Denaturation, annealing, and extension steps. It requires dNTP, Mg++, taq polymerase, and target-specific oligonucleotide primers.

42. Nested PCR, RT-PCR and quantitative PCR are modifications of the PCR protocol. bDNA and 3SR are alternative amplification techniques.

Career Avenues

43. All the above may be useful in preventing contamination of PCR.

44-a 45-b 46-c 47-c 48-a 49-a 50-d 51-c 52-c

53-1.b,2.c,3.a,4.e 54-b 55-d 56-b 57-a 58-a 59-a

60-b 61-d 62-a 63-c 64-a 65-b 66-d 67-c 68-a

69-b 70-c 71-c 72-d 73-d 74-a 75-a 76-d 77-d

78-c 79-b 80-a 81-c 82-c 83-b 84-c 85-c 86-c

87-a 88-c 89-d 90-c 91-b 92-b 93-d 94-b 95-a

96-a 97-b 98-b 99-c 100-a 101-a 102-b 103-d 104-b

105-a 106-d 107-b 108-a 109-1.b, 2.c, 3.d, 4.a 110-1.e,2.a,3.b,4.c

111-1.e,2.c,3.a,4.b 112-b 113-d 114-d

Career Avenues